PART VERIFICATION · REFERENCE COMPARISON

IDEEZA Part Comparison Log

6 submitted library parts checked against standard KiCad / EasyEDA packages and manufacturer datasheets, section by section: schematic symbol, footprint and 3D model.

5	23	20	26	74
HIGHEST	HIGH	MEDIUM	LOW	TOTAL ISSUES

PARTS

1. **TL16C2550PFBR** Texas Instruments, dual UART, TQFP-48 (PFB)
2. **74LVC1G99GT,115** Nexperia, low-power configurable multi-function gate, XSON8 (SOT833-1)
3. **AD9106BCPZ** Analog Devices, quad 12-bit DAC waveform generator, LFCSP-32 5 × 5 mm (CP-32-12)
4. **693072010801** Würth Elektronik, WR-CRD microSD card connector, hinge type, SMD, 8 contacts + 4 shield tabs
5. **ADV7180KST48Z** Analog Devices, 10-bit SDTV video decoder, 48-lead LQFP 7 × 7 mm (ST-48)
6. **PGA2311P** Texas Instruments, stereo audio volume control, PDIP-16

SEEN ON EVERY PART (FIX ONCE IN THE PART GENERATOR)

- Paste-mask shapes are tagged as copper (layerId F.Cu); nothing is on F.Paste.
- Solder-mask margin is declared 0.102 mm but drawn at 0.051 mm per side.
- The pin-1 marker is a dot outside the courtyard.
- No real 3D model is stored with the part.
- Pin labels come from the datasheet's formatted or grouped text instead of the per-pin name (grouped labels, scrambled subscripts, swapped GPOs).
- On 4 of 6 parts the footprint can't be soldered: a different package on 3 parts (wrong land pattern, or a through-hole DIP for an SMD part) and misplaced pad rows on TL16C2550.

Updated 2026-10-05 · Each issue shows the standard file and the submitted file side by side, with Jira fields.

PART	HIGHEST	HIGH	MEDIUM	LOW	TOTAL
TL16C2550PFBR Schematic 5 · Footprint 10 · 3D 1	1	5	5	5	**16**
74LVC1G99GT,115 Schematic 3 · Footprint 6 · 3D 1	1	2	4	3	**10**
AD9106BCPZ Schematic 5 · Footprint 7 · 3D 1	1	5	2	5	**13**
693072010801 Schematic 4 · Footprint 7 · 3D 1	1	4	3	4	**12**
ADV7180KST48Z Schematic 6 · Footprint 7 · 3D 1	1	5	3	5	**14**
PGA2311P Schematic 3 · Footprint 5 · 3D 1	–	2	3	4	**9**

PART 1 OF 6

TL16C2550PFBR

Part Texas Instruments TL16C2550PFBR, dual UART, TQFP-48 (PFB)

Submitted part library ID 24150

Reference SnapEDA KiCad package (symbol, footprint, STEP)

> **Not usable as submitted.** The footprint's top and bottom pad rows miss the leads, so 24 of 48 pins can't solder (FP-01), and there is no real 3D model, only a generated box of the wrong size (3D-01). The symbol has the right pin numbers, but 31 of its labels are wrong and none of its active-low pins are marked.

SCHEMATIC SYMBOL

Reference: TL16C2550PFBR.kicad_sym · Submitted: schematic.glb

PIN-BY-PIN COMPARISON

48 standard pins: 14 match, 1 differ in electrical type only, 33 with a naming, label or polarity problem

PIN	STANDARD NAME	SUBMITTED NAME	LABEL DRAWN	STANDARD TYPE	SUBMITTED TYPE	ACTIVE-LOW (STD / SUB)	RESULT
1	D5	D5	D5-D7	bidirectional	bidirectional	– / –	**drawn as "D5-D7"**
2	D6	D6	D5-D7	bidirectional	bidirectional	– / –	**drawn as "D5-D7"**
3	D7	D7	D5-D7	bidirectional	bidirectional	– / –	**drawn as "D5-D7"**
4	RXB	RXB	RXB	input	input	– / –	match
5	RXA	RXA	RXA	input	input	– / –	match
6	TXRDYB	TXRDYB	TXRDYA, TXRDYB	output	output	yes / –	**drawn as "TXRDYA, TXRDYB"; active-low not set**
7	TXA	TXA	TXA	output	output	– / –	match
8	TXB	TXB	TXB	output	output	– / –	match
9	OPB	OPB	OPA, OPB	output	output	yes / –	**drawn as "OPA, OPB"; active-low not set**
10	CSA	CSA	CSA, CSB	input	input	yes / –	**drawn as "CSA, CSB"; active-low not set**
11	CSB	CSB	CSA, CSB	input	input	yes / –	**drawn as "CSA, CSB"; active-low not set**
12	NC	NC	NC	no_connect	no_connect	– / –	match
13	XTAL1	XTAL1	XTAL1	input	input	– / –	match
14	XTAL2	XTAL2	XTAL2	output	output	– / –	match
15	IOW	IOW	IOW	input	input	yes / –	**active-low not set**
16	CDB	CDB	CDA, CDB	input	input	yes / –	**drawn as "CDA, CDB"; active-low not set**
17	GND	GND	GND	passive	power_in	– / –	type differs
18	RXRDYB	RXRDYB	RXRDYA, RXRDYB	output	output	yes / –	**drawn as "RXRDYA, RXRDYB"; active-low not set**
19	IOR	IOR	IOR	input	input	yes / –	**active-low not set**
20	DSRB	DSRB	DSRA, DSRB	input	input	yes / –	**drawn as "DSRA, DSRB"; active-low not set**
21	RIB	RIB	RIA, RIB	input	input	yes / –	**drawn as "RIA, RIB"; active-low not set**
22	RTSB	RTSB	RTSA, RTSB	output	output	yes / –	**drawn as "RTSA, RTSB"; active-low not set**
23	CTSB	CTSB	CTSA, CTSB	input	input	yes / –	**drawn as "CTSA, CTSB"; active-low not set**

PIN	STANDARD NAME	SUBMITTED NAME	LABEL DRAWN	STANDARD TYPE	SUBMITTED TYPE	ACTIVE-LOW (STD / SUB)	RESULT
24	NC	NC	NC	no_connect	no_connect	– / –	match
25	NC	NC	NC	no_connect	no_connect	– / –	match
26	A2	A2	A2	input	input	– / –	match
27	A1	A1	A1	input	input	– / –	match
28	A0	A0	A0	input	input	– / –	match
29	INTB	INTB	INTA, INTB	output	output	– / –	**drawn as "INTA, INTB"**
30	INTA	INTA	INTA, INTB	output	output	– / –	**drawn as "INTA, INTB"**
31	RXRDYA	RXRDYA	RXRDYA, RXRDYB	output	output	yes / –	**drawn as "RXRDYA, RXRDYB"; active-low not set**
32	OPA	OPA	OPA, OPB	output	output	yes / –	**drawn as "OPA, OPB"; active-low not set**
33	RTSA	RTSA	RTSA, RTSB	output	output	yes / –	**drawn as "RTSA, RTSB"; active-low not set**
34	DTRA	DTRA	DTRA, DTRB	output	output	yes / –	**drawn as "DTRA, DTRB"; active-low not set**
35	DTRB	DTRB	DTRA, DTRB	output	output	yes / –	**drawn as "DTRA, DTRB"; active-low not set**
36	RESET	RESET	RESET	input	input	– / –	match
37	NC	NC	NC	no_connect	no_connect	– / –	match
38	CTSA	CTSA	CTSA, CTSB	input	input	yes / –	**drawn as "CTSA, CTSB"; active-low not set**
39	DSRA (standard file misspells it DRSA)	DSRA	DSRA, DSRB	input	input	yes / –	**drawn as "DSRA, DSRB"; active-low not set**
40	CDA	CDA	CDA, CDB	input	input	yes / –	**drawn as "CDA, CDB"; active-low not set**
41	RIA	RIA	RIA, RIB	input	input	yes / –	**drawn as "RIA, RIB"; active-low not set**
42	VCC	VCC	V CC	power_in	power_in	– / –	**drawn as "V CC"**
43	TXRDYA	TXRDYA	TXRDYA, TXRDYB	output	output	yes / –	**drawn as "TXRDYA, TXRDYB"; active-low not set**
44	D0	D0	D0-D4	bidirectional	bidirectional	– / –	**drawn as "D0-D4"**
45	D1	D1	D0-D4	bidirectional	bidirectional	– / –	**drawn as "D0-D4"**
46	D2	D2	D0-D4	bidirectional	bidirectional	– / –	**drawn as "D0-D4"**
47	D3	D3	D0-D4	bidirectional	bidirectional	– / –	**drawn as "D0-D4"**
48	D4	D4	D0-D4	bidirectional	bidirectional	– / –	**drawn as "D0-D4"**

MATCHES THE REFERENCE

- All 48 pin numbers carry the right function (pinName matches the reference and datasheet on 48 of 48 pins).
- Electrical types match on 47 of 48 pins. Pin 17 GND is typed power_in, which is more correct than the reference's passive.
- The 4 NC pins (12, 24, 25, 37) are flagged no-connect.
- Pin length is 2.54 mm (100 mil) on all 48 pins, which meets the 100 mil schematic rule. The standard file uses 5.08 mm (200 mil), so here the submitted part is the correct one.

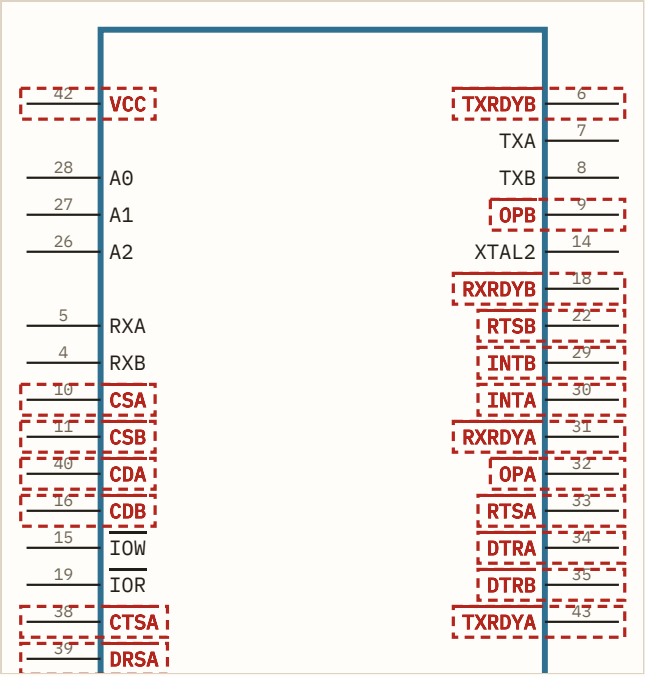
SCH-01 HIGH

31 of 48 pins display a grouped datasheet label instead of their own pin name

Each pin stores the right name in pinName (e.g. pin 10 = CSA, pin 11 = CSB), but the label drawn on the symbol comes from displayName, which holds the datasheet's grouped table row. Pins 10 and 11 both read "CSA, CSB", pins 1-3 all read "D5-D7", pins 44-48 all read "D0-D4", and pin 42 reads "V CC" with a space.

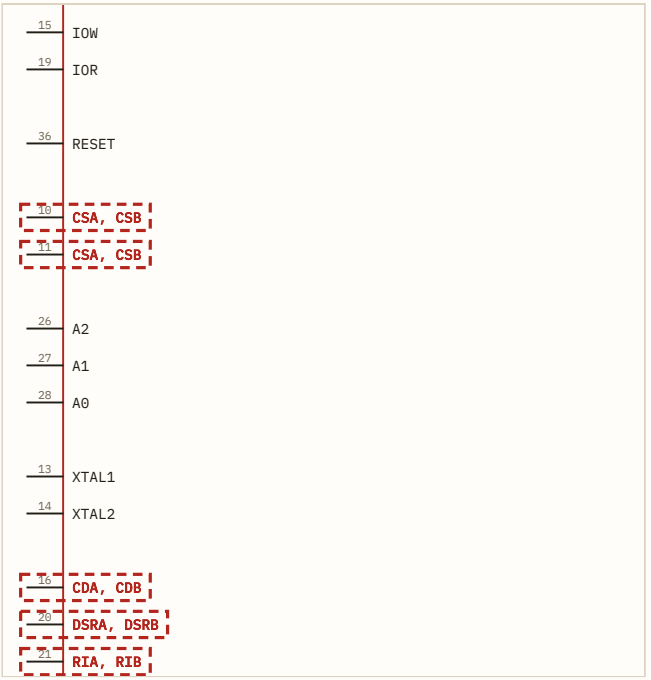
PIN(S)	CORRECT NAME (STORED IN PINNAME)	LABEL SHOWN ON SYMBOL
1, 2, 3	D5, D6, D7	D5-D7
44, 45, 46, 47, 48	D0, D1, D2, D3, D4	D0-D4
10, 11	CSA, CSB	CSA, CSB
6 / 43	TXRDYB / TXRDYA	TXRDYA, TXRDYB
9 / 32	OPB / OPA	OPA, OPB
18 / 31	RXRDYB / RXRDYA	RXRDYA, RXRDYB
16 / 40	CDB / CDA	CDA, CDB
20 / 39	DSRB / DSRA	DSRA, DSRB
21 / 41	RIB / RIA	RIA, RIB
23 / 38	CTSB / CTSA	CTSA, CTSB
22 / 33	RTSB / RTSA	RTSA, RTSB
29 / 30	INTB / INTA	INTA, INTB
34 / 35	DTRA / DTRB	DTRA, DTRB
42	VCC	V CC

STANDARD FILE
TL16C2550PFBR.kicad_sym



Standard: Every pin has its own name: pin 10 is CSA, pin 11 is CSB, pin 16 is CDB, pin 20 is DSRB.

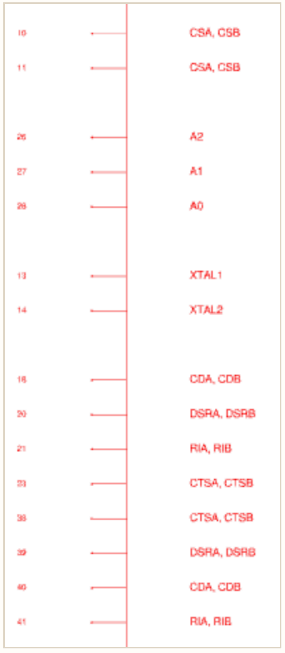
SUBMITTED FILE
schematic.glb



Submitted: Pins 10 and 11 both read "CSA, CSB", pin 16 reads "CDA, CDB", pin 20 "DSRA, DSRB", pin 21 "RIA, RIB" (upper left column shown).

> **Issue:** The channel A pin and the channel B pin carry the same label, so the symbol can't tell them apart.

Both drawn to the same scale, from the files' own pin data. ⛶ Affected pins.



Submitted symbol, left column as rendered: pins 10/11, 16/40, 20/39, 21/41 and 23/38 each show the same label twice.

IMPACT

A designer can't tell channel A from channel B, or which data bit a pin is, from the symbol. Wiring pin 10 vs 11 or D0 vs D4 has to be checked against the pin number every

FIX

- Render the pin label from pinName, not displayName.
- Fix the VCC label (no space).
- Check the same field mapping on other parts made the same way: any datasheet with grouped pin tables

time, which is how swapped-channel and swapped-bit errors get onto boards.

will hit this.

JIRA

Summary	TL16C2550 symbol: 31 pins show grouped labels ("CSA, CSB", "D0-D4") instead of own pin name
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, symbol-error, TL16C2550
Components	Part Creation / Symbol Library
Description	Submitted schematic symbol for TL16C2550PFBR (library ID 24150, schematic.glb). Pin names are stored correctly in pinName, but the visible label is taken from displayName, which holds the datasheet's grouped row text. 31 of 48 pins are affected: pins 10 and 11 both show "CSA, CSB", pins 1-3 show "D5-D7", pins 44-48 show "D0-D4", A/B pairs for TXRDY, OP, RXRDY, CD, DSR, RI, CTS, RTS, INT, DTR are duplicated, and pin 42 shows "V CC". Fix: render pin labels from pinName; correct VCC. Reference: SnapEDA KiCad symbol TL16C2550PFBR.kicad_sym.

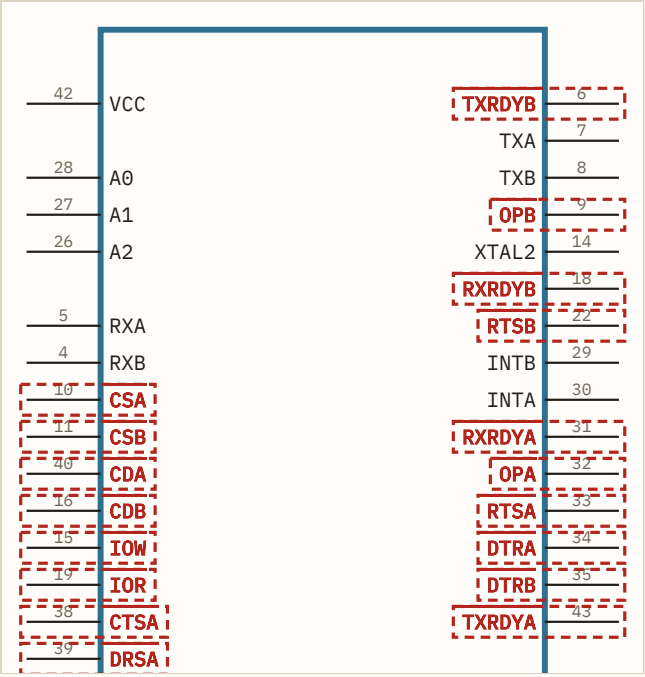
SCH-02 HIGH

No active-low marking on any pin (22 pins should have it)

activeLow is false on all 48 pins, and no label has an overbar. The reference symbol marks 22 pins active-low, matching the datasheet: CSA, CSB, IOR, IOW, CDA, CDB, CTSA, CTSB, DSRA, DSRB, RIA, RIB, DTRA, DTRB, RTSA, RTSB, OPA, OPB, TXRDYA, TXRDYB, RXRDYA, RXRDYB.

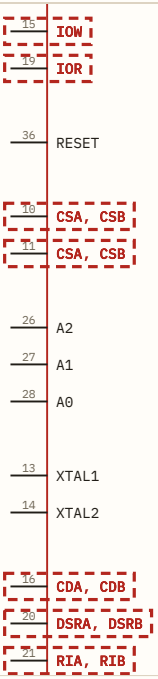
PIN(S)	SIGNAL	DATASHEET POLARITY
10, 11	CSA, CSB	chip selects, active low
15, 19	IOW, IOR	write / read strobes, active low
16, 40, 23, 38, 20, 39, 21, 41	CD, CTS, DSR, RI (A/B)	modem inputs, active low
22, 33, 34, 35, 9, 32	RTS, DTR, OP (A/B)	modem outputs, active low
6, 43, 18, 31	TXRDY, RXRDY (A/B)	DMA ready outputs, active low

STANDARD FILE
TL16C2550PFBR.kicad_sym



Standard: The 22 active-low pins carry an overbar (CSA, CSB, IOW, IOR, CDx, CTSx, DSRx, RIx, DTRx, RTSx, OPx, TXRDYx, RXRDYx).

SUBMITTED FILE
schematic.glb



Submitted: No pin has an overbar. activeLow is false on all 48 pins.

> **Issue:** CSA, CSB, IOR and IOW read as active-high, so they would be wired the wrong way.

Both drawn to the same scale, from the files' own pin data. ⛶ Affected pins.

IMPACT

Without the bar, CSA/CSB, IOR and IOW read as active-high. Tying a chip select high to enable the part, or inverting the read/write strobes, makes the UART unresponsive.

FIX

- Set activeLow = true on the 22 pins above and render the overbar (or a ~ prefix).
- Leave INTA, INTB and RESET active-high: they are active-high on this part.

JIRA

Summary	TL16C2550 symbol: active-low flag missing on 22 pins (CS, IOR, IOW, modem and DMA signals)
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, symbol-error, TL16C2550
Components	Part Creation / Symbol Library
Description	Submitted symbol for TL16C2550PFBR (ID 24150) has activeLow = false on all 48 pins and shows no overbars. 22 pins are active-low per the datasheet and the reference symbol: CSA, CSB, IOR, IOW, CDA, CDB, CTSA, CTSB, DSRA, DSRB, RIA, RIB, DTRA, DTRB, RTSA, RTSB, OPA, OPB, TXRDYA, TXRDYB, RXRDYA, RXRDYB. INTA, INTB and RESET are active-high and correct as is. Fix: set activeLow on those 22 pins and render the overbar.

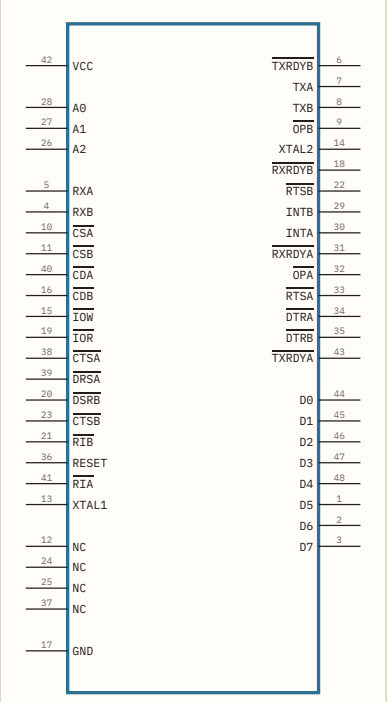
SCH-03 MEDIUM

Pins are not grouped by function: outputs on the input side, data bus split and out of order

34 pins sit on the left and 14 on the right (reference: 25 / 23). The left column mixes the outputs TXA (7), TXB (8) and XTAL2 (14) in with inputs. The data bus is split: D5, D6, D7 come first, then RXB, RXA, TXA, TXB, then D0 to D4, so D0-D7 is neither contiguous nor in order. VCC and GND sit mid-way down the right side instead of top and bottom.

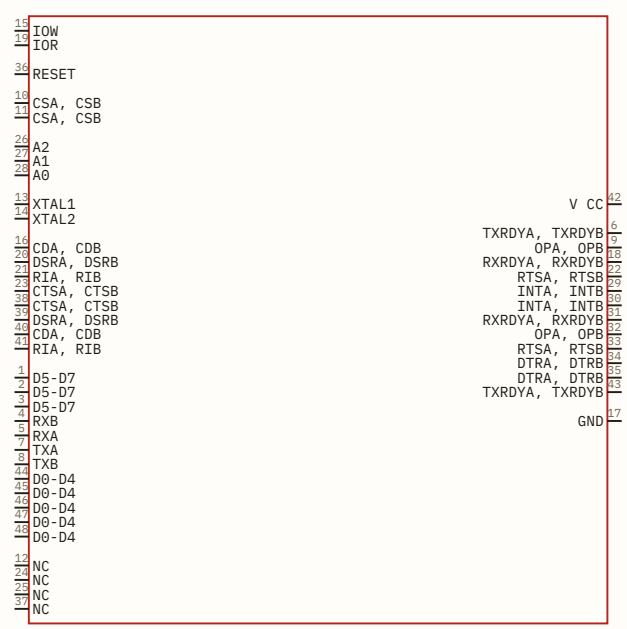
COLUMN	PIN ORDER
Left, top to bottom	IOW IOR RESET CSA CSB A2 A1 A0 XTAL1 XTAL2 CDB DSRB RIB CTSB CTSA DSRA CDA RIA D5 D6 D7 RXB RXA TXA TXB D0 D1 D2 D3 D4 NC NC NC NC
Right, top to bottom	VCC TXRDYB OPB RXRDYB RTSB INTB INTA RXRDYA OPA RTSA DTRA DTRB TXRDYA GND
Reference left	VCC A0 A1 A2 RXA RXB CSA CSB CDA CDB IOW IOR CTSA DSRA* DSRB CTSB RIB RESET RIA XTAL1 NC NC NC NC GND
Reference right	TXRDYB TXA TXB OPB XTAL2 RXRDYB RTSB INTB INTA RXRDYA OPA RTSA DTRA DTRB TXRDYA D0 D1 D2 D3 D4 D5 D6 D7

STANDARD FILE
TL16C2550PFBR.kicad_sym



Standard: Host bus and control inputs on the left (VCC top, GND bottom); outputs and D0–D7 in order on the right.

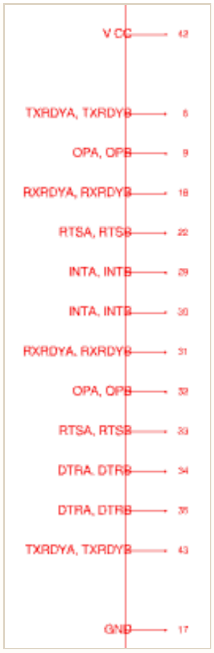
SUBMITTED FILE
schematic.glb



Submitted: 34 pins on the left, 14 on the right. TXA, TXB and XTAL2 (outputs) are on the input side. D5–D7 sit above RX/TX and D0–D4 below. VCC and GND are mid-column on the right.

> **Issue:** Functions are scattered, so the bus and channels are hard to follow on the schematic.

Each symbol drawn to fit its panel (different scales), from the files' own pin data.



Submitted symbol, right column: VCC near the middle, GND below the channel-A outputs.

IMPACT

Bus and channel wiring is harder to read and route on the schematic. It doesn't change connectivity.

FIX

- Inputs left, outputs right (or host bus left, serial/modem right).
- D0-D7 contiguous and in order.
- Keep channel A and channel B signals in their own blocks.
- VCC at the top, GND at the bottom.

JIRA

Summary	TL16C2550 symbol: pin layout not grouped (outputs on input side, D0-D7 split around RX/TX)
Issue Type	Improvement
Priority	Medium
Labels	part-verification, part-library, symbol-layout, TL16C2550
Components	Part Creation / Symbol Library
Description	Submitted symbol for TL16C2550PFBR (ID 24150) puts 34 pins on the left and 14 on the right. Outputs TXA (7), TXB (8) and XTAL2 (14) are on the input side. The data bus reads D5 D6 D7 RXB RXA TXA TXB D0 D1 D2 D3 D4, so D0-D7 is split and out of order. VCC and GND are mid-column on the right. Requested layout: inputs left / outputs right, D0-D7 contiguous and ordered, channel A and B grouped, VCC top and GND bottom.

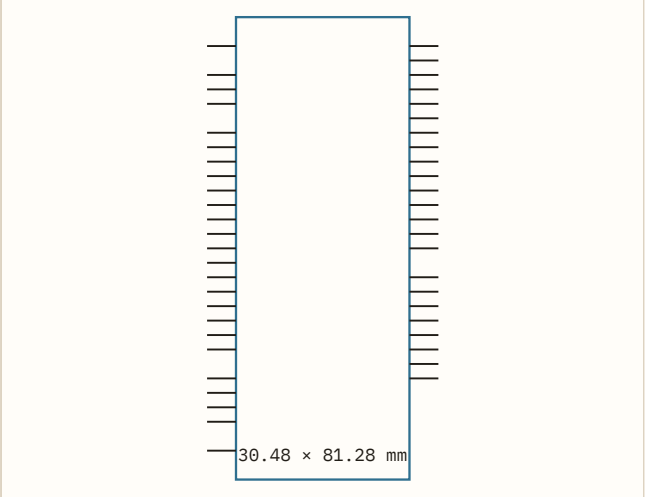
SCH-04 MEDIUM

Symbol body is 101.6 x 106.7 mm, more than 3x wider than the reference and mostly empty

The body outline measures 101.65 x 106.73 mm. The reference body is 30.48 x 81.28 mm. The longest pin label is 6 characters, so about 80 mm of the width is empty. On an A4 landscape sheet (210 mm tall) the body alone takes 51% of the sheet height.

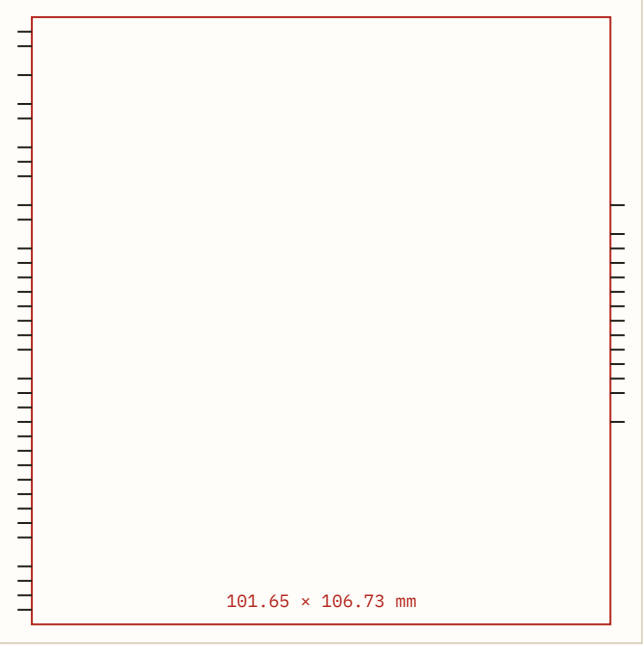
	REFERENCE	SUBMITTED
Body width	30.48 mm	101.65 mm
Body height	81.28 mm	106.73 mm
Pin length	5.08 mm (200 mil)	2.54 mm (100 mil), meets the 100 mil rule
Pin pitch	2.54 mm	2.54 mm

STANDARD FILE
TL16C2550PFBR.kicad_sym



Standard: Body 30.48 × 81.28 mm for 48 pins.

SUBMITTED FILE
schematic.glb



Submitted: Body 101.65 × 106.73 mm, mostly empty.

Issue: The submitted symbol is more than 3× wider than it needs to be and takes half an A4 sheet's height.

Both drawn to the same scale, from the files' own pin data.

IMPACT

The part crowds every sheet it is placed on and pushes the other circuitry onto more sheets.

FIX

- Size the body width from the longest left and right labels plus margin, about 25-30 mm here.
- Size the body height from the longer pin column.

JIRA

Summary	TL16C2550 symbol: body 101.6 x 106.7 mm, over 3x wider than needed
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, symbol-layout, TL16C2550
Components	Part Creation / Symbol Library
Description	Submitted symbol for TL16C2550PFBR (ID 24150) has a body outline of 101.65 x 106.73 mm (reference 30.48 x 81.28 mm). Labels are at most 6 characters, so most of the width is empty, and the body takes 51% of an A4 landscape sheet's height. Size the body width from the label lengths (about 25-30 mm) and the height from the longer pin column.

SCH-05 LOW

Value and MPN fields read TL16C2550 instead of the orderable TL16C2550PFBR

PackageValue on the symbol and manufacturerPartNumber on the footprint are "TL16C2550". The device field is correct (TL16C2550PFBR). The reference Value and MPN are TL16C2550PFBR. The base number TL16C2550 is also sold in other packages, such as 44-pin PLCC and 32-pin QFN, which have different pinouts.

FIELD	REFERENCE	SUBMITTED
Value (symbol)	TL16C2550PFBR	TL16C2550
MPN	TL16C2550PFBR	TL16C2550
Device	-	TL16C2550PFBR

STANDARD FILE
TL16C2550PFBR.kicad_sym

PART PROPERTIES

Value	TL16C2550PFBR
MPN	TL16C2550PFBR
Device	-
Footprint	TL16C2550PFBR:QFP50P900X900X120-48N

Standard: Value and MPN carry the full orderable number TL16C2550PFBR.

SUBMITTED FILE
schematic.glb

PART PROPERTIES

Value	**TL16C2550**
MPN	**TL16C2550**
Device	TL16C2550PFBR
Footprint	TL16C2550PFBR

Submitted: Value and MPN carry only TL16C2550; the package suffix PFBR is only in the device field.

> **Issue:** The BOM shows a part number that covers several packages with different pinouts.

IMPACT

The BOM can't tell which package to order, and other TL16C2550 packages have different pinouts.

FIX

- Use the full orderable part number TL16C2550PFBR for Value and MPN.

JIRA

Summary	TL16C2550 part: Value/MPN set to TL16C2550 instead of TL16C2550PFBR
Issue Type	Bug
Priority	Low
Labels	part-verification, part-library, metadata, TL16C2550
Components	Part Creation / Symbol Library
Description	Submitted part ID 24150: PackageValue and manufacturerPartNumber are "TL16C2550"; the device field is already TL16C2550PFBR. The base number covers several packages with different pinouts, so the BOM is ambiguous. Set Value and MPN to TL16C2550PFBR.

FOOTPRINT (2D)

Reference: QFP50P900X900X120-48N.kicad_mod · Submitted: twoDthreeD.glb

MATCHES THE REFERENCE

- 48 pads, 0.50 mm pitch on all four sides.
- Pin numbering is correct: pin 1 at top-left, counter-clockwise (1–12 left, 13–24 bottom, 25–36 right, 37–48 top).
- Left and right rows at x = ±4.20 mm (reference ±4.2164).

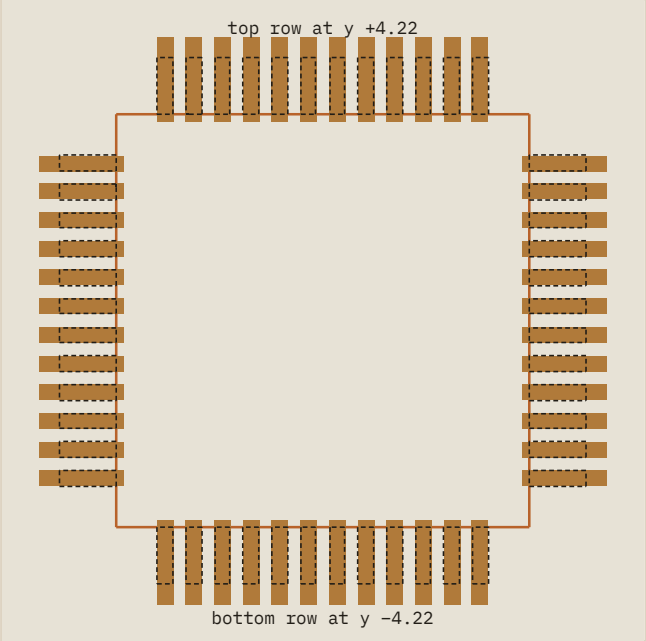
FP-01 HIGHEST

Top and bottom pad rows are 1.0 mm too close to centre: 24 of 48 pads can't meet the leads

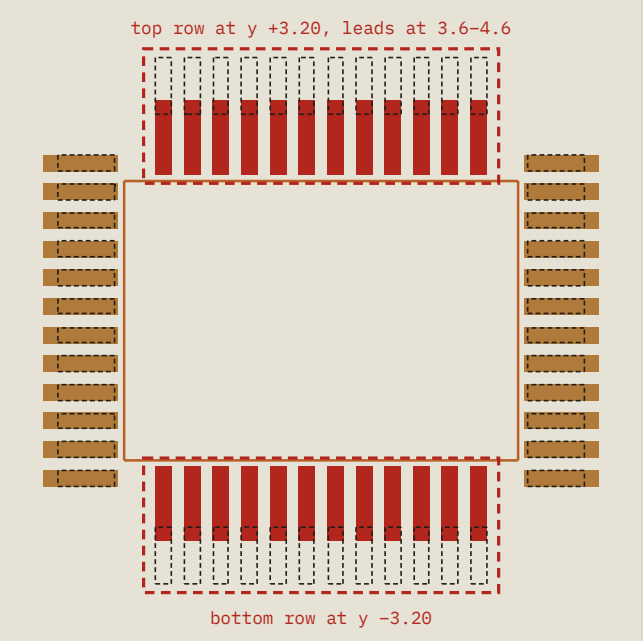
The left and right rows (pins 1-12, 25-36) are at x = ±4.20 mm, matching the standard file (±4.2164). The top and bottom rows (pins 13-24, 37-48) are at y = ±3.20 mm instead of ±4.2164. TL16C2550PFB is a square TQFP-48: the leads reach the same distance on all four sides (9.00 mm tip to tip in the STEP model; the standard footprint draws them to ±4.60 mm). So the submitted pad pattern is 9.7 x 7.7 mm instead of square.

PADS	REFERENCE CENTRE (MM)	SUBMITTED CENTRE (MM)	RESULT
Pins 1-12 (left)	x = -4.216	x = -4.20	OK
Pins 25-36 (right)	x = +4.216	x = +4.20	OK
Pins 13-24 (bottom)	y = -4.216	y = -3.20	1.02 mm short
Pins 37-48 (top)	y = +4.216	y = +3.20	1.02 mm short
Top/bottom pad span	3.48 to 4.95	2.55 to 3.85	lead foot at about 3.75 to 4.50

STANDARD FILE
QFP50P900X900X120-48N.kicad_mod



SUBMITTED FILE
twoDthreeD.glb



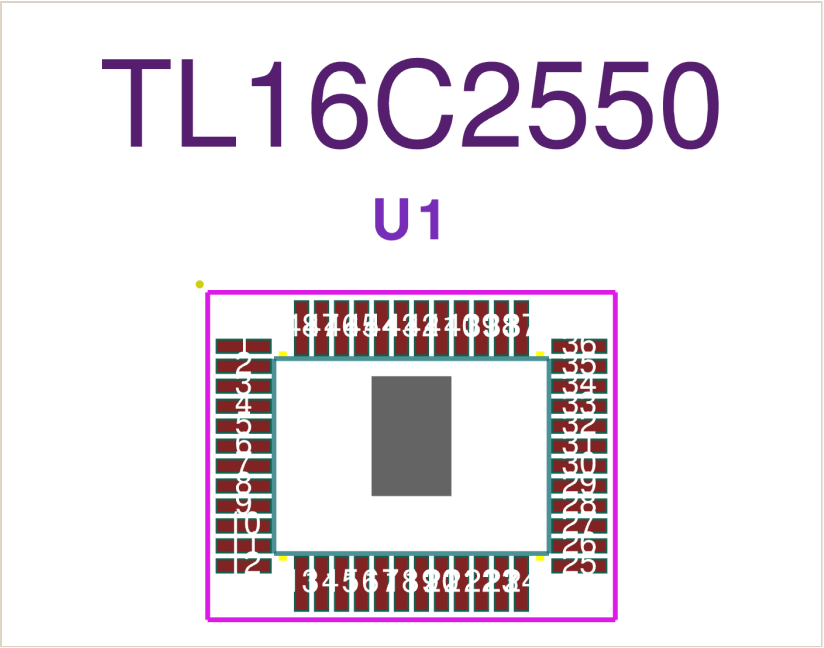
Standard: Pads 13–24 and 37–48 sit at y = ±4.2164 mm, the same distance from centre as the left/right rows. Every pad lies under its lead.

Submitted: Left/right rows are correct (x = ±4.20). Top/bottom rows are at y = ±3.20 mm. The pads end at 3.85 mm, but the lead foot that rests on the board is at about 3.75–4.50 mm.

Issue: On 24 pins the copper barely touches the lead: at most 0.1 mm of overlap, where 1.3 mm is needed. Those pins won't solder.

Copper pad · Misplaced pad · Real lead / terminal (standard file) · Body outline (F.Fab) · Problem area

· Top view, both drawn to the same scale from the files' own coordinates.



The submitted footprint as it renders in the editor: the top and bottom pad rows sit against the body, while the left and right rows extend out to the leads.

IMPACT

The pads end at 3.85 mm from centre and the lead tips are at 4.50 mm. On the top and bottom sides the copper sits under the part's plastic shoulder, not under the lead feet. 24 pins (address, data, modem, VCC/GND) can't solder, so the board doesn't work.

FIX

- Place the top and bottom rows at y = ±4.20 mm, the same span as left/right. The package is square.
- Add a check that QFP/QFN footprints have equal row spans on both axes when the package body is square.

JIRA

Summary	TL16C2550 footprint: top/bottom pad rows at ±3.20 mm instead of ±4.22 mm, 24 pads off the leads
Issue Type	Bug
Priority	Highest
Labels	part-verification, part-library, footprint-error, TL16C2550
Components	Part Creation / Footprint Library
Description	Submitted footprint TL16C2550PFBR (ID 24150, twoDthreeD.glb). Left/right rows are correct at x = ±4.20 mm. Top/bottom rows (pins 13-24 and 37-48) are at y = ±3.20 mm; reference QFP50P900X900X120-48N has them at ±4.2164 mm. The package is a square TQFP-48 with 9.00 mm lead span (reference STEP). Pads end at 3.85 mm while lead tips are at 4.50 mm, so 24 pins can't solder. Fix: put top/bottom rows at ±4.20 mm. Pitch (0.50 mm) and pin numbering order are correct.

FP-02 HIGH

Body outline is 6.88 x 4.88 mm instead of the 7.00 x 7.00 mm package

The submitted F.Fab body outline is a 6.88 x 4.88 mm rectangle. The standard file draws the body as a 7.21 x 7.21 mm square (±3.6068), and the datasheet body is 7.00 x 7.00 mm. Like FP-01, the Y dimension came out about 2 mm short.

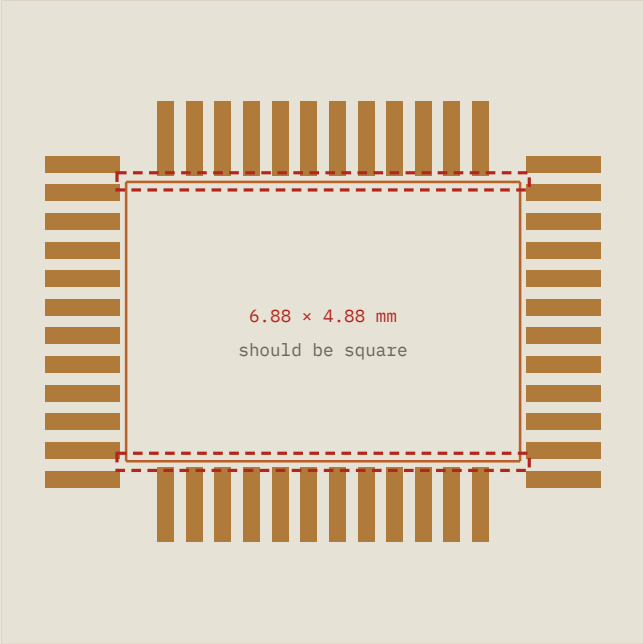
LAYER	REFERENCE (MM)	SUBMITTED (MM)
F.Fab body	7.21 x 7.21 (square)	6.88 x 4.88
Datasheet body	7.00 x 7.00	-
Silkscreen	corner marks at ±3.61 + pin-1 chamfer	6.88 x 5.12 full rectangle

STANDARD FILE
QFP50P900X900X120-48N.kicad_mod



Standard: The F.Fab body is a square at ±3.6068 mm (7.21 mm), sitting just inside all four pad rows.

SUBMITTED FILE
twoDthreeD.glb



Submitted: The F.Fab body is a 6.88 × 4.88 mm rectangle. Its top and bottom edges are 1 mm closer to centre than the real body edge.

Issue: The assembly and placement view shows a part 2 mm shorter than it is in Y. The error matches FP-01: the Y dimensions were built too small.

Copper pad Body outline (F.Fab) Problem area · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

Placement and the assembly drawing show a part 2 mm smaller than it is, so neighbouring parts can be placed into the real body.

FIX

- Draw the fab body as a square centred on the origin (7.00 mm nominal; the standard file uses 7.21 mm).

JIRA

Summary	TL16C2550 footprint: fab body 6.88 x 4.88 mm, package is 7.00 x 7.00 mm
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, footprint-error, TL16C2550
Components	Part Creation / Footprint Library
Description	Submitted footprint TL16C2550PFBR (ID 24150): F.Fab body outline is 6.88 x 4.88 mm. The standard footprint draws the body at 7.21 x 7.21 mm; the datasheet body is 7.00 x 7.00 mm. Likely the same Y-dimension error as the pad rows (see FP-01). Redraw as a 7.00 x 7.00 mm square.

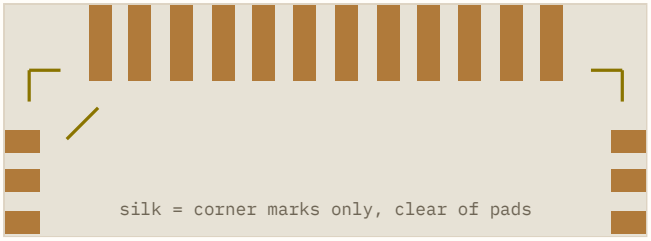
FP-03 HIGH

Silkscreen lines run over 24 pads

The silkscreen top and bottom edges are at y = ±2.56 mm. The top and bottom pads span y = 2.55 to 3.85 mm, so the silk line crosses the inner end of all 24 of those pads. The left and right silk edges are at x = ±3.44 mm, only 0.11 mm from the pad ends at ±3.55. The reference uses corner marks only, kept clear of the pads.

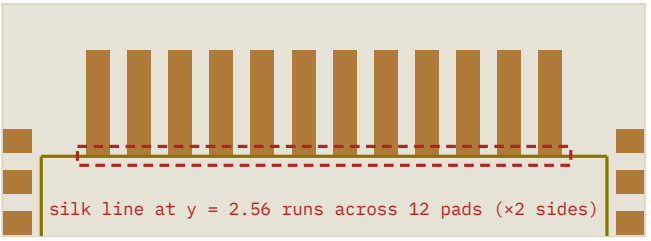
FEATURE	POSITION (MM)	PADS	RESULT
Silk top/bottom edge	y = ±2.56	on pad copper 2.55-3.85	24 pads
Silk left/right edge	x = ±3.44	pad inner end ±3.55	0.11 mm gap

STANDARD FILE
QFP50P900X900X120-48N.kicad_mod



Standard: Silkscreen is only corner marks at ±3.6068 mm plus a pin-1 chamfer line. The nearest pad is 0.34 mm away.

SUBMITTED FILE
twoDthreeD.glb



Submitted: The silkscreen is a full rectangle. Its top edge (y = +2.56) lies on pads 37–48 (copper from y = 2.55) and its bottom edge on pads 13–24. The side edges are 0.11 mm from the left/right pads.

Issue: Silk printed on 24 pads. Fabs either clip it or ink lands on the copper and blocks solder.

Copper pad Silkscreen Problem area · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

Fabs clip silkscreen off exposed copper or reject the file. If it is printed, ink on the pads blocks solder wetting.

FIX

- After FP-01/FP-02 are fixed, draw silk as corner marks outside the body, at least 0.2 mm from any pad, like the reference.
- Add a silk-to-pad clearance check (at least 0.15 mm) to footprint generation.

JIRA

Summary	TL16C2550 footprint: silkscreen crosses 24 pads (silk at y ±2.56, pads 2.55-3.85)
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, footprint-error, silkscreen, TL16C2550
Components	Part Creation / Footprint Library
Description	Submitted footprint TL16C2550PFBR (ID 24150): silkscreen rectangle 6.88 x 5.12 mm. Its top/bottom edges at y = ±2.56 mm cross the top/bottom pads (y 2.55-3.85 mm) on all 24 of them; side edges are 0.11 mm from the left/right pads. Redraw silk as corner marks clear of the pads (at least 0.2 mm) and add a silk-to-pad clearance check.

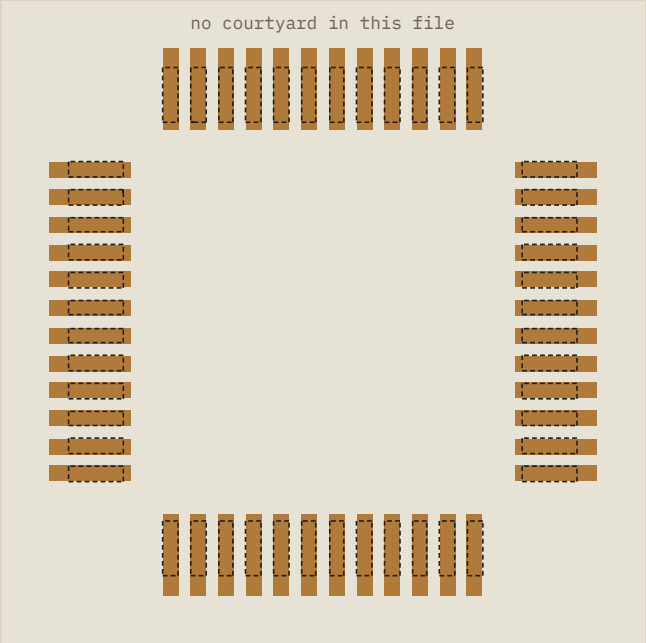
FP-04 MEDIUM

Courtyard 10.2 x 8.2 mm is smaller than the real part in Y

The submitted courtyard is a 10.2 x 8.2 mm rectangle, so ±4.10 mm in Y. The real leads reach ±4.5 to 4.6 mm on all four sides (STEP 9.00 mm span; the standard footprint draws the leads to ±4.60). So the courtyard ends 0.4 to 0.5 mm inside the part on the top and bottom. The standard file has no courtyard at all.

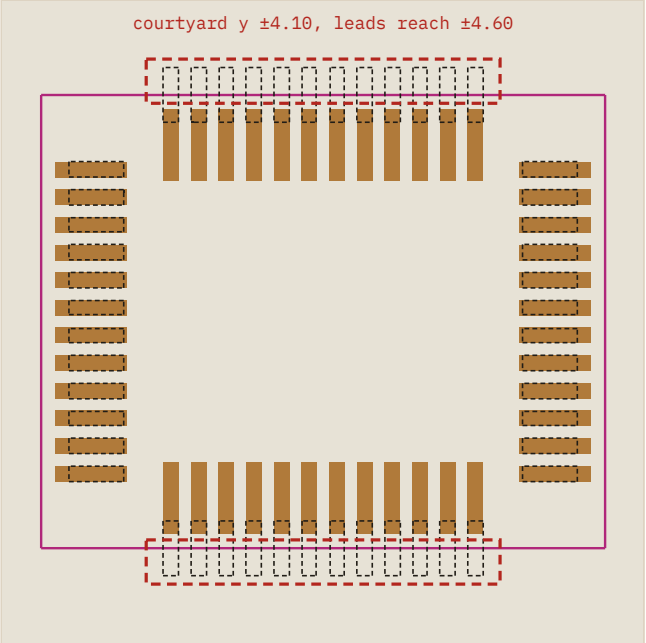
	REAL PART, FROM STANDARD FILE (MM)	SUBMITTED COURTYARD (MM)
X extent	±4.60 (lead drawings)	±5.10
Y extent	±4.60 (lead drawings)	±4.10

STANDARD FILE
QFP50P900X900X120-48N.kicad_mod



Standard: The standard file has no courtyard. Its lead drawings reach ±4.60 mm on all four sides, so a courtyard needs to be at least about 10.2 × 10.2 mm.

SUBMITTED FILE
twoDthreeD.glb



Submitted: The courtyard is 10.2 × 8.2 mm. In Y it stops at ±4.10 mm, while the real leads (drawn in dashed outline) continue to ±4.60 mm.

Issue: The top and bottom leads stick 0.5 mm out of the courtyard, so the overlap check can't catch a neighbour placed on them.

Copper pad · Real lead / terminal (standard file) · Courtyard · Problem area · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

The overlap check can't catch a neighbouring part placed over the top or bottom leads.

FIX

- Make the courtyard square, around 10.2 x 10.2 mm (pad toe extent plus 0.25 mm), once FP-01 moves the pads.

JIRA

Summary	TL16C2550 footprint: courtyard 10.2 x 8.2 mm, inside the real leads in Y
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, footprint-error, courtyard, TL16C2550
Components	Part Creation / Footprint Library
Description	Submitted footprint TL16C2550PFBR (ID 24150): F.CrtYd is 10.2 x 8.2 mm (±4.10 mm in Y). Leads reach ±4.50 mm on all sides (9.00 mm span), so the courtyard is 0.4 mm inside the part on top and bottom and the overlap check can't catch neighbours there. Make it square (about 10.2 x 10.2 mm) after the pad rows are fixed.

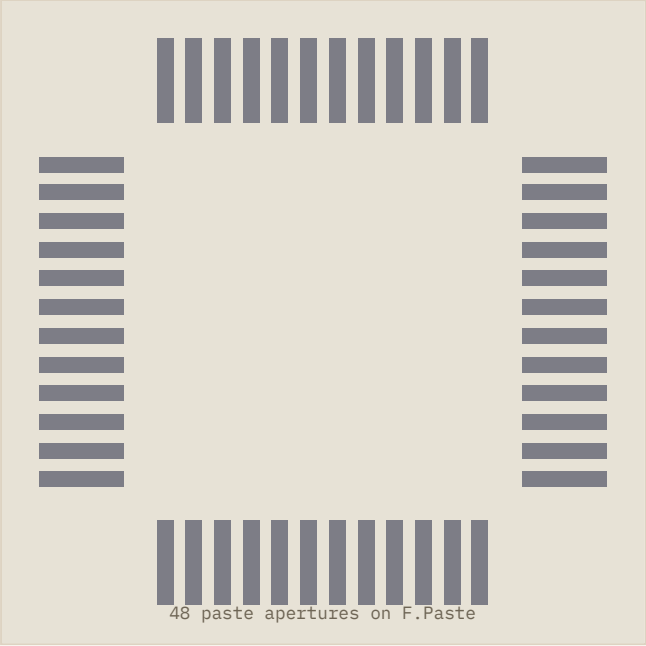
FP-05 MEDIUM

Paste-mask shapes are tagged as copper layer on all 48 pads

Each pad has a PasteMask node, but its data says name "CopperCirclePad" and layerId "F.Cu". It is a copy of the copper pad's record, so the paste aperture is labelled as copper. The copper node itself is also named CopperCirclePad, although every pad is a rectangle (pinShape: rectangle).

NODE	EXTRAS.NAME	EXTRAS.LAYERID	RESULT
SolderMask	SolderMask	F.Mask	OK
Copper	CopperCirclePad	F.Cu	name says circle, shape is rectangle
PasteMask	CopperCirclePad	F.Cu	should be F.Paste

STANDARD FILE
QFP50P900X900X120-48N.kicad_mod



Standard: Each pad lists F.Paste in its layers, so the paste layer has 48 apertures, one per pad.

SUBMITTED FILE
twoDthreeD.glb



Submitted: Each pad has a PasteMask node, but it is a copy of the copper record (name "CopperCirclePad", layerId "F.Cu"). Nothing is tagged F.Paste, so the paste layer drawn by layer comes out empty.

> **Issue:** If the export follows layerId, the stencil has no openings and no paste reaches the pads. It could also export duplicate copper. Confirm with a Gerber export.

Paste aperture · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

Depending on how export reads layerId, the paste layer may come out empty (no stencil openings, so pads aren't soldered) or with duplicate copper. Needs an export test to confirm which.

FIX

- Write layerId F.Paste (and a paste-specific name) on PasteMask nodes.
- Export Gerbers for this part and check the paste layer has 48 openings.

JIRA

Summary	TL16C2550 footprint: PasteMask nodes carry layerId F.Cu (paste tagged as copper) on all 48 pads
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, footprint-error, paste-mask, TL16C2550
Components	Part Creation / Footprint Library
Description	Submitted footprint TL16C2550PFBR (ID 24150, twoDthreeD.glb): every pad's PasteMask node has extras name "CopperCirclePad" and layerId "F.Cu", a copy of the copper pad record. The paste layer may export empty or as duplicate copper. Write layerId F.Paste on paste nodes, then confirm a Gerber export shows 48 paste openings. Copper nodes are also named CopperCirclePad although all pads are rectangles.

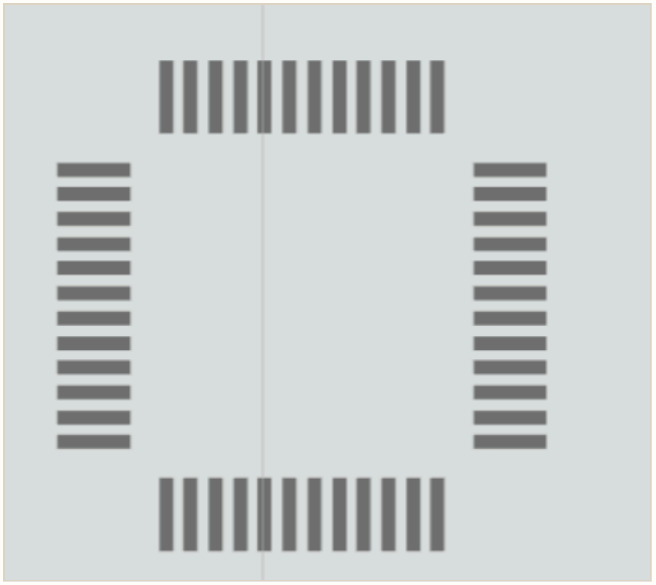
FP-09 MEDIUM

Layer views show red copper pads on every layer

In the IDEEZA layer view the standard footprint draws each layer in its own colour: grey paste, green solder mask, and so on. The submitted footprint shows red copper-coloured pads on the top paste, top and bottom silkscreen, bottom and bottom paste layers. The pads stay red even with every layer hidden. This is the visible side of FP-05: the paste nodes and extra pad nodes carry layerId F.Cu, so every layer view draws them as top copper.

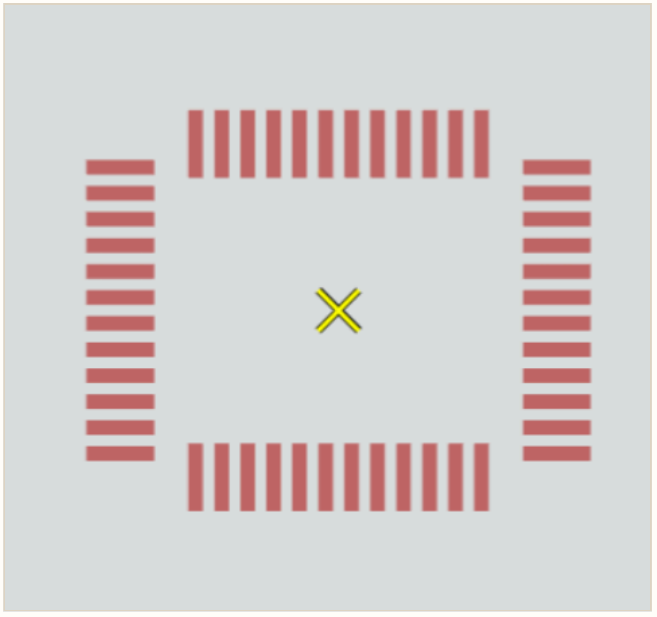
LAYER SHOWN	STANDARD FOOTPRINT	SUBMITTED FOOTPRINT
Top paste	grey pads	red pads
Top silkscreen only	silk marks only	silk marks + red pads
Bottom layer	bottom-layer colour	red pads
Bottom paste	bottom-paste colour	red pads
All layers hidden	nothing drawn	red pads still drawn

STANDARD FILE
IDEEZA layer view, top paste only



Standard: The paste layer shows grey apertures, one per pad.

SUBMITTED FILE
IDEEZA layer view, top paste only



Submitted: The paste layer shows red copper-coloured pads.

> **Issue:** The paste view draws copper, because the paste nodes are tagged F.Cu (FP-05).

Screenshots from the IDEEZA layer view, same layer selected on both.

IMPACT

Layer views can't be trusted to check paste, mask or silkscreen for this part, and Gerber layers built from the same data are suspect.

FIX

- Fix FP-05 (write the right layerId on every pad node), then re-check each layer view.
- Make hidden layers hide every node that belongs to them.

JIRA

Summary	TL16C2550 footprint: pads drawn red on paste, silk and bottom layers, and stay visible with all layers hidden
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, footprint-error, layer-display, TL16C2550
Components	Part Creation / Footprint Library
Description	Submitted footprint TL16C2550PFBR (ID 24150): in the layer view, top paste, top/bottom silkscreen, bottom and bottom paste layers all show red copper-coloured pads, and pads stay visible with every layer hidden. The standard footprint shows each layer in its own colour. Root cause is likely the PasteMask/extra pad nodes tagged layerId F.Cu (FP-05).

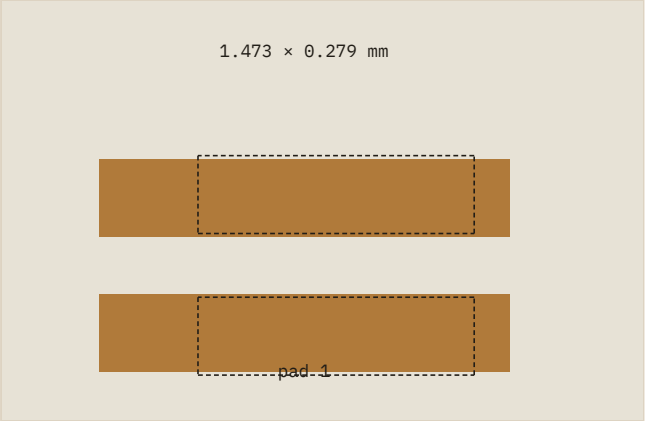
FP-06 LOW

Pads are 0.17 mm shorter than the reference land pattern

The submitted pads are 1.30 x 0.28 mm. The standard SnapEDA footprint uses 1.4732 x 0.2794 mm, and TI's datasheet land pattern for PFB is 1.35 x 0.25 mm. The submitted width (0.28) is wider than TI's 0.25, which narrows the copper gap at 0.5 mm pitch to 0.22 mm. The length is between TI's and SnapEDA's.

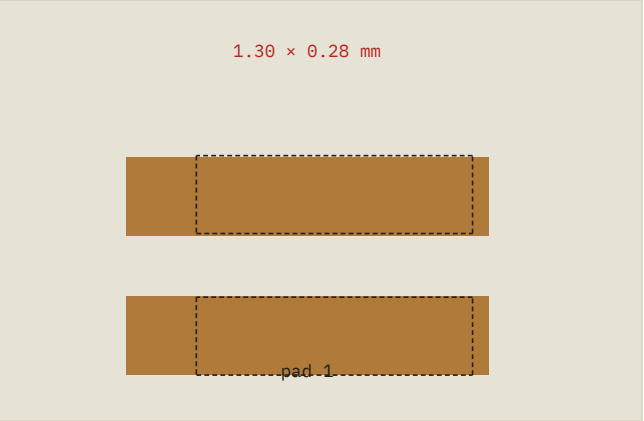
	SNAPEDA STANDARD (MM)	TI LAND PATTERN (MM)	SUBMITTED (MM)
Pad length	1.473	1.35	1.30
Pad width	0.279	0.25	0.28
Pitch	0.50	0.50	0.50

STANDARD FILE
QFP50P900X900X120-48N.kicad_mod



Standard: Pads are 1.4732 × 0.2794 mm. Each pad runs past the lead tip (toe) and back under the knee (heel), giving room for both fillets.

SUBMITTED FILE
twoDthreeD.glb



Submitted: Pads are 1.30 × 0.28 mm, 0.17 mm shorter, with the same width and pitch.

> **Issue:** A shorter pad leaves a smaller toe and heel fillet: weaker joints and harder inspection. Not a hard fault on its own.

Copper pad Real lead / terminal (standard file) · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

Weaker solder joints and harder inspection. Not a hard fault.

FIX

- Use TI's land pattern: 1.35 x 0.25 mm pads at 0.5 mm pitch.

JIRA

Summary	TL16C2550 footprint: pads 1.30 x 0.28 mm vs TI land pattern 1.35 x 0.25 mm
Issue Type	Improvement
Priority	Low
Labels	part-verification, part-library, footprint, TL16C2550
Components	Part Creation / Footprint Library
Description	Submitted footprint TL16C2550PFBR (ID 24150): pads 1.30 x 0.28 mm. TI datasheet land pattern (PFB) is 1.35 x 0.25 mm; SnapEDA standard is 1.4732 x 0.2794 mm. Pitch matches (0.50 mm). Use TI's 1.35 x 0.25 mm.

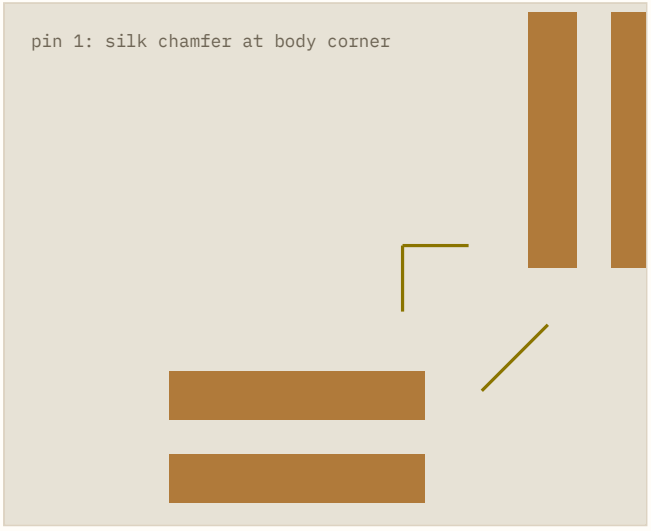
FP-07 LOW

Pin-1 marker sits outside the courtyard, and the value text is twice the footprint's width

The pin-1 dot (silk and fab) is at (-5.3, +4.3) mm, past the courtyard edge at x = -5.10. The standard file marks pin 1 with a silk chamfer inside the body corner. The value text "TL16C2550" is 15.4 x 2.3 mm over a 9.7 mm-wide pad pattern. It is on F.Fab only, so it does not print.

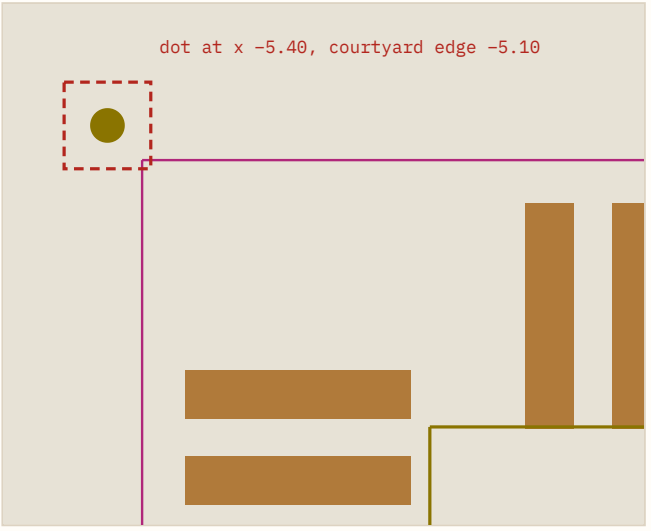
ITEM	STANDARD FILE	SUBMITTED
Pin-1 marker	diagonal silk chamfer at body corner	0.2 mm dot at x -5.40 to -5.20, courtyard edge -5.10
Value text	1.64 mm high, F.Fab	15.4 x 2.3 mm, F.Fab

STANDARD FILE
QFP50P900X900X120-48N.kicad_mod



Standard: Pin 1 is marked by a diagonal silk chamfer at the body corner next to pad 1, inside the part outline.

SUBMITTED FILE
twoDthreeD.glb



Submitted: Pin 1 is a 0.2 mm dot at (−5.3, +4.3) mm, past the courtyard edge at x = −5.10. Separately, the value text is 15.4 × 2.3 mm (F.Fab only).

Issue: The marker sits outside the part's own area, where a neighbour can cover it. Cosmetic.

Copper pad · Silkscreen · Courtyard · Pin-1 marker · Problem area · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

Cosmetic. The marker is easy to lose and the text overlaps neighbouring parts on the layout.

FIX

- Move the pin-1 dot inside the courtyard, next to pad 1.
- Scale the value text to the footprint (1.0 mm height is typical).

JIRA

Summary	TL16C2550 footprint: pin-1 marker outside courtyard; value text 15.4 mm wide
Issue Type	Improvement
Priority	Low
Labels	part-verification, part-library, footprint, TL16C2550
Components	Part Creation / Footprint Library
Description	Submitted footprint TL16C2550PFBR (ID 24150): pin-1 marker at (-5.3, +4.3) mm lies outside the courtyard (x edge -5.10). Value text is 15.4 x 2.3 mm (F.Fab only) over a 9.7 mm footprint. Move the marker next to pad 1 inside the courtyard and size the value text to the part.

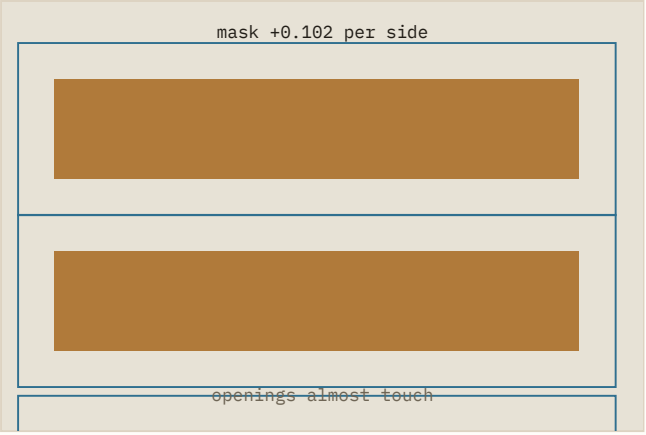
FP-08 LOW

Solder-mask opening applied at half the declared margin

Every submitted pad declares solder_mask_margin 0.102 mm, but the SolderMask shape drawn is the pad plus 0.051 mm per side (1.402 x 0.382 mm on a 1.30 x 0.28 mm pad). TI's land pattern specifies 0.05 mm max all around, so the applied value matches TI and the declared value does not. The SnapEDA standard file uses 0.102.

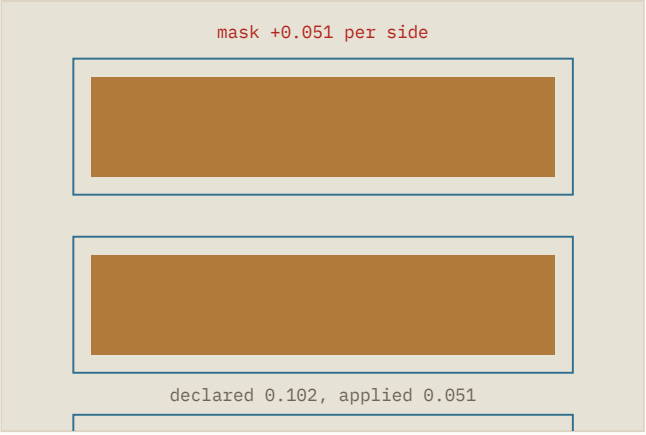
	SNAPEDA STANDARD (MM)	TI LAND PATTERN (MM)	SUBMITTED (MM)
Declared margin	0.102	0.05 max	0.102
Applied per side	0.102	-	0.051
Opening on pad 1	1.677 x 0.483	1.45 x 0.35	1.402 x 0.382

STANDARD FILE
QFP50P900X900X120-48N.kicad_mod



Standard: solder_mask_margin 0.102 mm, applied per side.

SUBMITTED FILE
twoDthreeD.glb



Submitted: Each pad declares solder_mask_margin 0.102 mm, but the SolderMask shape is the pad plus only 0.051 mm per side (1.402 × 0.382 mm on a 1.30 × 0.28 mm pad).

Issue: The declared and applied values disagree, so Gerbers and DRC can use different numbers. Pick one value and apply it per side.

Solder-mask opening Copper pad · Top view, both drawn to the same scale from the files' own coordinates.

FP-10 LOW

Pad numbers are larger than the pads and run into each other

The submitted pad numbers are drawn a fixed 0.58 mm tall on every part. On this footprint the pads are 0.28 mm wide at 0.50 mm pitch, so two-digit numbers (0.77 mm wide) are wider than the pitch, and the numbers on the top and bottom rows overlap into an unreadable strip. The standard draws each number small enough to sit inside its pad.

	STANDARD FOOTPRINT	SUBMITTED FOOTPRINT
Number height	fits inside the pad	0.58 mm
Two-digit number width	fits inside the pad	0.77 mm
Pad width / pitch	0.28 / 0.50 mm	0.28 / 0.50 mm

STANDARD FILE
IDEEZA 2D view

SUBMITTED FILE
IDEEZA 2D view

Reference: TL16C2550PFBR.step · Submitted: twoDthreeD.glb, general.glb

MATCHES THE REFERENCE

- The 3D view does draw the part (as a generated box) in the correct position over the footprint.

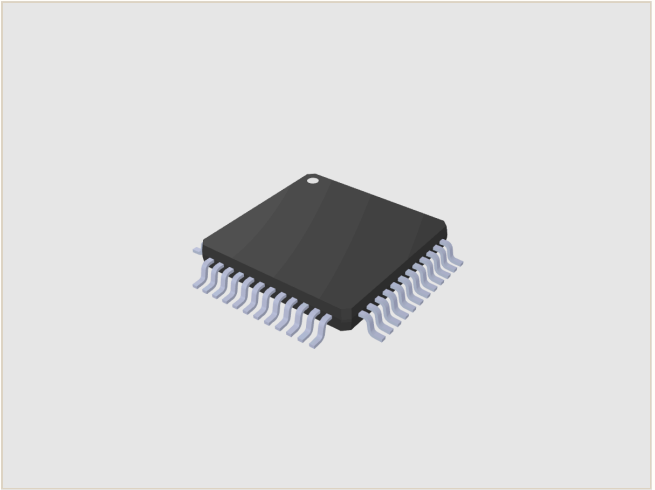
3D-01 HIGH

No real 3D model: the 3D view shows a generated box at the wrong size and colours

twoDthreeD.glb holds only 2D footprint data, and no 3D model file was submitted. In the 3D view the editor draws a generated placeholder instead: a light grey body, white leads and red pads. The box follows the wrong 6.88 x 4.88 mm body from FP-02, so it is short in Y. The left and right lead rows run past its ends, and the end pins sit partly outside the body. The standard file ships a real STEP model: TI's PFB package, 9.00 x 9.00 mm lead span, black body.

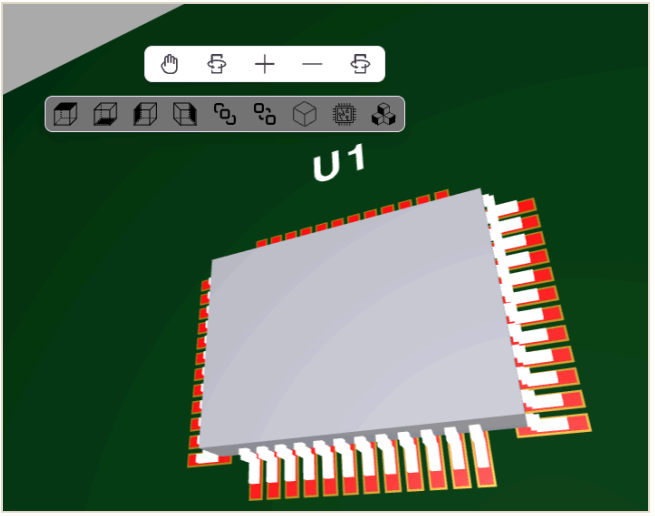
	STANDARD FILE	SUBMITTED
3D source	TL16C2550PFBR.step (TI PFB solid model)	none stored; generated box in the 3D view
Body	7.00 x 7.00 mm, black	follows 6.88 x 4.88 mm fab body, light grey
Leads	gull-wing, 9.00 mm span, metal	flat white strips
Fit	all 48 leads inside body ends	left/right end pins partly outside the body

STANDARD FILE
TL16C2550PFBR.step



Standard: TI's PFB model: black 7 × 7 mm body, 48 gull-wing leads, 9.00 mm lead span, 1.20 mm tall, pin-1 dot.

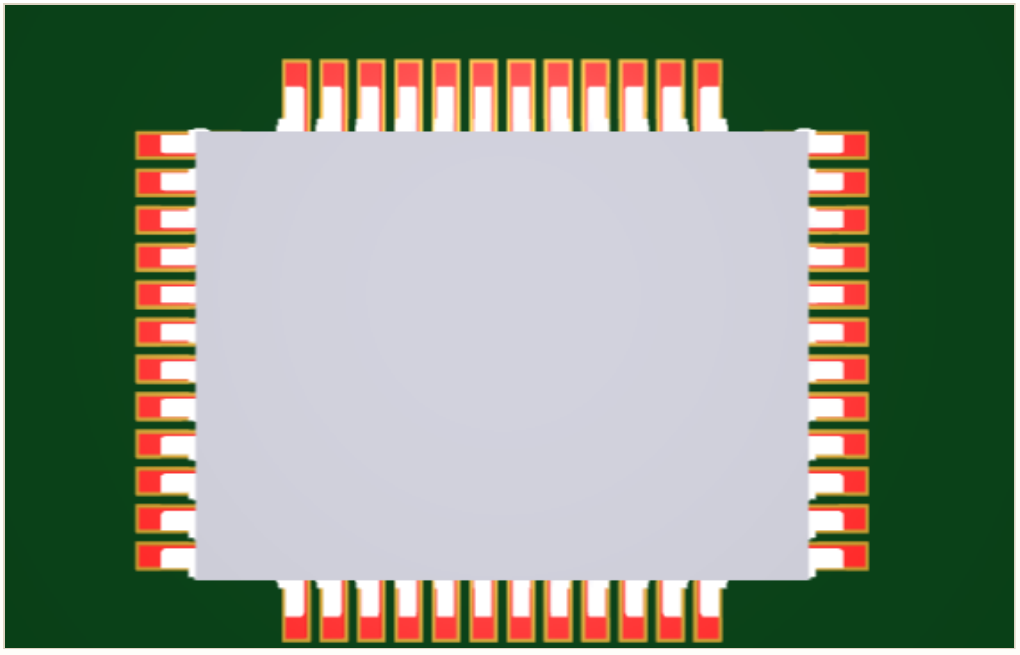
SUBMITTED FILE
IDEEZA 3D view (engineer screenshot)



Submitted: A generated light-grey box with flat white leads, sized from the wrong 6.88 × 4.88 mm fab body.

Issue: The 3D view shows a placeholder of the wrong size and look instead of the real package.

Left: the standard STEP file rendered. Right: the submitted part in the IDEEZA 3D view.



Submitted part in the IDEEZA 3D view, top-down (screenshot from the engineer review). The body is shorter than the left and right lead rows.

IMPACT

The 3D board view, enclosure fit and height checks use a box of the wrong size and look. The generated box hides FP-01, because it is built from the same wrong dimensions as the pads.

FIX

- Attach the real TQFP-48 PFB model (7 x 7 mm body, 9 x 9 mm lead span, 0.5 mm pitch, 1.20 mm max height). The standard STEP can be used once its height is checked (see the reference notes).
- If a generated placeholder stays as a fallback, build it from the datasheet package dimensions, not from the footprint, and colour it like a real part (black body, metal leads).
- Flag parts that have no attached 3D model.

JIRA

Summary	TL16C2550 part: no 3D model; 3D view shows generated box at wrong size (6.88 x 4.88) and colours
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, 3d-model, TL16C2550
Components	Part Creation / 3D Models
Description	Submitted part TL16C2550PFBR (ID 24150): no 3D model is stored (twoDthreeD.glb has 2D data only). The 3D view draws a generated box: light grey body, white leads, red pads, sized from the wrong 6.88 x 4.88 mm fab body (FP-02), so the left/right end pins sit partly outside it. The standard package ships TI's PFB STEP model (9.00 x 9.00 mm lead span, black body). Attach the real model; if a fallback box is kept, size it from datasheet package dimensions and colour it realistically.

NOTES ON THE REFERENCE

- Pin 39 is misspelled ~DRSA in the reference symbol; the datasheet name is DSRA. The submitted part has it right.
- The reference types GND (pin 17) as passive; power_in is the better type.
- The reference symbol uses 200 mil (5.08 mm) pins; the IDEEZA rule is 100 mil.
- The reference footprint has no courtyard layer and no 3D model link. Its pad positions are rounded to 0.4826 / 0.508

mm steps, at most 0.007 mm off the 0.5 mm grid, which is negligible.

- The standard footprint's fab layer draws the lead tips at ±4.5974 mm (9.19 mm span), slightly beyond the STEP model's 9.00 mm. Its silkscreen also has 4 short stray ticks outside the pad rows (e.g. x 0.74, y 5.31 to 6.32 mm) with no clear purpose.
- The STEP file is TI's PFB package model, saved under the product name DAC8734SPFB (same package). Rendered, it measures 9.00 × 9.00 mm over the leads and 1.20 mm tall, matching the footprint name.

PART 2 OF 6

74LVC1G99GT,115

Part Nexperia 74LVC1G99GT,115, low-power configurable multi-function gate, XSON8 (SOT833-1)

Submitted part library ID 24153

Reference Ultra Librarian KiCad v6 package (symbol, footprint; no 3D model)

> **Not usable as submitted.** The footprint is a land pattern for a different, larger package: all 8 pads sit outside the 1 × 2 mm XSON8 body (FP-01), and there is no 3D model (3D-01). The symbol's pin mapping is correct, but OE is drawn without its active-low bar and the part number is cut to 74LVC1G99GT,1.

SCHEMATIC SYMBOL

Reference: 2026-10-05_13-08-02.kicad_sym (74LVC1G99GT_115) · Submitted: schematic.glb

PIN-BY-PIN COMPARISON

8 standard pins: 6 match, 1 differ in electrical type only, 1 with a naming, label or polarity problem

PIN	STANDARD NAME	SUBMITTED NAME	LABEL DRAWN	STANDARD TYPE	SUBMITTED TYPE	ACTIVE-LOW (STD / SUB)	RESULT
1	OE	OE	OE	input	input	yes / yes (no bar)	**active-low set but no bar drawn**
2	A	A	A	input	input	– / –	match
3	B	B	B	input	input	– / –	match
4	GND	GND	GND	power_out	power_in	– / –	type differs
5	C	C	C	input	input	– / –	match
6	D	D	D	input	input	– / –	match
7	Y	Y	Y	output	output	– / –	match
8	VCC	VCC	VCC	power_in	power_in	– / –	match

MATCHES THE REFERENCE

- All 8 pin numbers carry the right function: 1 OE, 2 A, 3 B, 4 GND, 5 C, 6 D, 7 Y, 8 VCC.
- Pin 1 is stored as active-low (activeLow true). Only its drawing is wrong (SCH-01).
- GND is typed power_in, which is more correct than the reference's power_out.
- Pin length is 2.54 mm (100 mil), which meets the 100 mil rule. The standard file uses 7.62 mm (300 mil).

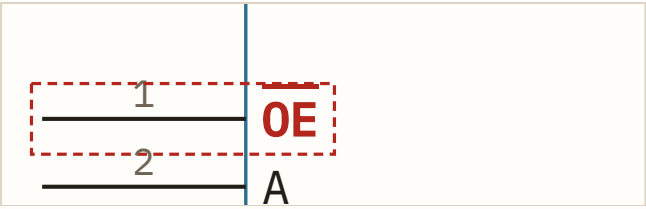
SCH-01 MEDIUM

OE is flagged active-low but drawn without the overbar

Pin 1 has activeLow = true and displayName "/OE", but the symbol draws the label as plain "OE" with no overbar. The datasheet and the standard symbol ("*OE") mark it as an active-low output enable.

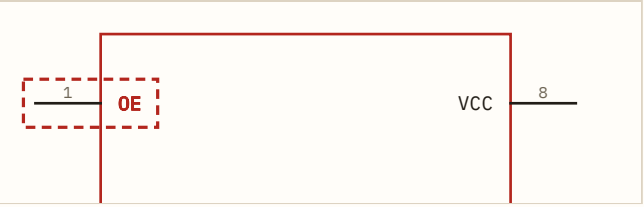
PIN	STANDARD FILE	SUBMITTED DATA	SUBMITTED DRAWING
1	*OE (active-low)	activeLow true, displayName /OE	drawn as "OE", no bar

STANDARD FILE
74LVC1G99GT_115 (.kicad_sym)



Standard: Pin 1 is drawn as OE with an overbar (the file stores it as "*OE"), so it reads as active-low.

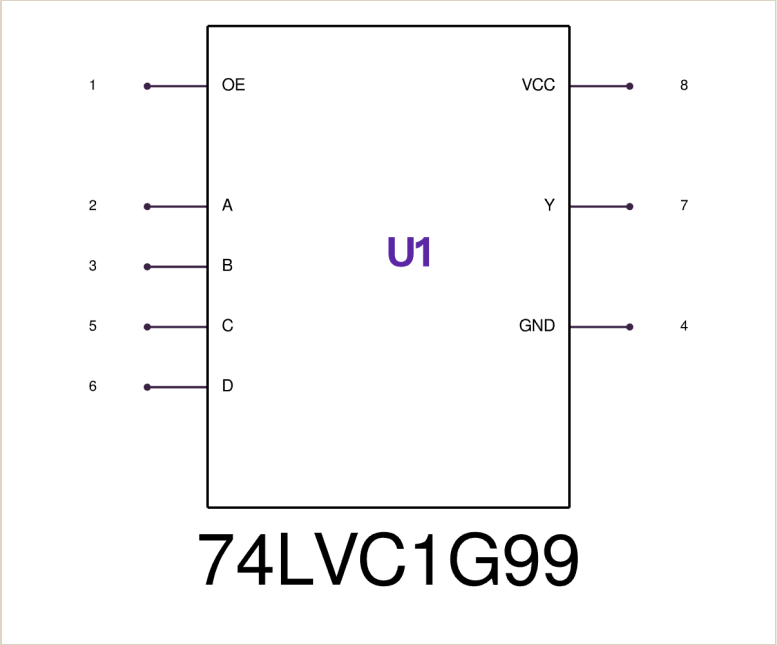
SUBMITTED FILE
schematic.glb



Submitted: Pin 1 is stored as activeLow = true ("/OE") but drawn as plain "OE" with no bar.

Issue: The drawing hides the active-low polarity that the data already has. Tying OE high would turn the output off.

Both drawn to the same scale, from the files' own pin data. Affected pins.



Submitted symbol as rendered: pin 1 reads "OE" with no overbar.

IMPACT

A designer reading the symbol will tie OE high to enable the output, which turns the output off (Y goes high-impedance).

FIX

- Render the overbar whenever activeLow is true (or when the name starts with / or ~).
- The data is already right on this part; only the drawing is wrong.

JIRA

Summary	74LVC1G99GT symbol: OE has activeLow=true but is drawn without overbar
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, symbol-error, 74LVC1G99
Components	Part Creation / Symbol Library
Description	Submitted symbol for 74LVC1G99GT,115 (library ID 24153, schematic.glb): pin 1 is stored with activeLow = true and displayName "/OE", but it renders as plain "OE" with no overbar. The standard Ultra Librarian symbol shows "*OE". Render the overbar from the activeLow flag.

SCH-02 MEDIUM

Part number cut short: 74LVC1G99GT,1 instead of 74LVC1G99GT,115

The orderable Nexperia part number is 74LVC1G99GT,115. The submitted part stores "74LVC1G99GT,1" as device, footprint name and record name, and "74LVC1G99" as value and MPN. The text after ",1" is missing everywhere, so the cut probably happens where the name is read.

FIELD	STANDARD FILE	SUBMITTED
Device / footprint / record name	74LVC1G99GT,115	74LVC1G99GT,1
Value / MPN	74LVC1G99GT_115 / 74LVC1G99GT,115	74LVC1G99

STANDARD FILE
74LVC1G99GT_115 (.kicad_sym)

PART PROPERTIES

Value	74LVC1G99GT_115
MPN / keywords	74LVC1G99GT,115
Device	-
Footprint	XSON8_SOT833-1_NEX

Standard: The full orderable number 74LVC1G99GT,115 is in the keywords, and the footprint is named after the package (SOT833-1).

SUBMITTED FILE
schematic.glb

PART PROPERTIES

Value	**74LVC1G99**
MPN / keywords	**74LVC1G99**
Device	**74LVC1G99GT,1**
Footprint	**74LVC1G99GT,1**

Submitted: Every field stops at "74LVC1G99GT,1" or "74LVC1G99". The ",115" suffix is gone.

> **Issue:** The BOM shows a part number that doesn't exist, and the package is not identified anywhere.

IMPACT

The BOM can't be ordered as is: 74LVC1G99GT,1 is not a valid part number, and 74LVC1G99 alone covers several packages with different footprints.

FIX

- Store the full orderable number 74LVC1G99GT,115 in device, MPN and value.
- Check how part names containing a comma are parsed; the truncation at ",1" suggests a split on the comma or a length limit.

JIRA

Summary	74LVC1G99GT part: part number truncated to 74LVC1G99GT,1 (should be 74LVC1G99GT,115)
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, metadata, 74LVC1G99
Components	Part Creation / Symbol Library
Description	Submitted part ID 24153: device, footprintName and recordName are "74LVC1G99GT,1", value and MPN are "74LVC1G99". The orderable Nexperia number is 74LVC1G99GT,115. Store the full number and check comma handling in part names.

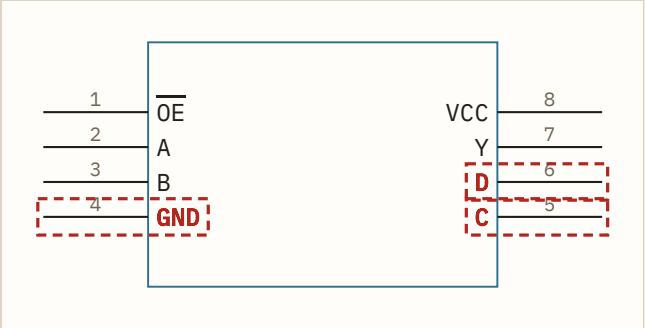
SCH-03 LOW

Pin order differs from the standard symbol

The standard symbol follows the package order: pins 1-4 (OE, A, B, GND) down the left and 8-5 (VCC, Y, D, C) down the right. The submitted symbol groups by function instead: all inputs (OE, A, B, C, D) on the left and VCC, Y, GND on the right. Every pin number still carries the right function.

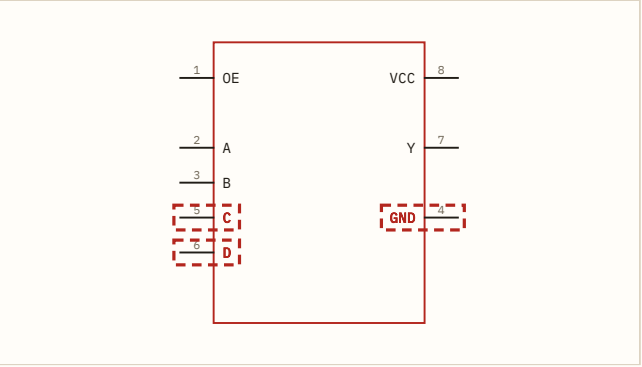
SIDE	STANDARD FILE	SUBMITTED
Left	*OE A B GND (1-4)	OE A B C D (1, 2, 3, 5, 6)
Right	VCC Y D C (8-5)	VCC Y GND (8, 7, 4)

STANDARD FILE
74LVC1G99GT_115 (.kicad_sym)



Standard: Package order: 1 OE, 2 A, 3 B, 4 GND on the left; 8 VCC, 7 Y, 6 D, 5 C on the right.

SUBMITTED FILE
schematic.glb



Submitted: Functional order: OE, A, B, C, D (all inputs) on the left; VCC, Y, GND on the right. Pins 4, 5 and 6 have moved sides.

Issue: The layout differs from the reference. Connectivity is correct; the house rule for pin arrangement needs to be decided.

Both drawn to the same scale, from the files' own pin data. Affected pins.

IMPACT

No effect on connectivity. It only matters if the IDEEZA rule is to match the reference layout. Grouping inputs left and outputs right is also a common convention.

FIX

- Decide one house rule (package order or functional grouping) and apply it to all generated symbols.

JIRA

Summary	74LVC1G99GT symbol: pin order differs from standard (C, D moved left, GND moved right)
Issue Type	Improvement
Priority	Low
Labels	part-verification, part-library, symbol-layout, 74LVC1G99
Components	Part Creation / Symbol Library
Description	Submitted symbol 74LVC1G99GT,115 (ID 24153) groups inputs OE, A, B, C, D on the left and VCC, Y, GND on the right. The standard symbol uses package order (1-4 left, 8-5 right). Connectivity is correct. Decide the house rule for pin arrangement and apply it consistently.

FOOTPRINT (2D)

Reference: XSON8_SOT833-1_NEX.kicad_mod (nominal density) · Submitted: twoDthreeD.glb

MATCHES THE REFERENCE

- 8 pads.
- Pin numbering order is correct: 1–4 down the left, 5–8 up the right, pin 1 top-left.

FP-01 HIGHEST

Wrong land pattern: all 8 pads lie outside the real package

XSON8 SOT833-1 is a leadless 1.0 x 1.95 mm package with 0.50 mm pitch. The standard footprint puts its pads in two columns at x = ±0.29 mm, under the body. The submitted footprint uses 0.65 mm pitch, 1.10 x 0.36 mm pads and columns at x = ±1.30 mm, which is the land pattern of a different, larger package. The nearest submitted copper is at x = ±0.75 mm, but the real package ends at ±0.52 mm.

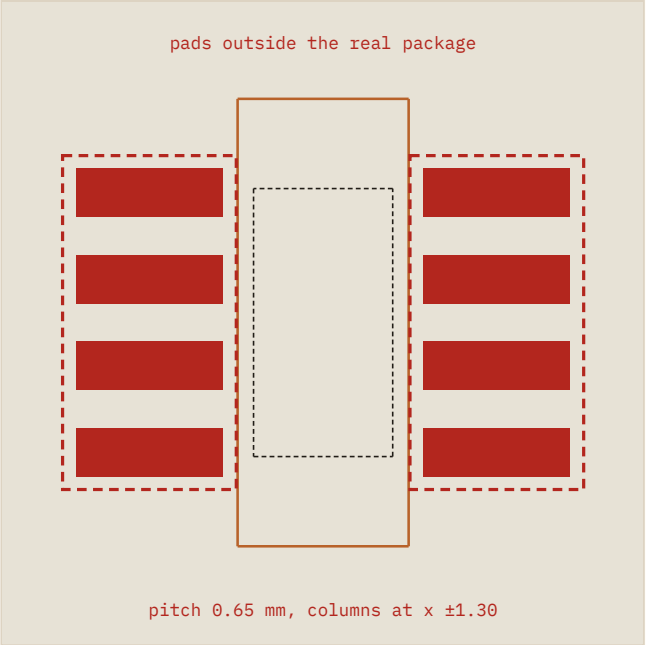
	STANDARD FILE (MM)	SUBMITTED (MM)
Pitch	0.50	0.65
Column centres (x)	±0.29	±1.30
Pad size	0.36 x 0.25 (pin 1: 0.41 x 0.25)	1.10 x 0.36
Row span (y, pad centres)	1.50	1.95
Copper vs package edge	pads inside ±0.47	pads start at ±0.75; package edge ±0.52

STANDARD FILE
XSON8_SOT833-1_NEX.kicad_mod



Standard: XSON8 (SOT833-1) is a leadless 1.0 × 1.95 mm package. Its 8 pads sit under the body in two columns at x = ±0.29 mm, 0.50 mm apart in Y. Pin 1 is drawn longer as a pin-1 indicator.

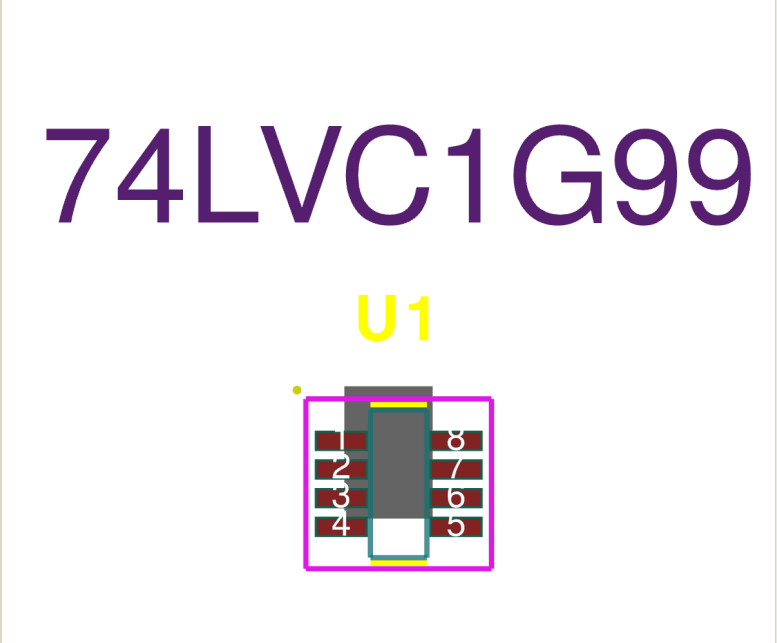
SUBMITTED FILE
twoDthreeD.glb



Submitted: The pads are 1.10 × 0.36 mm at x = ±1.30 mm with 0.65 mm pitch. The nearest copper is at x = ±0.75 mm, while the real package (dashed) ends at ±0.52 mm.

> **Issue:** None of the 8 terminals lands on copper. The part can't be soldered on this footprint. It is a land pattern for a different, larger package.

Copper pad · Misplaced pad · Real package outline (standard F.Fab) · Body outline (F.Fab) · Problem area · Top view, both drawn to the same scale from the files' own coordinates.



Submitted footprint as it renders in the editor.

IMPACT

No terminal touches copper, so the part can't be soldered. Every board using this part fails.

FIX

- Rebuild the footprint from the SOT833-1 package drawing (or use the standard XSON8_SOT833-1_NEX

footprint): 0.50 mm pitch, pads 0.36 x 0.25 mm at x = ±0.29 mm.

- Check that the package code (GT = XSON8 SOT833-1) is what drives land pattern selection. This pattern looks like it came from another 74LVC1G99 package variant.

JIRA

Summary	74LVC1G99GT footprint: wrong land pattern (0.65 mm pitch, pads at ±1.30 mm), all pads outside the XSON8 package
Issue Type	Bug
Priority	Highest
Labels	part-verification, part-library, footprint-error, 74LVC1G99
Components	Part Creation / Footprint Library
Description	Submitted footprint 74LVC1G99GT,115 (ID 24153, twoDthreeD.glb): pads 1.10 x 0.36 mm, 0.65 mm pitch, columns at x = ±1.30 mm. The part is XSON8 SOT833-1 (1.0 x 1.95 mm, 0.50 mm pitch); the standard footprint XSON8_SOT833-1_NEX has 0.36 x 0.25 mm pads at x = ±0.29 mm under the body. Submitted copper starts at x = ±0.75 mm while the package ends at ±0.52 mm, so no terminal lands on a pad. Rebuild from the SOT833-1 drawing.

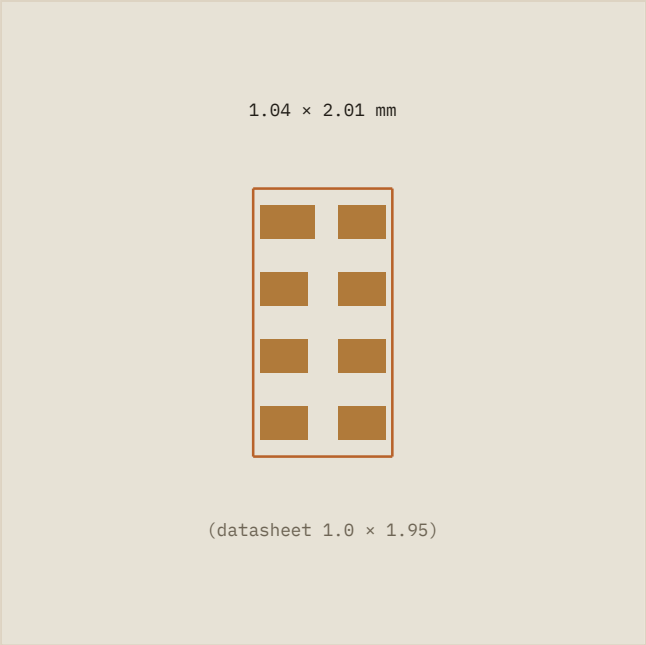
FP-02 HIGH

Body outline is 1.28 x 3.35 mm instead of 1.0 x 1.95 mm

The submitted F.Fab body is 1.28 x 3.35 mm. The standard file draws 1.04 x 2.01 mm (the SOT833-1 outline at maximum tolerance), and the datasheet body is 1.0 x 1.95 mm.

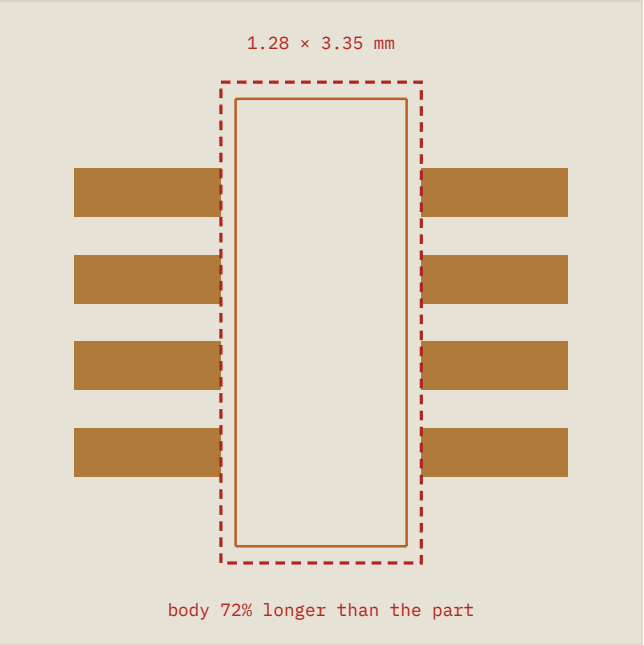
	STANDARD FILE (MM)	SUBMITTED (MM)
F.Fab body	1.04 x 2.01	1.28 x 3.35
Datasheet body	1.00 x 1.95	-

STANDARD FILE
XSON8_SOT833-1_NEX.kicad_mod



Standard: The F.Fab body is 1.04 × 2.01 mm (±0.5207 × ±1.0033), the SOT833-1 outline at maximum tolerance, with a pin-1 notch arc at the top.

SUBMITTED FILE
twoDthreeD.glb



Submitted: The F.Fab body is 1.28 × 3.35 mm.

Issue: Placement and the assembly drawing show a part 1.4 mm longer and 0.24 mm wider than it is.

Copper pad · Body outline (F.Fab) · Problem area · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

The layout shows a part 1.4 mm longer than it is, so the space around the real part is misjudged.

FIX

- Draw the fab body at 1.0 x 1.95 mm (or 1.05 x 2.0 mm at max tolerance).

JIRA

Summary	74LVC1G99GT footprint: body outline 1.28 x 3.35 mm, package is 1.0 x 1.95 mm
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, footprint-error, 74LVC1G99
Components	Part Creation / Footprint Library
Description	Submitted footprint 74LVC1G99GT,115 (ID 24153): F.Fab body 1.28 x 3.35 mm. Standard footprint draws 1.04 x 2.01 mm; datasheet SOT833-1 body is 1.0 x 1.95 mm. Redraw to the package size.

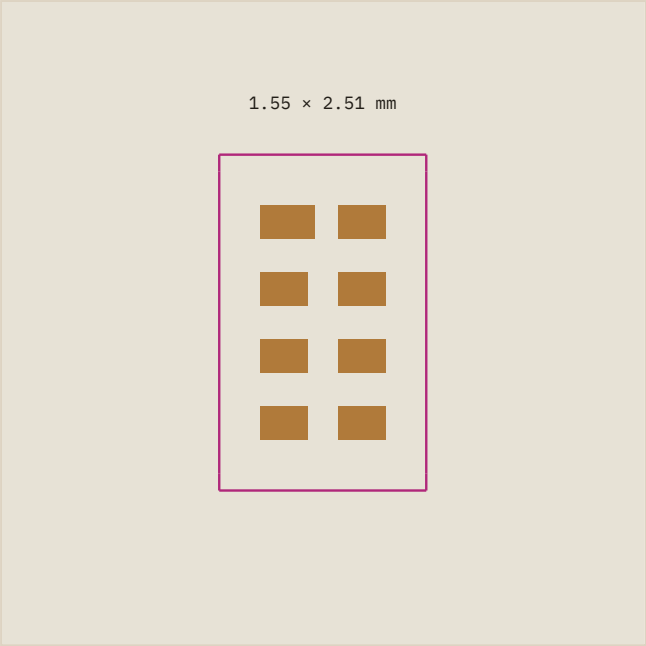
FP-03 MEDIUM

Courtyard is 4.2 x 3.85 mm, four times the standard

The courtyard is 4.20 x 3.85 mm because it wraps the oversized pads from FP-01. The standard courtyard is 1.55 x 2.51 mm.

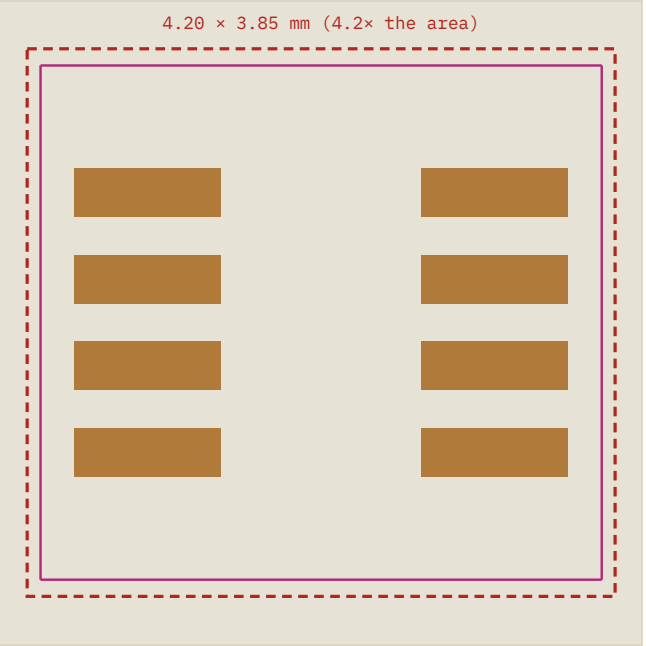
	STANDARD FILE (MM)	SUBMITTED (MM)
Courtyard	1.55 x 2.51 (3.9 mm²)	4.20 x 3.85 (16.2 mm²)

STANDARD FILE
XSON8_SOT833-1_NEX.kicad_mod



Standard: The courtyard hugs the part: 1.55 × 2.51 mm, with steps around the pad rows.

SUBMITTED FILE
twoDthreeD.glb



Submitted: The courtyard is 4.20 × 3.85 mm, sized around the wrong pads from FP-01.

Issue: The part reserves about 16 mm² of board instead of 3.9 mm², which blocks dense placement. This is fixed by fixing FP-01.

Copper pad Courtyard Problem area · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

The part blocks about 12 mm² of board that a 1 x 2 mm part doesn't need.

FIX

- Recompute the courtyard after FP-01 is fixed: pad extents plus 0.25 mm.

JIRA

Summary	74LVC1G99GT footprint: courtyard 4.20 x 3.85 mm vs 1.55 x 2.51 mm standard
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, footprint-error, courtyard, 74LVC1G99
Components	Part Creation / Footprint Library
Description	Submitted footprint 74LVC1G99GT,115 (ID 24153): F.CrtYd 4.20 x 3.85 mm (16.2 mm²), sized around the wrong pads. Standard courtyard is 1.55 x 2.51 mm (3.9 mm²). Recompute after the land pattern is fixed.

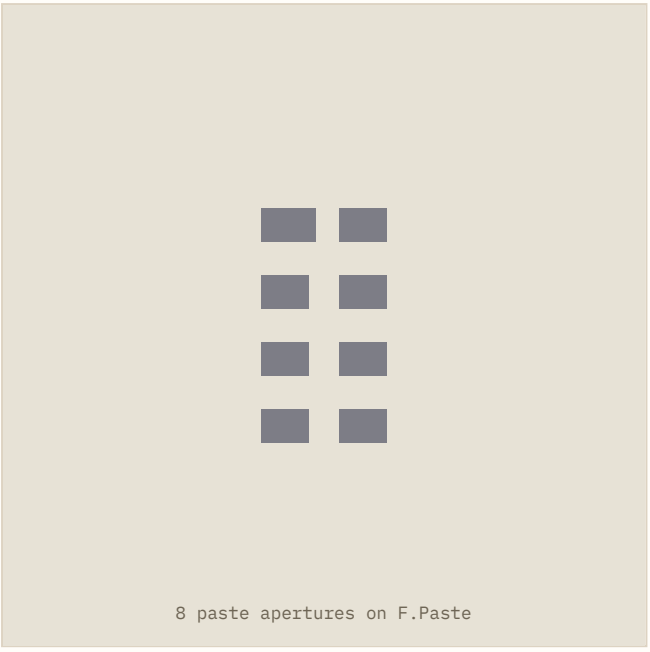
FP-04 MEDIUM

Paste-mask shapes are tagged as copper layer on all 8 pads

As on TL16C2550 (FP-05 there): each pad's PasteMask node has name "CopperCirclePad" and layerId "F.Cu". Nothing is tagged F.Paste. This looks like a generator-wide defect, not specific to one part.

NODE	EXTRAS.NAME	EXTRAS.LAYERID	RESULT
SolderMask	SolderMask	F.Mask	OK
Copper	CopperCirclePad	F.Cu	name says circle, shape is rectangle
PasteMask	CopperCirclePad	F.Cu	should be F.Paste

STANDARD FILE
XSON8_SOT833-1_NEX.kicad_mod



Standard: Every pad lists F.Paste, so the stencil gets 8 openings.

SUBMITTED FILE
twoDthreeD.glb



Submitted: Every pad's PasteMask node is a copy of the copper record (name "CopperCirclePad", layerId "F.Cu"). Nothing is tagged F.Paste.

Issue: Same defect as TL16C2550 FP-05. If export follows layerId, the stencil has no openings for this part.

Paste aperture · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

The paste layer may export empty for this part. It also explains the engineer review's report that paste and silk views show red pads.

FIX

- Fix once in the generator: write layerId F.Paste on PasteMask nodes.

JIRA

Summary	74LVC1G99GT footprint: PasteMask nodes carry layerId F.Cu (same defect as TL16C2550)
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, footprint-error, paste-mask, 74LVC1G99
Components	Part Creation / Footprint Library
Description	Submitted footprint 74LVC1G99GT,115 (ID 24153): every PasteMask node has name CopperCirclePad and layerId F.Cu. Same as TL16C2550 (ID 24150), so likely a generator-wide defect. Write F.Paste on paste nodes.

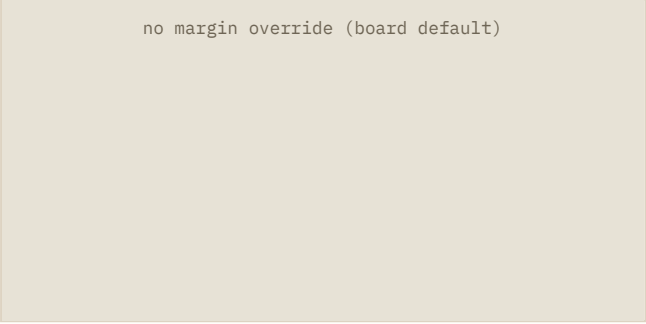
FP-05 LOW

Solder-mask margin declared 0.102 mm, applied 0.051 mm

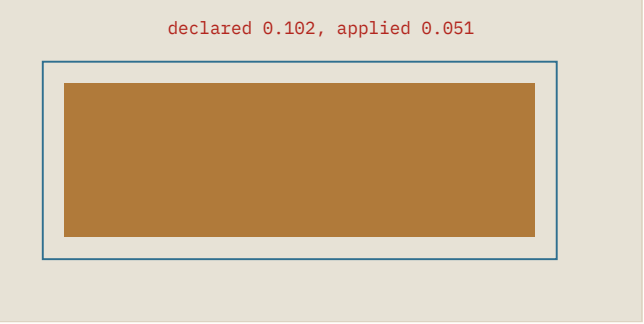
Pads declare solder_mask_margin 0.102 mm, but the SolderMask shape is the pad plus 0.051 mm per side (1.20 x 0.46 mm on a 1.10 x 0.36 mm pad). The standard footprint sets no margin and uses the board default. Same mismatch as on TL16C2550.

	STANDARD FILE (MM)	SUBMITTED (MM)
Declared margin	none (board default)	0.102
Applied per side	board default	0.051

STANDARD FILE
XSON8_SOT833-1_NEX.kicad_mod



SUBMITTED FILE
twoDthreeD.glb



Standard: The footprint sets no solder_mask_margin, so the board's global mask rule applies.

Submitted: Pads declare solder_mask_margin 0.102 mm, but the SolderMask shape is the pad plus 0.051 mm per side.

Issue: Same declared-vs-applied mismatch as TL16C2550 FP-08.

Solder-mask opening Copper pad · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

The declared and drawn values disagree, so export and DRC can use different numbers.

FIX

- Store the margin that is actually applied, per side.

JIRA

Summary	74LVC1G99GT footprint: solder mask applied at 0.051 mm/side while 0.102 mm declared
Issue Type	Bug
Priority	Low
Labels	part-verification, part-library, footprint, solder-mask, 74LVC1G99

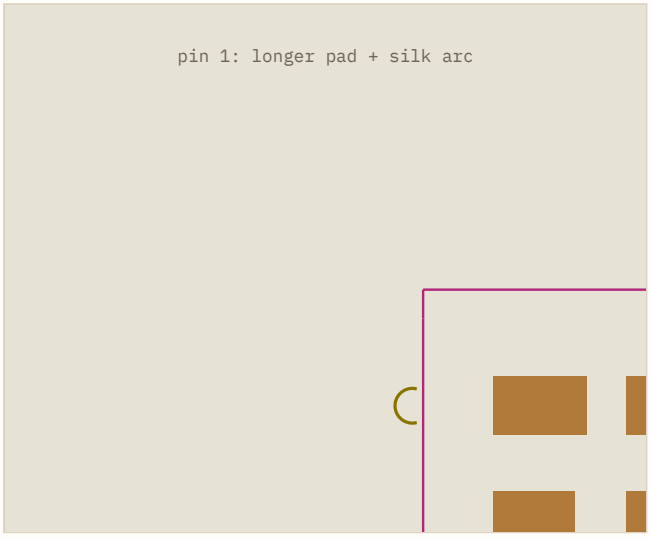
FP-06 LOW

Pin 1 is marked only by a dot outside the courtyard

In the standard footprint, pad 1 is longer than the other pads (0.41 vs 0.36 mm) and a silk arc sits beside it. In the submitted footprint all 8 pads are identical. The only pin-1 mark is a 0.2 mm dot at (-2.30, +2.13) mm, outside the courtyard corner. The value text is 15.9 x 2.3 mm on F.Fab.

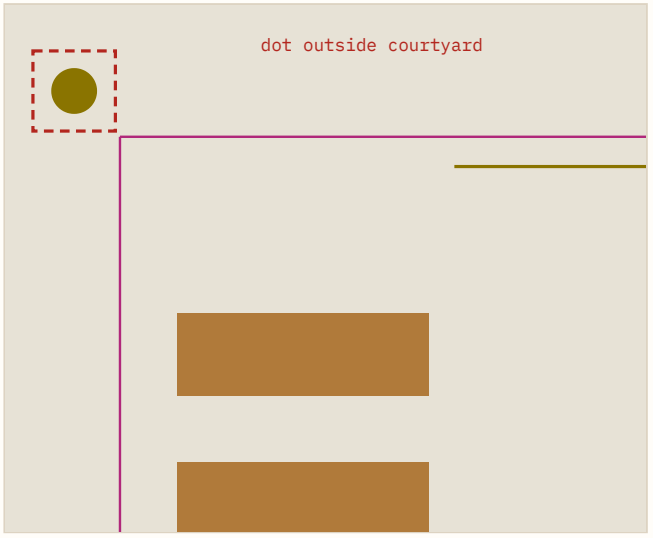
	STANDARD FILE	SUBMITTED
Pin-1 indication	longer pad 1 + silk arc	0.2 mm dot only
Marker position	beside pad 1	(-2.30, +2.13), courtyard ±2.10 x ±1.93

STANDARD FILE
XSON8_SOT833-1_NEX.kicad_mod



Standard: Pin 1 is marked twice: pad 1 is longer than the others (0.41 vs 0.36 mm), and a silk arc sits just left of it.

SUBMITTED FILE
twoDthreeD.glb



Submitted: All 8 pads are the same size. The pin-1 dot is at (−2.30, +2.13) mm, outside the courtyard corner (±2.10, ±1.93). The value text is 15.9 × 2.3 mm (F.Fab only).

Issue: Pin 1 can only be found from a marker that sits off the part's own area. Cosmetic, but orientation errors on 8-pin leadless parts are common.

Copper pad · Silkscreen · Courtyard · Pin-1 marker · Problem area · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

The orientation mark is easy to lose. On a leadless 8-pin part a 180° rotation is a common assembly error.

FIX

- Move the pin-1 mark next to pad 1, inside the courtyard.
- Consider a pin-1 pad shape cue as in the standard.

3D-01 HIGH

No 3D model; any generated body follows the wrong footprint

twoDthreeD.glb holds only 2D data (no 3D body or terminal geometry). On TL16C2550 the editor's 3D view builds a placeholder box from the footprint's fab body. Here that body is 1.28 x 3.35 mm instead of 1.0 x 1.95 mm, so any such box is wrong too. The engineer review says the same: "2D side of both ICs are not same, that's why 3D side also not same". The standard Ultra Librarian KiCad pack contains no STEP model either.

	STANDARD FILE	SUBMITTED
3D model file	none in the KiCad pack	none
Body used for 3D	-	1.28 x 3.35 mm fab body (wrong)
Real package	XSON8 SOT833-1, 1.0 x 1.95 x 0.5 mm max	-

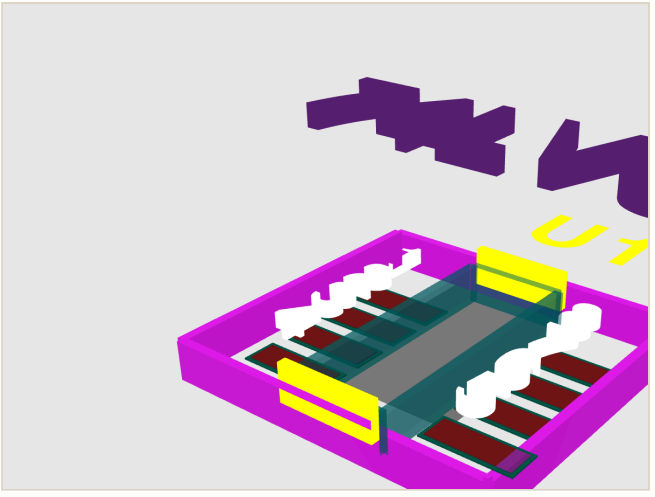
STANDARD FILE
Ultra Librarian KiCad pack

No 3D model in the standard pack.

Datasheet package SOT833-1: 1.0 × 1.95 × 0.5 mm, 8 bottom terminals, 0.5 mm pitch.

Standard: The Ultra Librarian pack ships symbol and footprints only. The model has to come from Nexperia's SOT833-1 package page.

SUBMITTED FILE
twoDthreeD.glb rendered in 3D



Submitted: Only flat layer geometry: pads, mask, silk, courtyard and text. There is no package body or terminals. Any body the editor generates would follow the wrong 1.28 × 3.35 mm outline.

Issue: Neither side has a real 3D model. The submitted footprint would also put any model in the wrong place.

Right: the submitted GLB rendered at an angle, showing it has no 3D body.

IMPACT

3D view, height checks and enclosure fit show the wrong part.

FIX

- Get the SOT833-1 3D model from Nexperia's package page, or build one: 1.0 x 1.95 x 0.5 mm, 8 bottom terminals at 0.5 mm pitch.
- Same fallback rule as TL16C2550: size any generated box from the datasheet package, not the footprint.

JIRA

Summary	74LVC1G99GT part: no 3D model; generated body would follow the wrong 1.28 x 3.35 mm footprint body
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, 3d-model, 74LVC1G99
Components	Part Creation / 3D Models
Description	Submitted part 74LVC1G99GT,115 (ID 24153): twoDthreeD.glb contains 2D data only. The 3D view's generated body is built from the fab outline, here 1.28 x 3.35 mm instead of the 1.0 x 1.95 mm SOT833-1 body. Neither the submitted nor the standard pack has a STEP model. Source the SOT833-1 model from Nexperia and size any fallback from datasheet dimensions.

NOTES ON THE REFERENCE

- The standard symbol uses 300 mil (7.62 mm) pins; the IDEEZA rule is 100 mil.
- The standard symbol types GND (pin 4) as power_out, which is wrong for a ground pin; power_in or passive is expected.
- The Ultra Librarian pack has three density variants (NEX nominal, NEX-M most, NEX-L least). This log compares against the nominal one. The pad pitch and column positions are the same in all three; only pad size changes.
- The pack contains no 3D model.
- The submitted GLB places the whole part 0.23 mm right and 0.21 mm down on its default board. Fab, silk, courtyard and pads all move together, so the figures use the part's own local coordinates. This is not a defect.

AD9106BCPZ

Part Analog Devices AD9106BCPZ, quad 12-bit DAC waveform generator, LFCSP-32 5 × 5 mm (CP-32-12)

Submitted part library ID 24154

Reference Ultra Librarian KiCad v6 package (symbol, footprint; no 3D model)

> **Not usable as submitted.** The footprint is a through-hole DIP-32 instead of a 5 × 5 mm surface-mount LFCSP-32, and it has no exposed pad (FP-01, FP-02). The symbol is missing the exposed-pad pin and none of its three active-low pins are marked. All 32 pin names are correct.

SCHEMATIC SYMBOL

Reference: 2026-10-05_13-46-48.kicad_sym (AD9106BCPZ) · Submitted: schematic.glb

PIN-BY-PIN COMPARISON

33 standard pins: 21 match, 8 differ in electrical type only, 4 with a naming, label or polarity problem

PIN	STANDARD NAME	SUBMITTED NAME	LABEL DRAWN	STANDARD TYPE	SUBMITTED TYPE	ACTIVE-LOW (STD / SUB)	RESULT
1	SCLK	SCLK	SCLK	input	input	– / –	match
2	SDIO	SDIO	SDIO	bidirectional	bidirectional	– / –	match
3	DGND	DGND	DGND	power_in	power_in	– / –	match
4	DLDO2	DLDO2	DLDO2	output	power_out	– / –	type differs
5	DVDD	DVDD	DVDD	power_in	power_in	– / –	match
6	DLDO1	DLDO1	DLDO1	output	power_out	– / –	type differs
7	SDO/SDI2/DOUT	SDO/SDI2/DOUT	SDO/SDI2/DOUT	bidirectional	bidirectional	– / –	match
8	CS	CS	CS	unspecified	input	yes / –	**active-low not set** type differs
9	RESET	RESET	RESET	unspecified	input	yes / –	**active-low not set** type differs
10	IOUTP4	IOUTP4	IOUTP4	output	output	– / –	match
11	IOUTN4	IOUTN4	IOUTN4	output	output	– / –	match
12	AVDD2	AVDD2	AVDD2	power_in	power_in	– / –	match
13	IOUTN3	IOUTN3	IOUTN3	output	output	– / –	match
14	IOUTP3	IOUTP3	IOUTP3	output	output	– / –	match
15	AGND	AGND	AGND	power_in	power_in	– / –	match
16	FSADJ3	FSADJ3	FSADJ3	output	passive	– / –	type differs
17	FSADJ4	FSADJ4	FSADJ4	output	passive	– / –	type differs
18	REFIO	REFIO	REFIO	bidirectional	bidirectional	– / –	match
19	CLKGND	CLKGND	CLKGND	power_in	power_in	– / –	match
20	CLKN	CLKN	CLKN	input	input	– / –	match
21	CLKP	CLKP	CLKP	input	input	– / –	match
22	CLDO	CLDO	CLDO	output	unspecified	– / –	type differs
23	CLKVDD	CLKVDD	CLKVDD	input	power_in	– / –	type differs
24	FSADJ2/CAL_SENSE	FSADJ2/CAL_SENSE	FSADJ2/CAL_SENSE	bidirectional	input	– / –	type differs
25	FSADJ1	FSADJ1	FSADJ1	output	passive	– / –	type differs
26	AGND	AGND	AGND	power_in	power_in	– / –	match

PIN	STANDARD NAME	SUBMITTED NAME	LABEL DRAWN	STANDARD TYPE	SUBMITTED TYPE	ACTIVE-LOW (STD / SUB)	RESULT
27	IOUTP1	IOUTP1	IOUTP1	output	output	– / –	match
28	IOUTN1	IOUTN1	IOUTN1	output	output	– / –	match
29	AVDD1	AVDD1	AVDD1	power_in	power_in	– / –	match
30	IOUTN2	IOUTN2	IOUTN2	output	output	– / –	match
31	IOUTP2	IOUTP2	IOUTP2	output	output	– / –	match
32	TRIGGER	TRIGGER	TRIGGER	input	input	yes / –	**active-low not set**
HTAB	EPAD	–	–	unspecified	–	– / –	**missing in submitted**

MATCHES THE REFERENCE

- All 32 package pin names match the standard and the datasheet.
- Pin length is 2.54 mm (100 mil), which meets the 100 mil rule. The standard uses 7.62 mm (300 mil).
- Pin lines are 0.15 mm, matching the KiCad default.
- Value and MPN carry the full part number AD9106BCPZ.

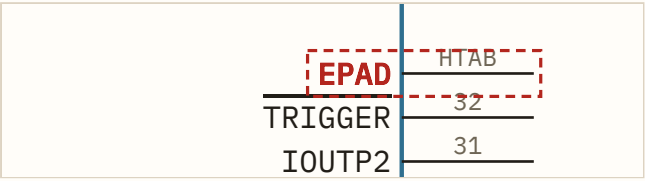
SCH-01 HIGH

Exposed-pad pin (EPAD) is missing from the symbol

The standard symbol has 33 pins: 32 package pins plus pin "HTAB" named EPAD for the exposed pad. The submitted symbol has only 32. The AD9106 datasheet requires the exposed pad to be connected to ground.

	STANDARD FILE	SUBMITTED
Pin count	33 (1-32 + HTAB/EPAD)	32
EPAD pin	HTAB, right side top	missing

STANDARD FILE
AD9106BCPZ (.kicad_sym)



Standard: Pin "HTAB" (EPAD) is the exposed pad, at the top of the right side.

SUBMITTED FILE
schematic.glb



Submitted: The symbol has 32 pins and no EPAD pin. The GND group on the right ends at AGND (26).

> **Issue:** The exposed pad can't be connected in the schematic, so the netlist never ties it to ground.

Both drawn to the same scale, from the files' own pin data. ⛶ Affected pins.

IMPACT

The exposed pad never gets a net, so it isn't tied to ground in the layout. The chip loses its main thermal path and ground return.

FIX

- Add the exposed pad as a pin (EPAD) and map it to the footprint's exposed pad.
- Make part creation add an EPAD pin whenever the package has an exposed pad.

JIRA

Summary	AD9106BCPZ symbol: exposed pad pin (EPAD) missing, 32 of 33 pins
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, symbol-error, AD9106
Components	Part Creation / Symbol Library
Description	Submitted symbol for AD9106BCPZ (library ID 24154, schematic.glb) has 32 pins. The standard Ultra Librarian symbol has 33, including pin HTAB (EPAD) for the exposed pad, which the datasheet requires to be grounded. Add the EPAD pin and map it to the footprint's exposed pad.

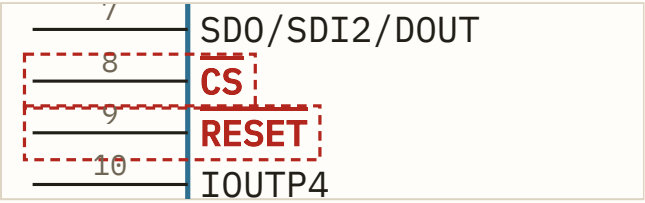
SCH-02 HIGH

No active-low marking on CS, RESET and TRIGGER

The standard symbol marks CS (8), RESET (9) and TRIGGER (32) as active-low ("*CS", "*RESET", "*TRIGGER"), matching the datasheet. The submitted symbol stores activeLow = false on all three and draws them as plain labels.

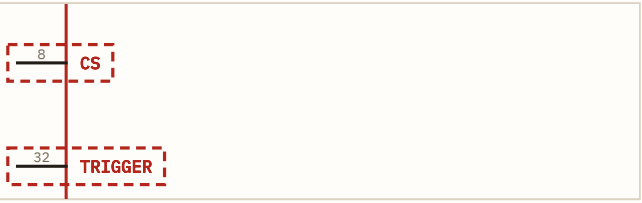
PIN	STANDARD FILE	SUBMITTED
8	*CS	CS, activeLow false
9	*RESET	RESET, activeLow false
32	*TRIGGER	TRIGGER, activeLow false

STANDARD FILE
AD9106BCPZ (.kicad_sym)



Standard: CS (8), RESET (9) and TRIGGER (32) carry an overbar: all three are active-low.

SUBMITTED FILE
schematic.glb



Submitted: CS, RESET and TRIGGER are plain labels with activeLow = false.

> **Issue:** Tying CS or RESET high to enable the chip holds it deselected or in reset.

Both drawn to the same scale, from the files' own pin data. Affected pins.

IMPACT

Read as active-high, CS and RESET would be tied the wrong way: the chip stays deselected or held in reset.

FIX

- Set activeLow = true on pins 8, 9 and 32 and draw the overbar.

JIRA

Summary	AD9106BCPZ symbol: CS, RESET, TRIGGER not marked active-low
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, symbol-error, AD9106
Components	Part Creation / Symbol Library
Description	Submitted symbol AD9106BCPZ (ID 24154): pins 8 CS, 9 RESET and 32 TRIGGER have activeLow = false and no overbar. Standard symbol and datasheet mark all three active-low. Set the flag and draw the bar.

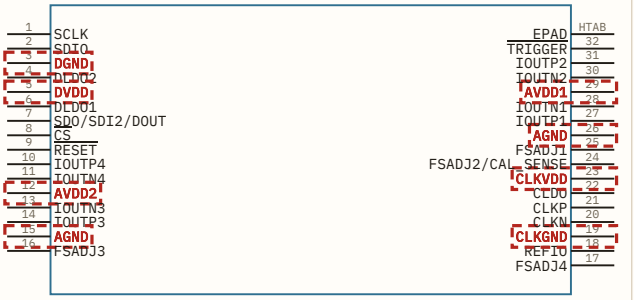
SCH-03 LOW

Pin order and shape differ from the standard symbol

The standard symbol follows package order: pins 1-16 down the left and 17-32 up the right, EPAD at the top right, on a 91.4 x 50.8 mm wide rectangle. The submitted symbol puts 21 pins on the left (clock and SPI at the top, DAC outputs below) and 11 power and ground pins on the right, on a 66.2 x 71.3 mm near-square body. All pin numbers carry the right names.

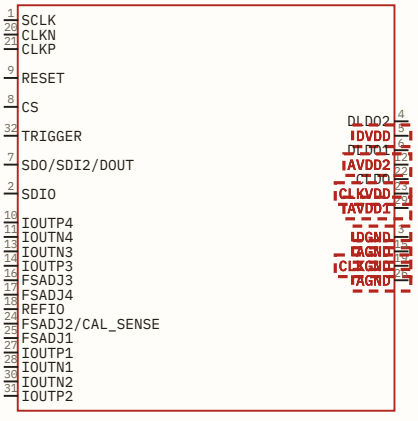
	STANDARD FILE	SUBMITTED
Left / right pins	16 / 17	21 / 11
Body	91.44 x 50.80 mm	66.19 x 71.27 mm
Body area	4,645 mm²	4,717 mm²
Pin length	7.62 mm (300 mil)	2.54 mm (100 mil), meets the 100 mil rule

STANDARD FILE
AD9106BCPZ (.kicad_sym)



Standard: Package order: pins 1–16 down the left, 17–32 up the right, EPAD at the top right. A 91.4 × 50.8 mm wide rectangle.

SUBMITTED FILE
schematic.glb



Submitted: 21 pins on the left (clock and SPI at the top, DAC outputs below), 11 power/ground pins on the right. A 66.2 × 71.3 mm near-square body.

> **Issue:** The layout differs from the reference (power pins highlighted). Connectivity is unaffected; the house rule for pin arrangement needs to be decided.

Both drawn to the same scale, from the files' own pin data. Affected pins.

IMPACT

FIX

No effect on connectivity. The submitted layout is lopsided: 11 pins on a 71 mm side. Both symbols are about the same area, so "too big" applies to both.

- Decide one house rule for pin arrangement and apply it to all generated symbols.
- Balance the two sides so the body height follows the longer column, not a 21-pin one.

JIRA

Summary	AD9106BCPZ symbol: pin order and shape differ from standard (21/11 split, near-square body)
Issue Type	Improvement
Priority	Low
Labels	part-verification, part-library, symbol-layout, AD9106
Components	Part Creation / Symbol Library
Description	Submitted symbol AD9106BCPZ (ID 24154) uses functional grouping: 21 pins left, 11 power/ground pins right, body 66.19 x 71.27 mm. Standard uses package order on a 91.44 x 50.80 mm body (same area). Connectivity is correct. Decide the house rule and balance the sides.

SCH-04 LOW

Part filed under ADC, but the AD9106 is a DAC

The submitted part's category is "Analog to Digital Converters (ADC)". The AD9106 is a quad, 12-bit digital-to-analog converter (DAC) waveform generator. The value and MPN are correct (AD9106BCPZ). The footprint is named "AD9106BCPZ", which gives no package; the standard calls it CP_32_12_ADI.

FIELD	STANDARD FILE	SUBMITTED
Category	-	Analog to Digital Converters (ADC)
Value / MPN	AD9106BCPZ	AD9106BCPZ
Footprint name	CP_32_12_ADI	AD9106BCPZ

STANDARD FILE
AD9106BCPZ (.kicad_sym)

PART PROPERTIES

Value	AD9106BCPZ
MPN	AD9106BCPZ
Footprint	CP_32_12_ADI
Category	-

Standard: The footprint is named after the package (CP-32-12, 5 × 5 mm LFCSP).

SUBMITTED FILE
schematic.glb

PART PROPERTIES

Value	AD9106BCPZ
MPN	AD9106BCPZ
Footprint	**AD9106BCPZ**
Category	**Analog to Digital Converters (ADC)**

Submitted: Part number is complete. The category says ADC, but the AD9106 is a quad DAC (digital-to-analog) waveform generator. The footprint name gives no package.

Issue: Library search by category will file the part under the wrong function.

IMPACT

FIX

Searching the library for DACs won't find it, and the footprint name doesn't reveal the package.

- Set the category to Digital to Analog Converters (DAC).
- Name the footprint after the package (LFCSP-32 / CP-32-12).

JIRA

Summary	AD9106BCPZ part: category ADC (should be DAC); footprint name gives no package
Issue Type	Bug
Priority	Low
Labels	part-verification, part-library, metadata, AD9106
Components	Part Creation / Symbol Library
Description	Submitted part ID 24154: category "Analog to Digital Converters (ADC)"; AD9106 is a quad 12-bit DAC. Footprint name "AD9106BCPZ" doesn't identify the package (standard: CP_32_12_ADI). Value and MPN are correct.

SCH-05 LOW

Two electrical types are inconsistent

9 pins have a different electrical type from the standard. Seven of the submitted types are the better choice: the LDO outputs as power_out, the FSADJ resistor pins as passive, CLKVDD as power_in, and CS and RESET as input. Two are not: CLDO (22), a regulator output like DLDO1/DLDO2, is typed unspecified; FSADJ2/CAL_SENSE (24), which also senses calibration, is typed input only.

PIN	STANDARD TYPE	SUBMITTED TYPE	ASSESSMENT
22 CLDO	output	unspecified	should be power_out, like DLDO1/2
24 FSADJ2/CAL_SENSE	bidirectional	input	passive or bidirectional
4, 6 DLDO2/DLDO1	output	power_out	submitted is better
16, 17, 25 FSADJx	output	passive	submitted is better
23 CLKVDD	input	power_in	submitted is better
8, 9 CS, RESET	unspecified	input	submitted is better

IMPACT

ERC can't check CLDO as a supply output, so a missing decoupling or load connection there won't be flagged.

FIX

- Type CLDO (22) as power_out and FSADJ2/CAL_SENSE (24) as passive or bidirectional.

JIRA

Summary	AD9106BCPZ symbol: CLDO typed unspecified, FSADJ2/CAL_SENSE typed input
Issue Type	Improvement
Priority	Low
Labels	part-verification, part-library, symbol-error, AD9106
Components	Part Creation / Symbol Library
Description	Submitted symbol AD9106BCPZ (ID 24154): pin 22 CLDO is unspecified though DLDO1/DLDO2 are power_out; pin 24 FSADJ2/CAL_SENSE is input only. Seven other type changes vs the standard are improvements. Set CLDO to power_out and pin 24 to passive or bidirectional.

FOOTPRINT (2D)

Reference: CP_32_12_ADI.kicad_mod (nominal density) · Submitted: twoDthreeD.glb

MATCHES THE REFERENCE

- Pin numbering runs 1–32 in order with pin 1 marked by a square pad.

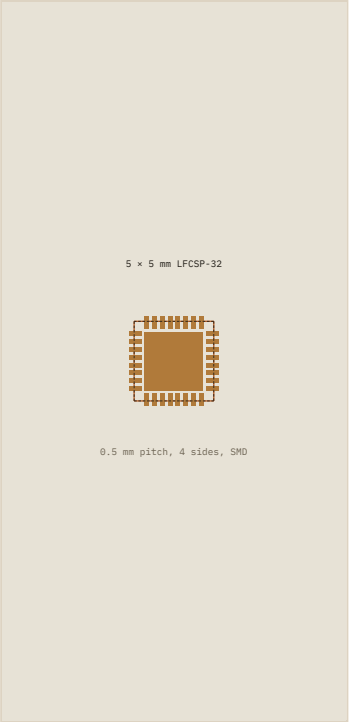
FP-01 HIGHEST

Wrong package: a through-hole DIP-32 instead of a 5 x 5 mm LFCSP-32

The AD9106BCPZ is a 5 x 5 mm LFCSP-32 (CP-32-12): 8 SMD pads per side at 0.50 mm pitch and a 3.75 mm exposed pad. The submitted footprint is a 600-mil through-hole DIP-32: two rows of 16 pads, 1.53 mm round (pin 1 square), 0.83 mm drills, 2.54 mm pitch, rows 15.24 mm apart.

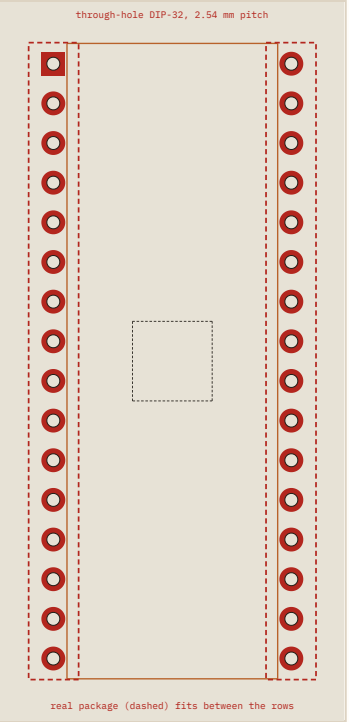
	STANDARD FILE (MM)	SUBMITTED (MM)
Mount type	SMD	through-hole (32 drilled holes)
Pitch	0.50	2.54
Layout	8 pads on each of 4 sides	2 rows of 16
Pad	0.27 x 0.80 rect	1.53 round, 0.83 drill
Pad span	5.71 x 5.71	16.77 x 39.63

STANDARD FILE
CP_32_12_ADI.kicad_mod



Standard: The AD9106BCPZ is a 5 × 5 mm LFCSP-32: 8 SMD pads per side at 0.50 mm pitch, plus a 3.75 mm exposed pad in the middle. Both panels are drawn at the same scale.

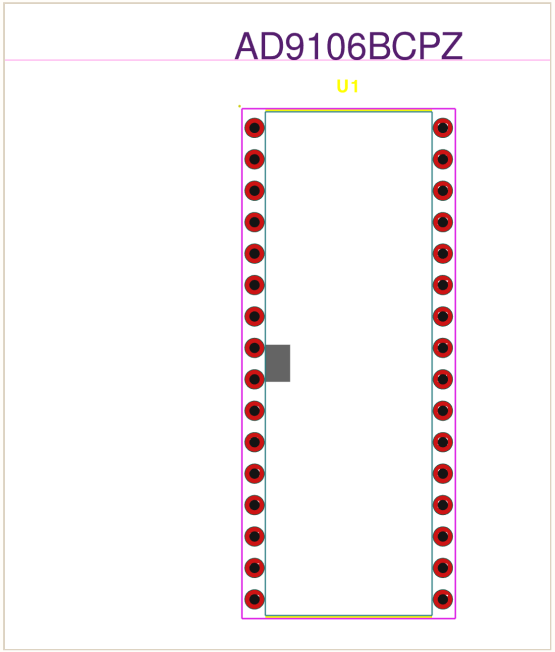
SUBMITTED FILE
twoDthreeD.glb



Submitted: Two rows of 16 through-hole pads, 1.53 mm round (pin 1 square) with 0.83 mm drills, 2.54 mm pitch, rows 15.24 mm apart: a 600-mil DIP-32.

> **Issue:** This is a different package type. No pad lines up with the chip, and the board would need drilled holes that the part can't use.

Copper pad · Misplaced pad · Real package outline (standard F.Fab) · Body outline (F.Fab) · Problem area · Top view, both drawn to the same scale from the files' own coordinates.



The submitted footprint as it renders in the editor.

IMPACT

FIX

The chip can't be mounted. The board would also be drilled with 32 unneeded holes.

- Rebuild from the CP-32-12 package drawing, or use the standard CP_32_12_ADI footprint.
- Find out why an SMD part (suffix CPZ = LFCSP, lead-free) produced a DIP. The package code should drive land-pattern selection.

JIRA

Summary	AD9106BCPZ footprint: through-hole DIP-32 generated instead of 5x5 mm LFCSP-32
Issue Type	Bug
Priority	Highest
Labels	part-verification, part-library, footprint-error, AD9106
Components	Part Creation / Footprint Library
Description	Submitted footprint AD9106BCPZ (ID 24154, twoDthreeD.glb) is a 600-mil DIP-32: 1.53 mm round through-hole pads, 0.83 mm drills, 2.54 mm pitch, rows 15.24 mm apart. The part is a 5 x 5 mm LFCSP-32 (CP-32-12) with 0.50 mm pitch SMD pads on 4 sides and a 3.75 mm exposed pad (standard: CP_32_12_ADI). Rebuild from the package drawing.

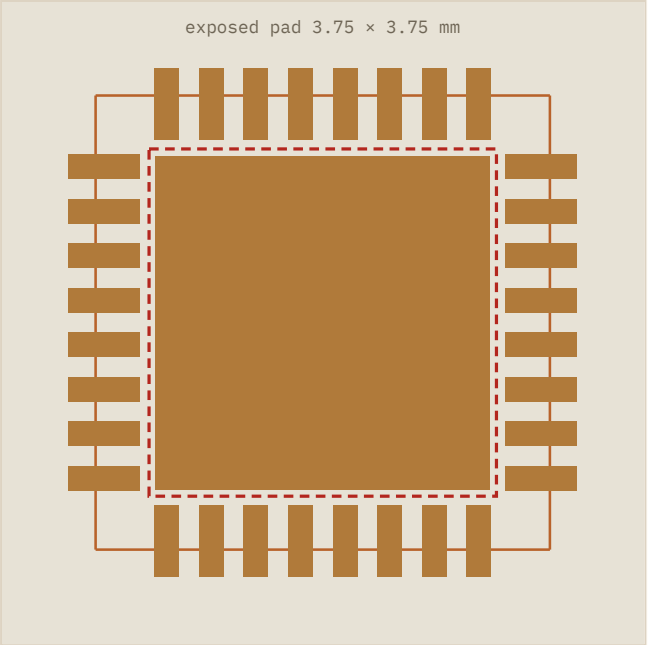
FP-02 HIGH

No exposed pad

The standard footprint has a 3.75 x 3.75 mm exposed pad ("HTAB") under the chip, with a 4-window paste pattern. The submitted footprint has 32 pads and nothing under the package centre. This matches SCH-01, where the EPAD pin is also missing.

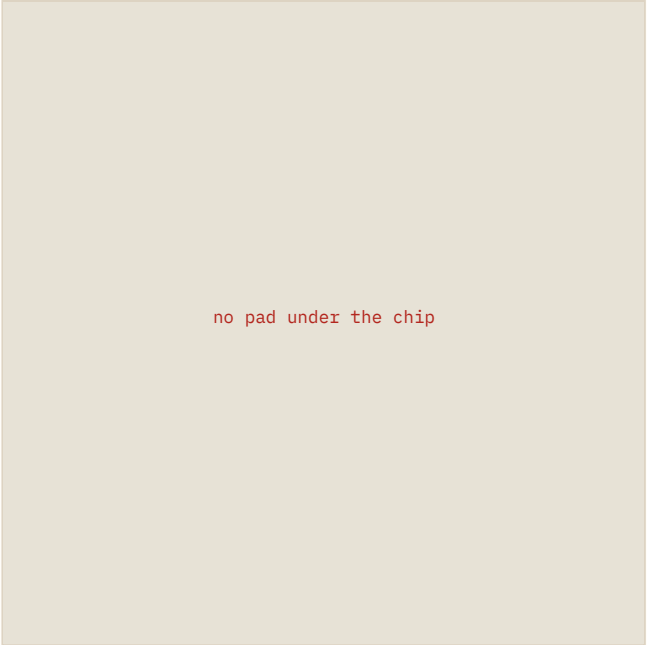
	STANDARD FILE	SUBMITTED
Exposed pad	3.75 x 3.75 mm (HTAB)	none
Pad count	33	32

STANDARD FILE
CP_32_12_ADI.kicad_mod



Standard: Pad "HTAB" is the 3.75 × 3.75 mm exposed pad under the chip. The datasheet requires it to be soldered to ground for heat and electrical return.

SUBMITTED FILE
twoDthreeD.glb



Submitted: There are only 32 pads. Nothing is under the package centre (same 7.2 mm window at the same scale).

Issue: Without the exposed pad the chip has no thermal path and no ground on its underside.

Copper pad · Body outline (F.Fab) · Problem area · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

No thermal or ground connection under the chip, even after FP-01 is fixed, unless the pad is added.

FIX

- Add the 3.75 x 3.75 mm exposed pad with a windowed paste pattern, mapped to the EPAD pin.

JIRA

Summary	AD9106BCPZ footprint: exposed pad (3.75 x 3.75 mm) missing
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, footprint-error, AD9106
Components	Part Creation / Footprint Library
Description	Submitted footprint AD9106BCPZ (ID 24154): 32 pads, no exposed pad. Standard CP_32_12_ADI has pad HTAB 3.75 x 3.75 mm with 4-window paste. Add it and map it to the EPAD pin (see symbol issue).

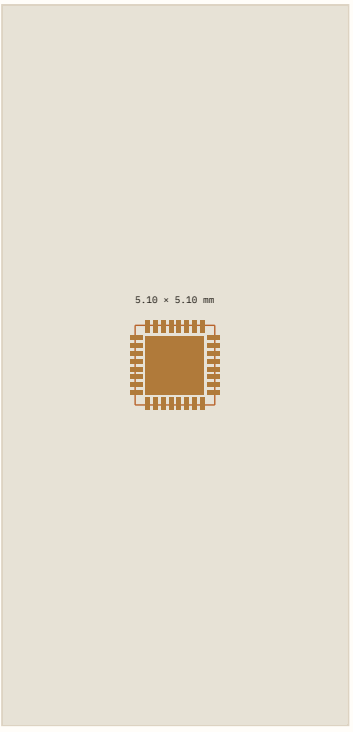
FP-03 HIGH

Body outline is 13.5 x 40.7 mm instead of 5 x 5 mm

The submitted F.Fab body is 13.49 x 40.70 mm, a DIP-32 outline. The standard draws 5.10 x 5.10 mm; the datasheet body is 5.00 x 5.00 mm.

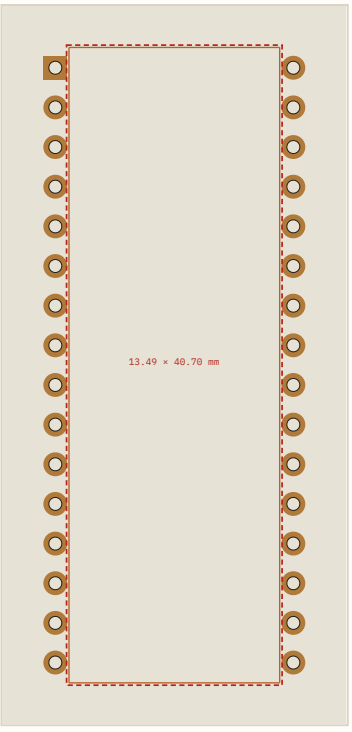
	STANDARD FILE (MM)	SUBMITTED (MM)
F.Fab body	5.10 x 5.10	13.49 x 40.70
Datasheet body	5.00 x 5.00	-

STANDARD FILE
CP_32_12_ADI.kicad_mod



Standard: F.Fab body 5.10 × 5.10 mm (datasheet 5.00 × 5.00 mm).

SUBMITTED FILE
twoDthreeD.glb



Submitted: F.Fab body 13.49 × 40.70 mm, the outline of a DIP-32.

Issue: The placement view shows a part 8 times longer than the real chip.

Copper pad · Body outline (F.Fab) · Problem area · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

The layout reserves the space of a part 8 times longer than the real chip.

FIX

- Redraw the body at 5.00 x 5.00 mm with the LFCSP footprint.

JIRA

Summary	AD9106BCPZ footprint: body outline 13.49 x 40.70 mm, package is 5 x 5 mm
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, footprint-error, AD9106
Components	Part Creation / Footprint Library
Description	Submitted footprint AD9106BCPZ (ID 24154): F.Fab body 13.49 x 40.70 mm (DIP-32). Standard 5.10 x 5.10 mm; datasheet 5.00 x 5.00 mm.

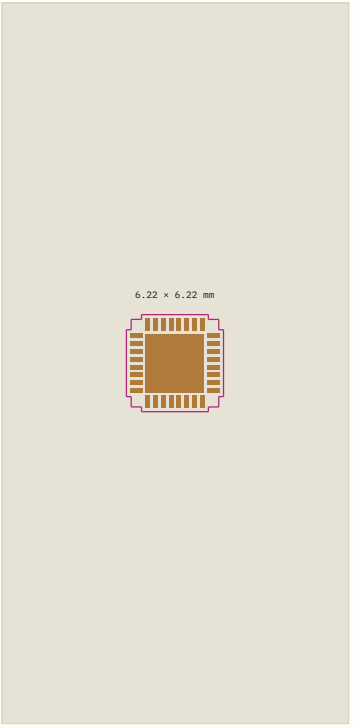
FP-04 MEDIUM

Courtyard is 17.3 x 41.2 mm, 18 times the standard

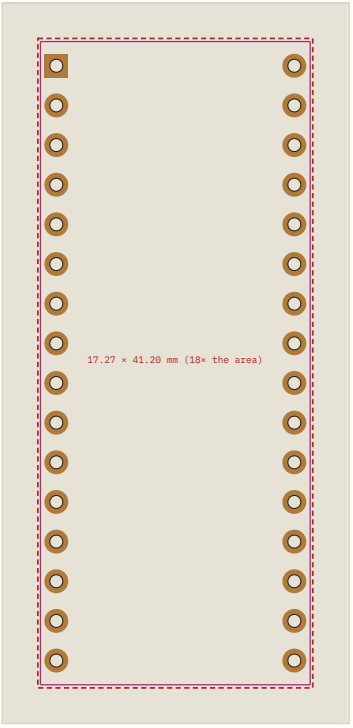
The courtyard is 17.27 x 41.20 mm (711 mm²), around the DIP pads. The standard courtyard is 6.22 x 6.22 mm (38.7 mm²).

	STANDARD FILE (MM)	SUBMITTED (MM)
Courtyard	6.22 x 6.22 (38.7 mm²)	17.27 x 41.20 (711 mm²)

STANDARD FILE
CP_32_12_ADI.kicad_mod



SUBMITTED FILE
twoDthreeD.glb



Standard: Courtyard 6.22 × 6.22 mm (38.7 mm²).

Submitted: Courtyard 17.27 × 41.20 mm (711 mm²), around the DIP pads.

Issue: The part blocks about 670 mm² of board it doesn't need. This is fixed by fixing FP-01.

Copper pad Courtyard Problem area · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

The part blocks about 670 mm² of extra board area.

FIX

- Recompute after FP-01: pad extents plus 0.25 mm.

JIRA

Summary	AD9106BCPZ footprint: courtyard 17.27 x 41.20 mm vs 6.22 x 6.22 mm standard
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, footprint-error, courtyard, AD9106
Components	Part Creation / Footprint Library
Description	Submitted footprint AD9106BCPZ (ID 24154): F.CrtYd 17.27 x 41.20 mm (711 mm²). Standard 6.22 x 6.22 mm (38.7 mm²). Recompute after the land pattern is fixed.

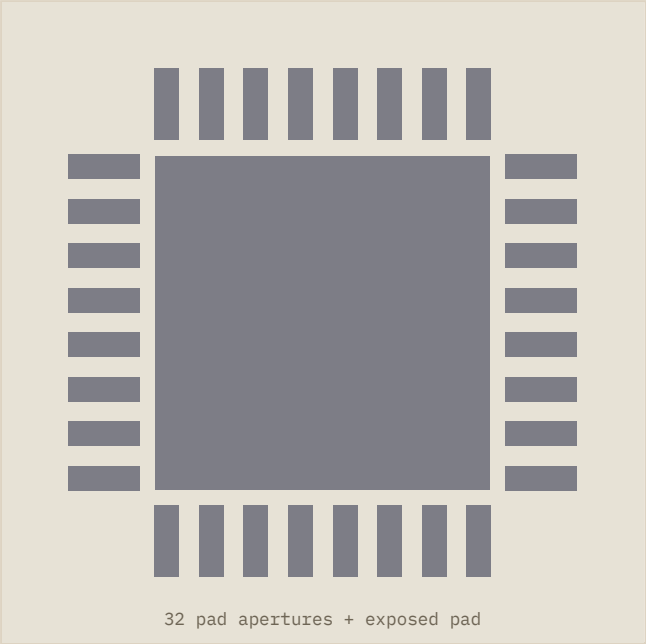
FP-05 MEDIUM

Paste-mask shapes are tagged as copper layer

As on the other two parts, each pad's PasteMask nodes are copies of the copper record (name "CopperCirclePad", layerId "F.Cu"). Nothing is tagged F.Paste. Each through-hole pad also carries two copper, two mask and two paste nodes (one per side).

NODE	EXTRAS.NAME	EXTRAS.LAYERID	RESULT
PasteMask	CopperCirclePad	F.Cu	should be F.Paste
Copper	CopperCirclePad	F.Cu	x2 per pad (both sides)
SolderMask	SolderMask	F.Mask	x2 per pad

STANDARD FILE
CP_32_12_ADI.kicad_mod



Standard: Every pad, including the exposed pad, is on F.Paste. The exposed pad also carries a 4-window paste pattern.

SUBMITTED FILE
twoDthreeD.glb



Submitted: The paste nodes are copies of the copper record (layerId F.Cu), as on the other two parts. Paste on through-hole pads would be wrong anyway.

Issue: Same generator defect as TL16C2550 FP-05 and 74LVC1G99GT FP-04.

Paste aperture · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

Same generator-wide defect on all three parts checked so far.

FIX

- Fix once in the generator: write layerId F.Paste on PasteMask nodes.

JIRA

Summary	AD9106BCPZ footprint: PasteMask nodes carry layerId F.Cu (same as TL16C2550, 74LVC1G99GT)
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, footprint-error, paste-mask, AD9106
Components	Part Creation / Footprint Library
Description	Submitted footprint AD9106BCPZ (ID 24154): PasteMask nodes are named CopperCirclePad with layerId F.Cu. Third part with the same defect (IDs 24150, 24153, 24154), so it is in the generator.

FP-06 LOW

Solder-mask margin declared 0.102 mm, applied 0.051 mm

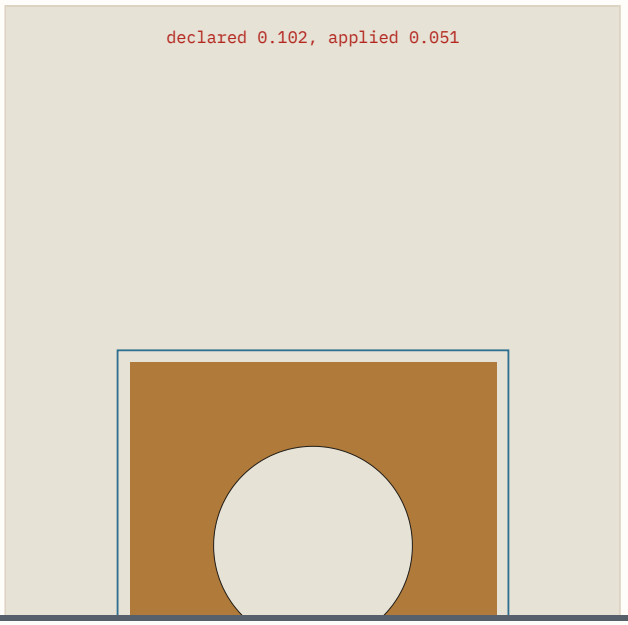
Pads declare solder_mask_margin 0.102 mm, but the SolderMask shape is the pad plus 0.051 mm per side (1.63 mm on a 1.53 mm pad). The standard sets no margin and uses the board default.

	STANDARD FILE (MM)	SUBMITTED (MM)
Declared margin	none (board default)	0.102
Applied per side	board default	0.051

STANDARD FILE
CP_32_12_ADI.kicad_mod



SUBMITTED FILE
twoDthreeD.glb



FP-07 LOW

Pin-1 marker sits outside the courtyard

Pin 1 is a square pad, which is a good cue, plus a 0.2 mm dot at (-8.83, +20.80) mm, outside the courtyard corner (±8.64, ±20.60). The standard marks pin 1 with a silk "*" next to pad 1. The value text is 18.4 x 2.3 mm on F.Fab.

	STANDARD FILE	SUBMITTED
Pin-1 marker	silk "*" beside pad 1	square pad + dot outside courtyard

STANDARD FILE
CP_LFCSP_ADI.kicad_mod

SUBMITTED FILE
twoDthreeD.glb

Reference: none in the KiCad pack (shown: standard footprint in the IDEEZA 3D view) · Submitted: twoDthreeD.glb

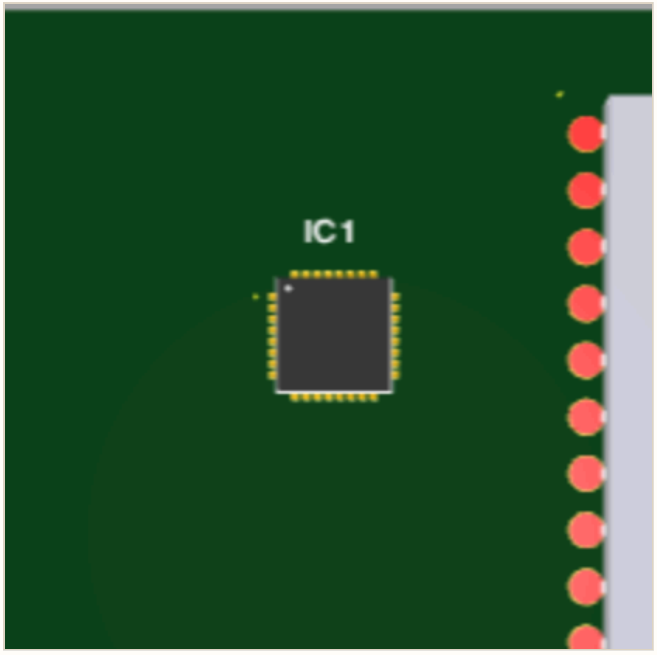
3D-01 HIGH

3D view shows a DIP-32 box instead of a 5 x 5 mm LFCSP

twoDthreeD.glb holds only 2D data plus 32 drill cylinders; there is no package model. In the IDEEZA 3D view the part appears as a long grey 13.5 x 40.7 mm DIP box. The standard footprint placed next to it renders as a small black 5 x 5 mm chip. The Ultra Librarian pack itself contains no STEP model.

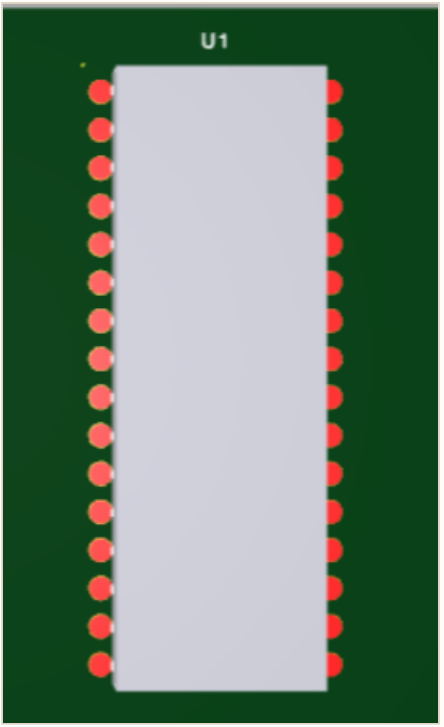
	STANDARD	SUBMITTED
3D source	none in the KiCad pack	none; generated from the DIP outline
Shown as	5 x 5 mm black chip	13.5 x 40.7 mm grey DIP box

STANDARD FILE
IDEEZA 3D view, standard footprint (IC1)



Standard: The standard footprint renders as a small black 5 × 5 mm QFN-style chip. The Ultra Librarian pack itself has no STEP model.

SUBMITTED FILE
IDEEZA 3D view, submitted part (U1)



Submitted: A long grey DIP-32 box (13.5 × 40.7 mm) with round pads down both sides.

Issue: The 3D view shows a through-hole DIP instead of a 5 × 5 mm LFCSP.

Both crops are from the same engineer-review screenshot, so they share one scale.

IMPACT

Height, enclosure and collision checks use the wrong part.

FIX

- Attach the CP-32-12 (5 x 5 x 0.9 mm LFCSP) model from Analog Devices.
- Fix FP-01 first; the generated body follows the footprint.

JIRA

Summary	AD9106BCPZ part: 3D view shows 13.5 x 40.7 mm DIP box instead of 5 x 5 mm LFCSP
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, 3d-model, AD9106
Components	Part Creation / 3D Models
Description	Submitted part AD9106BCPZ (ID 24154): no 3D model; the 3D view renders a grey DIP-32 box (13.5 x 40.7 mm) generated from the wrong footprint. Standard footprint renders as a 5 x 5 mm chip. Attach the CP-32-12 LFCSP model after the footprint is fixed.

NOTES ON THE REFERENCE

- The standard symbol uses 300 mil (7.62 mm) pins; the IDEEZA rule is 100 mil.
- Several standard pin types are questionable: CLKVDD is typed input (it is a supply), FSADJ1-4 output (they connect to a resistor), and CS/RESET unspecified. The submitted types are better on these pins (see SCH-05).
- The standard footprint puts pin 1 at the top-right, with pins 1–8 along the top edge. That is the datasheet pin-out rotated 90°, which is valid but means a 90° placement offset against the datasheet drawing.
- The standard footprint's exposed pad (HTAB) lists F.Paste on the whole pad and also has a 4-window paste pattern. Normally only the windows should print; check the paste layer before using it as is.
- The Ultra Librarian pack has three density variants (ADI, ADI-M, ADI-L) with the same pad positions; this log compares against the nominal one. The pack contains no 3D model.
- The submitted GLB places the whole part 5.74 mm right and 0.47 mm up on its default board. Pads and outlines move together, so the figures use the part's own coordinates.

693072010801

Part Würth Elektronik 693072010801, WR-CRD microSD card connector, hinge type, SMD, 8 contacts + 4 shield tabs

Submitted part library ID 24152

Reference SnapEDA KiCad package (symbol, footprint, STEP)

> **Not usable as submitted.** The footprint is a through-hole DIP-8 instead of the SMD microSD socket, and it has no shield pads (FP-01, FP-02). The symbol has no shield pins, and there is no 3D model. All 8 contact names are correct.

SCHEMATIC SYMBOL

Reference: 693072010801.kicad_sym · Submitted: schematic.glb

PIN-BY-PIN COMPARISON

12 standard pins: 0 match, 8 differ in electrical type only, 4 with a naming, label or polarity problem

PIN	STANDARD NAME	SUBMITTED NAME	LABEL DRAWN	STANDARD TYPE	SUBMITTED TYPE	ACTIVE-LOW (STD / SUB)	RESULT
1	DAT2	DAT2	DAT2	bidirectional	unspecified	– / –	type differs
2	CD/DAT3	CD/DAT3	CD/DAT3	bidirectional	unspecified	– / –	type differs
3	CMD	CMD	CMD	bidirectional	unspecified	– / –	type differs
4	VDD	VDD	VDD	bidirectional	power_in	– / –	type differs
5	CLK	CLK	CLK	bidirectional	unspecified	– / –	type differs
6	VSS	VSS	VSS	bidirectional	power_in	– / –	type differs
7	DAT0	DAT0	DAT0	bidirectional	unspecified	– / –	type differs
8	DAT1	DAT1	DAT1	bidirectional	unspecified	– / –	type differs
S1	SHIELD	–	–	bidirectional	–	– / –	**missing in submitted**
S2	SHIELD	–	–	bidirectional	–	– / –	**missing in submitted**
S3	SHIELD	–	–	bidirectional	–	– / –	**missing in submitted**
S4	SHIELD	–	–	bidirectional	–	– / –	**missing in submitted**

MATCHES THE REFERENCE

- All 8 contact pins carry the right names: DAT2, CD/DAT3, CMD, VDD, CLK, VSS, DAT0, DAT1.
- Reference designator prefix is J, as in the standard.
- Pin length is 2.54 mm (100 mil), which meets the 100 mil rule. The standard uses 5.08 mm (200 mil).
- Pin lines are 0.15 mm, matching the KiCad default.

SCH-01 HIGH

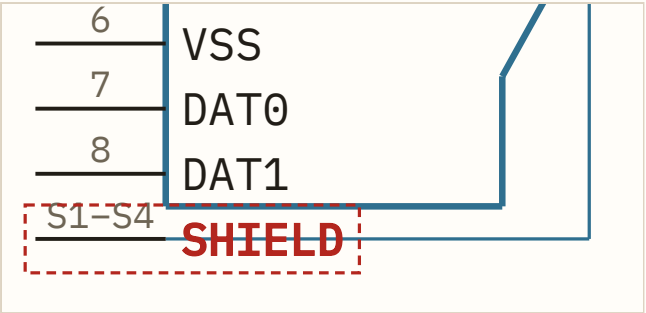
Shield pins S1–S4 are missing from the symbol

The standard symbol has 12 pins: contacts 1-8 plus S1-S4 for the four shield tabs of the metal shell. The submitted symbol has only pins 1-8.

	STANDARD FILE	SUBMITTED
Pin count	12 (1-8 + S1-S4)	8

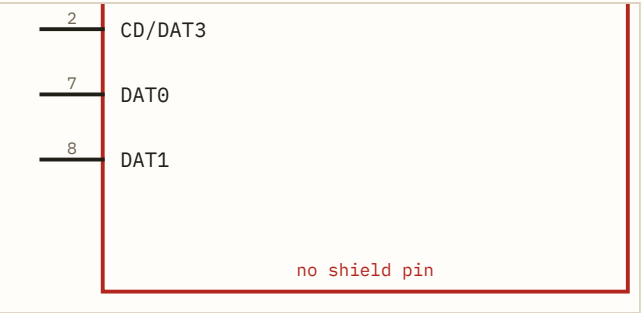
	STANDARD FILE	SUBMITTED
Shield / shell	S1-S4, bottom of the pin column	missing

STANDARD FILE
693072010801.kicad_sym



Standard: Shield pins S1–S4 (the metal shell) sit at the bottom of the pin column, stacked on one connection point.

SUBMITTED FILE
schematic.glb



Submitted: Only pins 1–8 exist. There is no pin for the shell.

> **Issue:** The shell can't be connected to ground in the schematic.

Both drawn to the same scale, from the files' own pin data. ⛶ Affected pins.

IMPACT

The shell can't be tied to ground in the schematic, so the board gets no ESD path through the connector shell, and the netlist doesn't match a footprint with shield pads.

FIX

- Add pins S1-S4 (or one SHIELD pin mapped to all four tabs).

JIRA

Summary	693072010801 symbol: shield pins S1-S4 missing (8 of 12 pins)
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, symbol-error, 693072010801
Components	Part Creation / Symbol Library
Description	Submitted symbol for Würth 693072010801 microSD connector (ID 24152) (schematic.glb) has pins 1-8 only. The standard SnapEDA symbol has 12: contacts 1-8 plus S1-S4 for the shield tabs. Add the shield pins so the shell can be grounded.

SCH-04 MEDIUM

MPN and value hold the whole product description

The MPN and value fields both read "Würth Elektronik 693072010801 WR-CRD Micro SD Card Connector". The standard has 693072010801 in both. The category is a generic placeholder rather than a connector category, and the value text renders 41 mm wide on the footprint.

FIELD	STANDARD FILE	SUBMITTED
MPN	693072010801	Würth Elektronik 693072010801 WR-CRD Micro SD Card Connector
Value	693072010801	same full description

FIELD	STANDARD FILE	SUBMITTED
Category	-	generic placeholder

STANDARD FILE
693072010801.kicad_sym

PART PROPERTIES

Reference	J
Value	693072010801
MPN	693072010801
Description	CardConnector_SMT_MicroSD_Hinge_8pins
Category	-

Standard: Value and MPN hold the part number; the description says what it is.

SUBMITTED FILE
schematic.glb

PART PROPERTIES

Reference	J1
Value	**Würth Elektronik 693072010801 WR-CRD Micro SD Card Connector**
MPN	**Würth Elektronik 693072010801 WR-CRD Micro SD Card Connector**
Description	**693072010801 (created from datasheet)**
Category	**generic placeholder, not a connector category**

Submitted: Value and MPN hold the whole product description. The category is a generic placeholder.

Issue: BOM matching and supplier lookups on the MPN field fail, and the part can't be found under connectors.

IMPACT

BOM tools and supplier lookups match on the MPN field and won't find this part. It also can't be found under connectors in the library.

FIX

- Put 693072010801 in MPN and value; keep the description in the description field.
- Set the category to connectors / memory card.

JIRA

Summary	693072010801 part: MPN and value contain the full product description; category is a placeholder
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, metadata, 693072010801
Components	Part Creation / Symbol Library
Description	Submitted part 693072010801 (ID 24152): manufacturerPartNumber and value = "Würth Elektronik 693072010801 WR-CRD Micro SD Card Connector". Standard has 693072010801. Category is a generic placeholder. Store the bare part number and a connector category.

SCH-02 LOW

Signal pins typed unspecified

The six signal pins (DAT0-DAT2, CD/DAT3, CMD, CLK) are typed unspecified. The standard types them bidirectional. VDD and VSS are power_in in the submitted part, which is better than the standard's bidirectional.

PINS	STANDARD TYPE	SUBMITTED TYPE
1, 2, 3, 5, 7, 8 (signals)	bidirectional	unspecified
4 VDD, 6 VSS	bidirectional	power_in (better)

STANDARD FILE
693072010801.kicad_sym

PART	PROPERTIES
1 DAT2	bidirectional
2 CD/DAT3	bidirectional
3 CMD	bidirectional
5 CLK	bidirectional
7 DAT0	bidirectional
8 DAT1	bidirectional
4 VDD	bidirectional
6 VSS	bidirectional

Standard: Every pin is typed bidirectional, including VDD and VSS.

SUBMITTED FILE
schematic.glb

PART	PROPERTIES
1 DAT2	**unspecified**
2 CD/DAT3	**unspecified**
3 CMD	**unspecified**
5 CLK	**unspecified**
7 DAT0	**unspecified**
8 DAT1	**unspecified**
4 VDD	power_in
6 VSS	power_in

Submitted: The six signal pins are unspecified. VDD and VSS are power_in, which is better than the standard.

> **Issue:** With signal pins unspecified, ERC can't check them at all.

IMPACT

ERC can't check the SD bus pins at all.

FIX

- Type the data and CMD pins bidirectional and CLK input (or passive for a connector).

JIRA

Summary	693072010801 symbol: signal pins typed unspecified
Issue Type	Improvement
Priority	Low
Labels	part-verification, part-library, symbol-error, 693072010801
Components	Part Creation / Symbol Library
Description	Submitted symbol 693072010801 (ID 24152): DAT0-DAT2, CD/DAT3, CMD and CLK typed unspecified; standard types them bidirectional. VDD/VSS power_in is fine. Set signal pin types so ERC can check them.

SCH-03 LOW

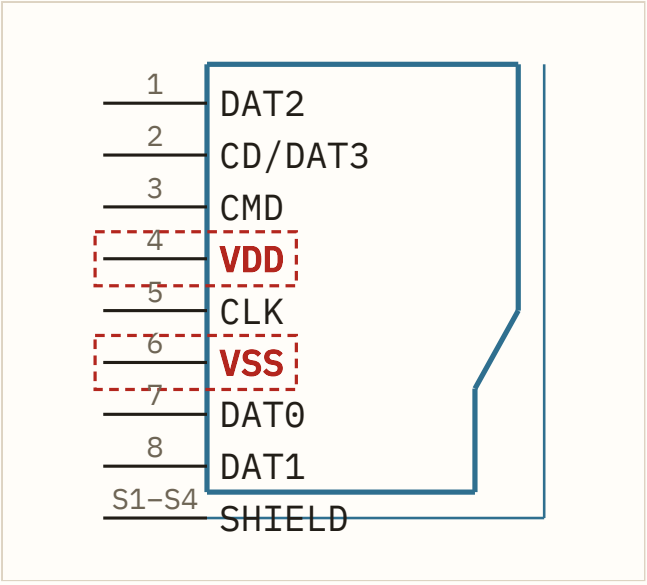
Pin order, side and shape differ from the standard

The standard symbol lists the contacts 1-8 in order down the left, with the shield at the bottom, inside a card-shaped outline (about 16.5 x 22.2 mm). The submitted symbol reorders the signal pins (CMD, CLK, DAT2, CD/DAT3, DAT0, DAT1), moves VDD and VSS to the right, and uses a plain 20.5 x 25.6 mm box, about 1.4 times the standard's area.

	STANDARD FILE	SUBMITTED
Left side	1-8 in order + shield	3, 5, 1, 2, 7, 8

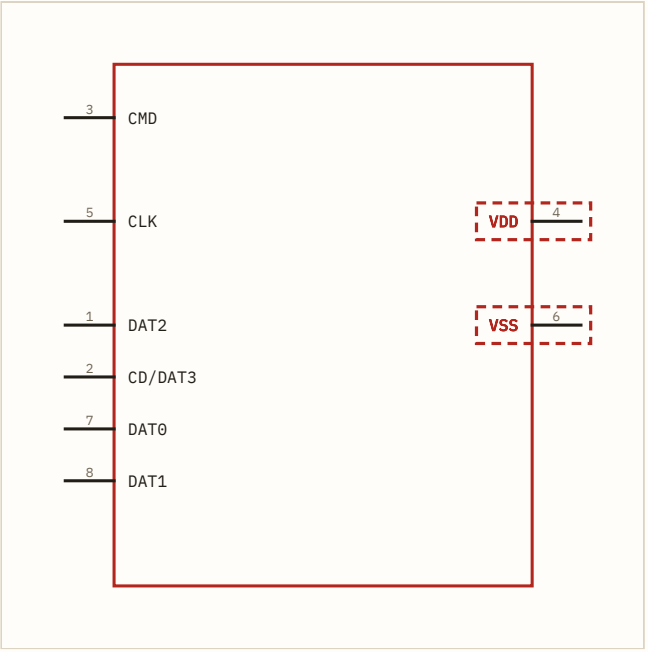
	STANDARD FILE	SUBMITTED
Right side	none	4 VDD, 6 VSS
Body	about 16.5 x 22.2 mm, card-shaped	20.47 x 25.55 mm box

STANDARD FILE
693072010801.kicad_sym



Standard: All pins in order 1–8 down the left, shield at the bottom, inside a card-shaped outline. Body about 16.5 × 22.2 mm.

SUBMITTED FILE
schematic.glb



Submitted: Signal pins on the left in a different order (CMD, CLK, DAT2, CD/DAT3, DAT0, DAT1), VDD and VSS moved to the right. Plain 20.5 × 25.6 mm box.

Issue: For a connector, the pin order should follow the contacts so the schematic matches the physical card. The submitted symbol reorders them and is about 1.4× the standard's area.

Both drawn to the same scale, from the files' own pin data. ⛶ Affected pins.

IMPACT

For a connector, a pin order that follows the contacts makes the schematic easy to check against the card. Connectivity is unaffected.

FIX

- For connectors, keep pins in contact order on one side, as the standard does.

JIRA

Summary	693072010801 symbol: pins reordered and split across two sides; plain box 1.4x the standard area
Issue Type	Improvement
Priority	Low
Labels	part-verification, part-library, symbol-layout, 693072010801
Components	Part Creation / Symbol Library
Description	Submitted symbol 693072010801 (ID 24152): signal pins reordered (CMD, CLK, DAT2, CD/DAT3, DAT0, DAT1) on the left, VDD/VSS on the right, 20.47 x 25.55 mm box. Standard keeps contacts 1-8 in order on the left in a 16.5 x 22.2 mm card-shaped outline. For connectors keep contact order on one side.

FOOTPRINT (2D)

Reference: 693072010801.kicad_mod · Submitted: twoDthreeD.glb

MATCHES THE REFERENCE

- Pins 1–8 exist and pin 1 is marked by a square pad.

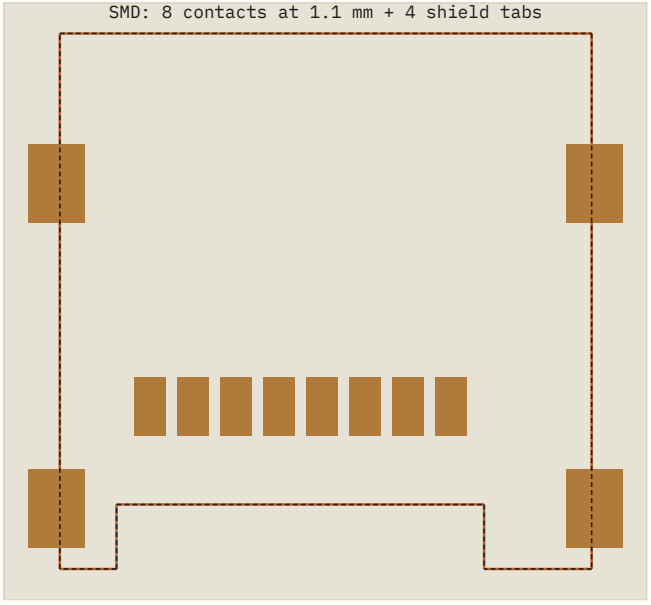
FP-01 HIGHEST

Wrong footprint: a through-hole DIP-8 instead of the SMD microSD socket

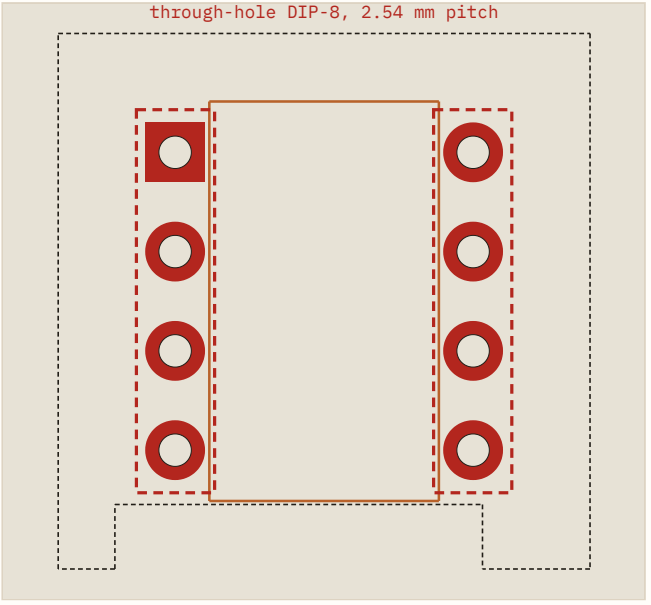
The WR-CRD hinge-type microSD socket is a 13.6 x 13.7 mm SMD part: 8 contact pads (0.8 x 1.5 mm) in one row at 1.1 mm pitch, plus 4 shield tabs (1.45 x 2.0 mm). The submitted footprint is a 300-mil DIP-8: two rows of 4 through-hole pads, 1.53 mm round (pin 1 square), 0.83 mm drills, 2.54 mm pitch, rows 7.62 mm apart.

	STANDARD FILE (MM)	SUBMITTED (MM)
Mount type	SMD	through-hole (8 drilled holes)
Contacts	1 row of 8 at 1.1 mm	2 rows of 4 at 2.54 mm
Contact pad	0.8 x 1.5 rect	1.53 round, 0.83 drill
Shield tabs	4 x (1.45 x 2.0)	none

STANDARD FILE
693072010801.kicad_mod



SUBMITTED FILE
twoDthreeD.glb

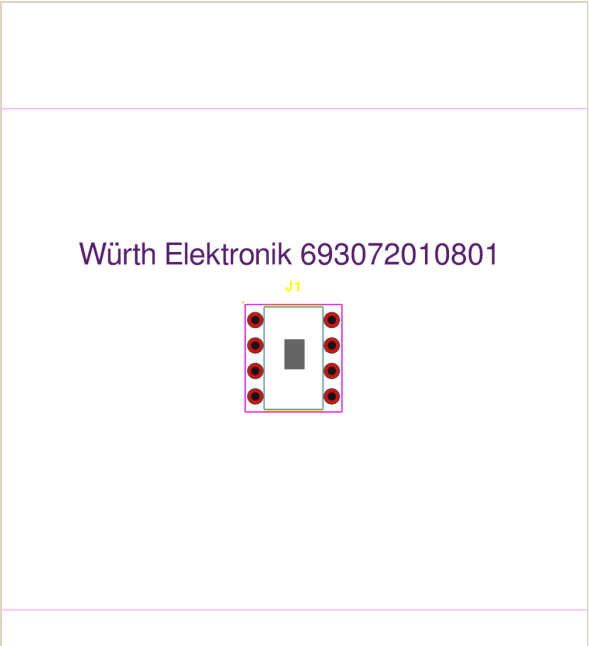


Standard: The WR-CRD hinge-type microSD socket is a 13.6 × 13.7 mm SMD part: 8 contact pads (0.8 × 1.5 mm, 1.1 mm pitch) in one row near the card slot, plus 4 shield tabs (1.45 × 2.0 mm) at the corners.

Submitted: 8 through-hole pads in two rows of 4: 1.53 mm round (pin 1 square), 0.83 mm drills, 2.54 mm pitch, rows 7.62 mm apart. This is a 300-mil DIP-8. The real connector outline is drawn dashed.

Issue: The connector can't be mounted on this footprint: wrong mount type, wrong pad positions, and no shield tabs.

Copper pad · Misplaced pad · Real connector outline (standard F.Fab) · Body outline (F.Fab) · Problem area · Top view, both drawn to the same scale from the files' own coordinates.



The submitted footprint as it renders in the editor.

IMPACT

The connector can't be mounted, and the board would get 8 drilled holes it doesn't need.

FIX

- Rebuild from Würth's recommended land pattern for 693072010801, or use the standard footprint.
- Connectors need the manufacturer's land pattern; a generic package template can't produce them.

JIRA

Summary	693072010801 footprint: through-hole DIP-8 generated instead of SMD microSD socket land pattern
Issue Type	Bug
Priority	Highest
Labels	part-verification, part-library, footprint-error, 693072010801
Components	Part Creation / Footprint Library
Description	Submitted footprint 693072010801 (ID 24152) (twoDthreeD.glb) is a 300-mil DIP-8: 1.53 mm round through-hole pads, 0.83 mm drills, 2.54 mm pitch. The part is an SMD microSD socket: 8 contacts 0.8 x 1.5 mm at 1.1 mm pitch plus 4 shield tabs 1.45 x 2.0 mm (standard SnapEDA footprint). Rebuild from Würth's land pattern.

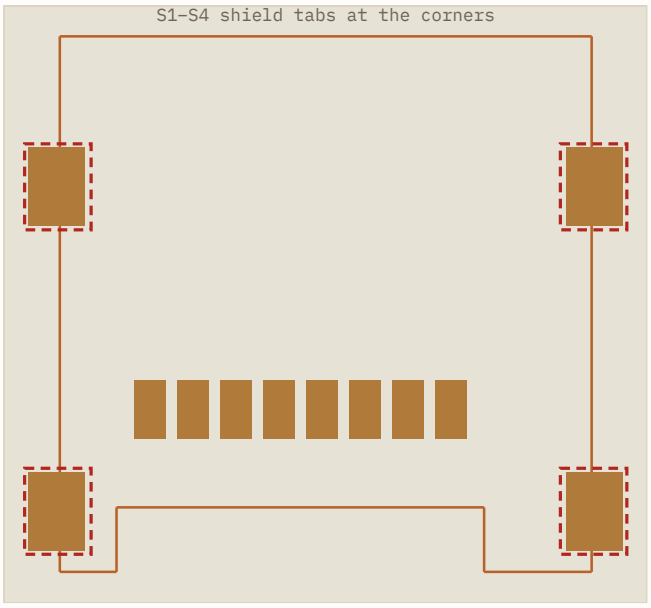
FP-02 HIGH

No shield pads

The standard footprint has four shield pads S1-S4 (1.45 x 2.0 mm) at the corners for the metal shell. The submitted footprint has none. This matches SCH-01, where the shield pins are also missing.

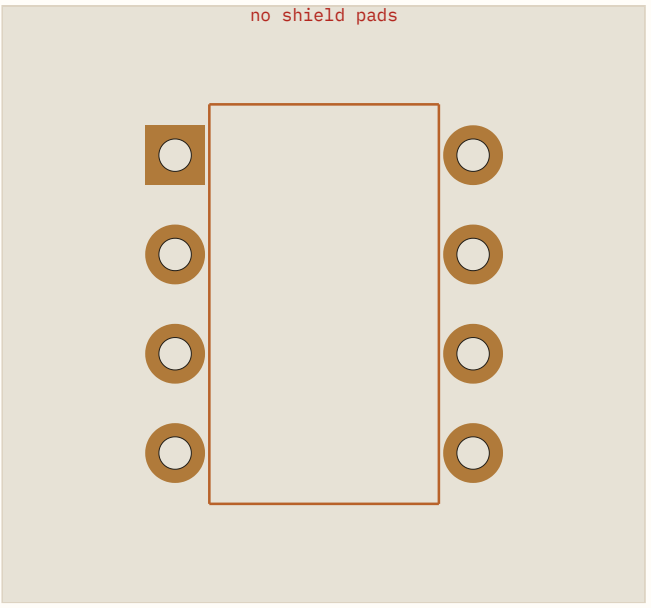
	STANDARD FILE	SUBMITTED
Shield pads	S1-S4, 1.45 x 2.0 mm	none
Pad count	12	8

STANDARD FILE
693072010801.kicad_mod



Standard: Four shield pads S1–S4 (1.45 × 2.0 mm) hold the metal shell to the board and ground it.

SUBMITTED FILE
twoDthreeD.glb



Submitted: Only the 8 numbered pads exist. There is nothing to solder the shell to.

Issue: The connector's shell isn't held or grounded, so the socket can tear off when a card is inserted, and it loses its ESD path.

Copper pad Body outline (F.Fab) Problem area · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

The shell isn't soldered down or grounded. Inserting a card puts the force on the 8 contacts, which can tear the socket off.

FIX

- Add S1-S4 and map them to the shield pins.

JIRA

Summary	693072010801 footprint: shield pads S1-S4 missing
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, footprint-error, 693072010801
Components	Part Creation / Footprint Library
Description	Submitted footprint 693072010801 (ID 24152): 8 pads, no shield pads. Standard has S1-S4 (1.45 x 2.0 mm) for the metal shell. Add them and map to the shield pins.

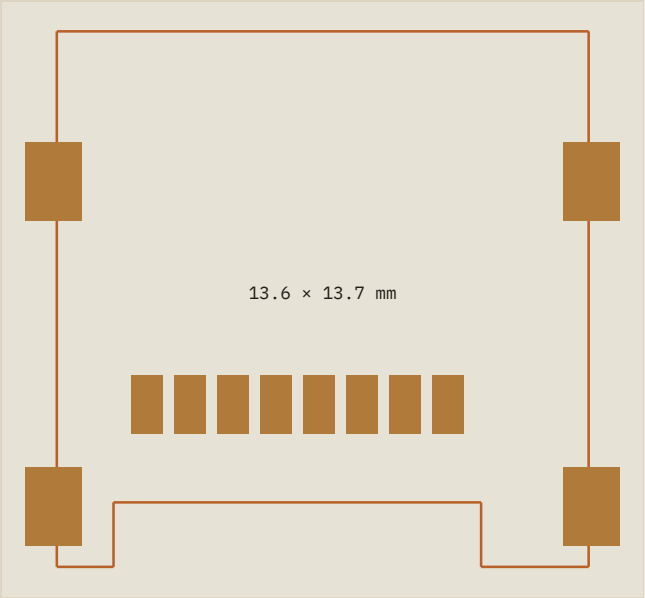
FP-03 HIGH

Body outline is 5.9 x 10.2 mm instead of 13.6 x 13.7 mm

The submitted F.Fab body is 5.87 x 10.22 mm, a DIP-8 outline. The standard draws the connector at 13.6 x 13.7 mm, with the card-slot notch.

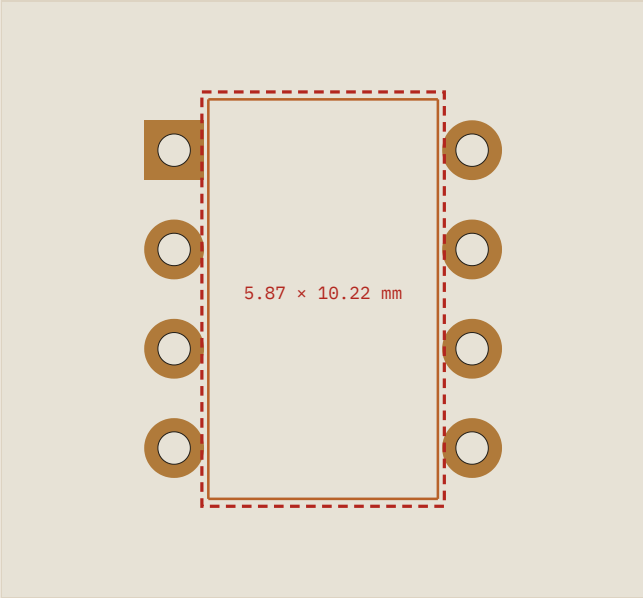
	STANDARD FILE (MM)	SUBMITTED (MM)
F.Fab body	13.6 x 13.7 (with slot notch)	5.87 x 10.22

STANDARD FILE
693072010801.kicad_mod



Standard: F.Fab outline 13.6 × 13.7 mm, with the card-slot notch on one side.

SUBMITTED FILE
twoDthreeD.glb



Submitted: F.Fab outline 5.87 × 10.22 mm, a DIP-8 body.

Issue: The layout shows a part less than half the connector's width, so neighbours get placed into its real area.

Copper pad · Body outline (F.Fab) · Problem area · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

The layout shows a part less than half the connector's width, so other parts get placed into its real area, including where the card goes.

FIX

- Draw the connector outline and card-slot side from the datasheet.

JIRA

Summary	693072010801 footprint: body outline 5.87 x 10.22 mm, connector is 13.6 x 13.7 mm
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, footprint-error, 693072010801
Components	Part Creation / Footprint Library
Description	Submitted footprint 693072010801 (ID 24152): F.Fab body 5.87 x 10.22 mm (DIP-8). Standard draws 13.6 x 13.7 mm with the card-slot notch.

FP-04 MEDIUM

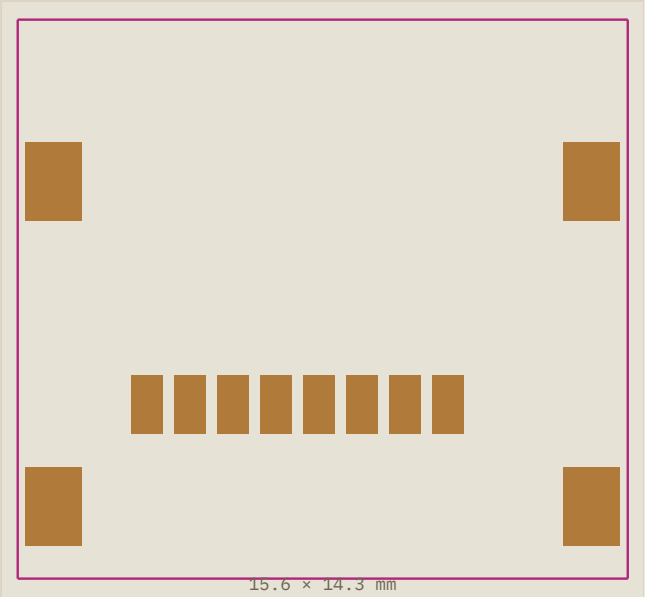
Courtyard is smaller than the real connector

The submitted courtyard is 9.65 x 10.72 mm, sized around the DIP pads. The real connector is 13.6 x 13.7 mm, and the standard courtyard is 15.6 x 14.3 mm.

	STANDARD FILE (MM)	SUBMITTED (MM)
Courtyard	15.6 x 14.3	9.65 x 10.72

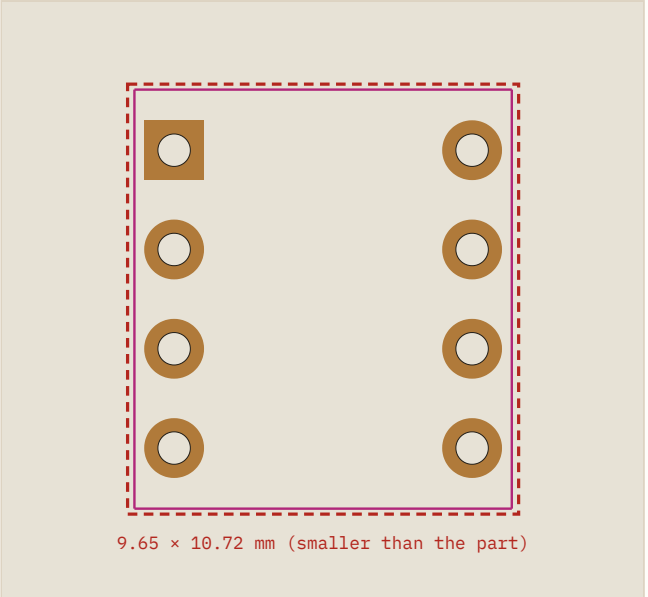
	STANDARD FILE (MM)	SUBMITTED (MM)
Connector body	13.6 x 13.7	-

STANDARD FILE
693072010801.kicad_mod



Standard: Courtyard 15.6 × 14.3 mm, around the whole connector.

SUBMITTED FILE
twoDthreeD.glb



Submitted: Courtyard 9.65 × 10.72 mm, sized around the DIP pads, smaller than the real connector body.

Issue: The overlap check can't protect the real connector area.

Copper pad Courtyard Problem area · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

The overlap check can't keep other parts out of the connector's real area.

FIX

- Recompute after FP-01 and FP-03: outline plus 0.25-0.5 mm, including the card-insertion side.

JIRA

Summary	693072010801 footprint: courtyard 9.65 x 10.72 mm, smaller than the 13.6 x 13.7 mm connector
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, footprint-error, courtyard, 693072010801
Components	Part Creation / Footprint Library
Description	Submitted footprint 693072010801 (ID 24152): F.CrtYd 9.65 x 10.72 mm around DIP pads. Real connector 13.6 x 13.7 mm; standard courtyard 15.6 x 14.3 mm.

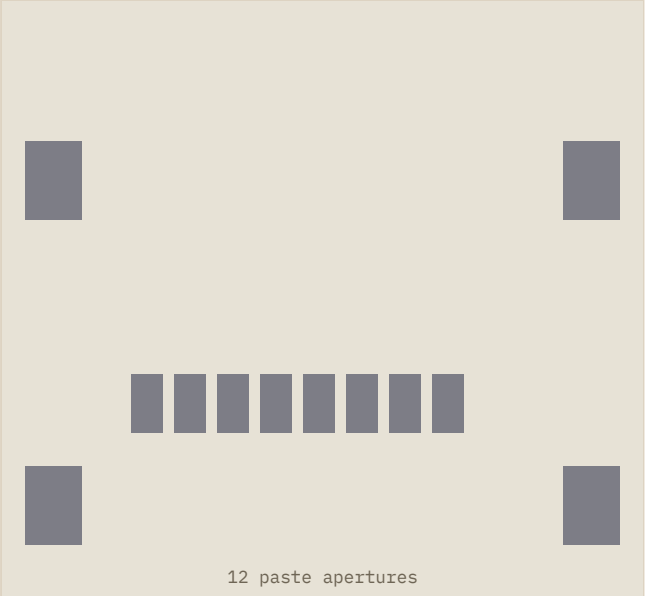
FP-05 MEDIUM

Paste-mask shapes are tagged as copper layer

As on the other parts, each pad's PasteMask nodes are copies of the copper record (layerId F.Cu). Here they also sit on through-hole pads, which shouldn't have paste at all.

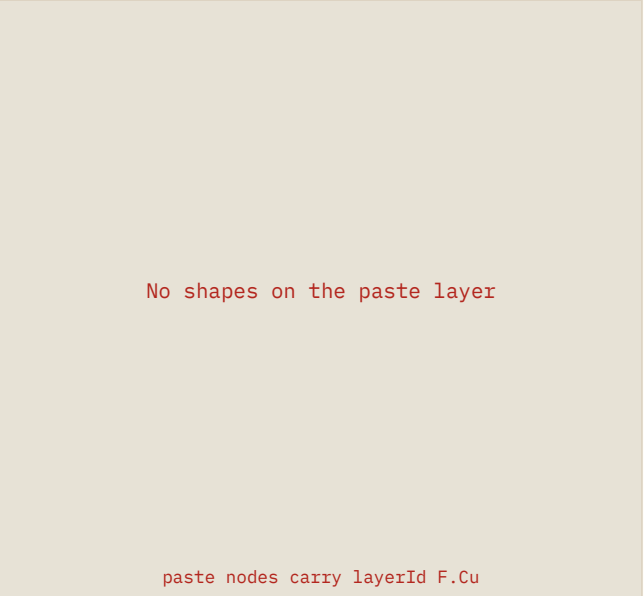
NODE	EXTRAS.NAME	EXTRAS.LAYERID	RESULT
PasteMask	CopperCirclePad	F.Cu	should be F.Paste (and absent on TH pads)

STANDARD FILE
693072010801.kicad_mod



Standard: All 12 SMD pads have paste apertures.

SUBMITTED FILE
twoDthreeD.glb



Submitted: The paste nodes are copies of the copper record (layerId F.Cu), and the pads are through-hole, which should have no paste at all.

Issue: Same generator defect as on the other three parts.

Paste aperture · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

Fourth part with the same generator defect.

FIX

- Fix once in the generator: write layerId F.Paste on PasteMask nodes, and skip paste on through-hole pads.

JIRA

Summary	693072010801 footprint: PasteMask nodes carry layerId F.Cu (4th part with this defect)
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, footprint-error, paste-mask, 693072010801
Components	Part Creation / Footprint Library
Description	Submitted footprint 693072010801 (ID 24152): PasteMask nodes are named CopperCirclePad with layerId F.Cu, and are present on through-hole pads. Same defect as parts 24150, 24153, 24154.

FP-06 LOW

Solder-mask margin declared 0.102 mm, applied 0.051 mm

Pads declare solder_mask_margin 0.102 mm, but the SolderMask shape is the pad plus 0.051 mm per side. The standard applies 0.102 mm.

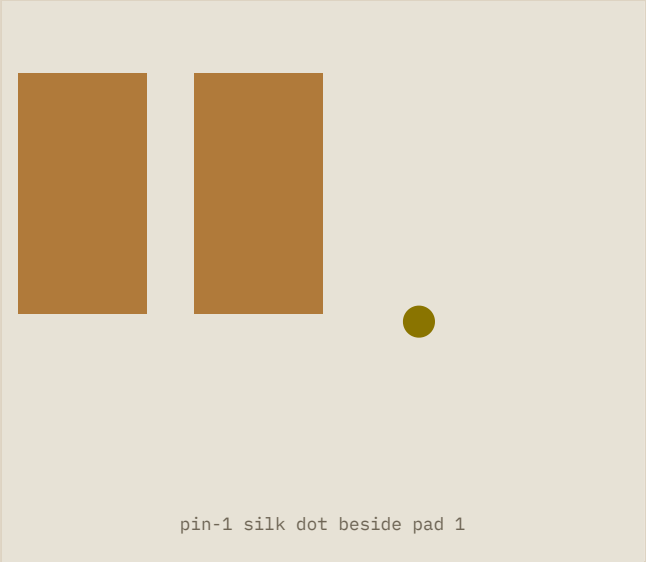
FP-07 LOW

Pin-1 marker sits outside the courtyard

Pin 1 is a square pad plus a 0.2 mm dot at (-5.03, +5.56) mm, outside the courtyard corner (±4.83, ±5.36). The standard marks pin 1 with a silk dot next to pad 1. The value text is 41 mm wide on F.Fab (see SCH-04).

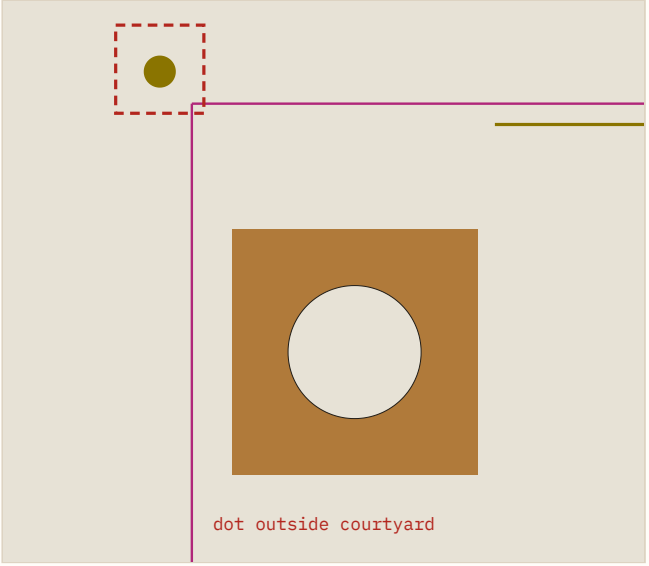
	STANDARD FILE	SUBMITTED
Pin-1 marker	silk dot beside pad 1	square pad + dot outside courtyard

STANDARD FILE
693072010801.kicad_mod



Standard: Pin 1 is marked by a silk dot next to pad 1, inside the outline.

SUBMITTED FILE
twoDthreeD.glb



Submitted: Pin 1 is a square pad plus a 0.2 mm dot at (−5.03, +5.56) mm, outside the courtyard corner (±4.83, ±5.36).

Issue: The marker sits off the part's own area. Cosmetic compared with FP-01.

Copper pad · Silkscreen · Courtyard · Pin-1 marker · Problem area · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

Cosmetic compared with FP-01.

FIX

- Place the pin-1 mark beside pad 1, inside the courtyard.

JIRA

Summary	693072010801 footprint: pin-1 dot outside courtyard
Issue Type	Improvement
Priority	Low
Labels	part-verification, part-library, footprint, 693072010801
Components	Part Creation / Footprint Library
Description	Submitted footprint 693072010801 (ID 24152): pin-1 dot at (-5.03, +5.56) mm, outside the courtyard (±4.83 x ±5.36 mm). Move it beside pad 1.

3D-01 HIGH

No 3D model

twoDthreeD.glb holds only flat footprint layers and 8 drill cylinders; there is no connector body. The standard package includes Würth's STEP model of the socket (metal shell, black housing, gold contacts, shield tabs).

	STANDARD FILE	SUBMITTED
3D model	693072010801.step	none

NOTES ON THE REFERENCE

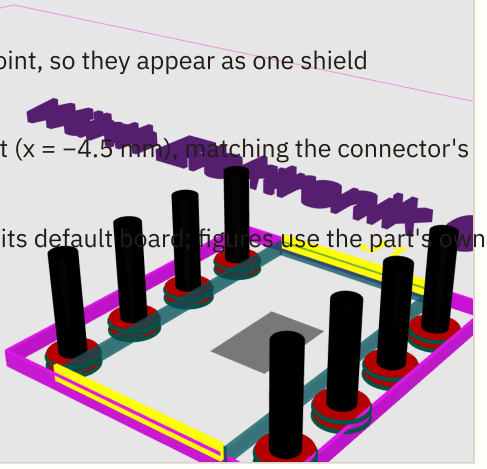
- The standard symbol uses 200 mil (5.08 mm) pins; the IDEEZA rule is 100 mil.
- The standard symbol types every pin bidirectional, including VDD and VSS. power_in for the supply pins, as in the submitted part, is better.
- In the standard symbol S1–S4 are four separate pins placed on the same point, so they appear as one shield connection. That is intended: all four tabs are the shell.
- The standard footprint runs pin 1 at the right (x = 3.2 mm) to pin 8 at the left (x = −4.5 mm), matching the connector's contact order.
- The submitted GLB places the whole part 0.10 mm left and 0.11 mm up on its default board; figures use the part's own coordinates.

STANDARD FILE
693072010801.step



Standard: Würth's model of the hinge-type microSD socket: metal shell, black housing, gold contacts and shield tabs.

SUBMITTED FILE
twoDthreeD.glb rendered in 3D



Submitted: Only flat footprint layers and 8 drill cylinders. There is no connector body.

> **Issue:** The 3D view has no connector, so the card slot direction, height and clearance can't be checked.

Left: the standard STEP file rendered. Right: the submitted GLB rendered at an angle.

IMPACT

The 3D view can't show the connector, so card-slot direction, height and enclosure clearance can't be checked.

FIX

- Attach Würth's STEP model (the standard package already has it), after the footprint is fixed.

JIRA

Summary	693072010801 part: no 3D model
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, 3d-model, 693072010801
Components	Part Creation / 3D Models
Description	Submitted part 693072010801 (ID 24152): twoDthreeD.glb contains only 2D layers and drill cylinders. The standard SnapEDA package includes 693072010801.step. Attach the model once the footprint is fixed.

PART 5 OF 6

ADV7180KST48Z

Part Analog Devices ADV7180KST48Z, 10-bit SDTV video decoder, 48-lead LQFP 7 × 7 mm (ST-48)

Submitted part library ID 24155

Reference Ultra Librarian KiCad v6 package (symbol, footprint; no 3D model), checked against the ADV7180 datasheet (Rev. I, Figure 11 / Table 12)

> **Not usable as submitted.** The symbol has wrong pins: pin 24 is named AGND instead of ELPF, pin 25 is drawn as ELPF instead of PVDD, and GPO0/GPO1 and GPO2/GPO3 are swapped (SCH-01, SCH-02). The footprint is the right shape but its pads sit 0.23 mm too far out and the silkscreen runs over 24 pads.

SCHEMATIC SYMBOL

Reference: 2026-10-05_14-12-29.kicad_sym (ADV7180KST48Z) + datasheet · Submitted: schematic.glb

PIN-BY-PIN COMPARISON

48 standard pins: 7 match, 20 differ in electrical type only, 21 with a naming, label or polarity problem

PIN	STANDARD NAME	SUBMITTED NAME	LABEL DRAWN	STANDARD TYPE	SUBMITTED TYPE	ACTIVE-LOW (STD / SUB)	RESULT
1	DGND	DGND	DGND	power_out	power_in	– / –	type differs
2	DVDDIO	DVDDIO	DVDDIO	power_in	power_in	– / –	match
3	SFL	SFL	SFL	unspecified	output	– / –	type differs
4	DVDDIO	DVDDIO	DVDDIO	power_in	power_in	– / –	match
5	GPO1	GPO0	GPO0 to GPO3	bidirectional	output	– / –	**name differs; drawn as "GPO0 to GPO3"** type differs
6	GPO0	GPO1	GPO0 to GPO3	bidirectional	output	– / –	**name differs; drawn as "GPO0 to GPO3"** type differs
7	P7	P7	P7 to P2	bidirectional	output	– / –	**drawn as "P7 to P2"** type differs
8	P6	P6	P7 to P2	bidirectional	output	– / –	**drawn as "P7 to P2"** type differs
9	P5	P5	P7 to P2	bidirectional	output	– / –	**drawn as "P7 to P2"** type differs
10	P4	P4	P7 to P2	bidirectional	output	– / –	**drawn as "P7 to P2"** type differs
11	P3	P3	P7 to P2	bidirectional	output	– / –	**drawn as "P7 to P2"** type differs
12	P2	P2	P7 to P2	bidirectional	output	– / –	**drawn as "P7 to P2"** type differs
13	DGND	DGND	DGND	power_out	power_in	– / –	type differs
14	LLC	LLC	LLC	unspecified	output	– / –	type differs
15	NC	NC	NC	no_connect	no_connect	– / –	match
16	XTAL1	XTAL1	XTAL1	unspecified	output	– / –	type differs
17	XTAL	XTAL	XTAL	unspecified	input	– / –	type differs
18	DVDD	DVDD	DVDD	power_in	power_in	– / –	match
19	DGND	DGND	DGND	power_out	power_in	– / –	type differs
20	P1	P1	P1	bidirectional	output	– / –	type differs

PIN	STANDARD NAME	SUBMITTED NAME	LABEL DRAWN	STANDARD TYPE	SUBMITTED TYPE	ACTIVE-LOW (STD / SUB)	RESULT
21	PWRDWN	PWRDWN	PWRDWN	unspecified	input	yes / –	**active-low not set** type differs
22	P0	P0	P0	bidirectional	output	– / –	type differs
23	AGND	AGND	AGND	power_out	power_in	– / –	type differs
24	ELPF	AGND	AGND	unspecified	power_in	– / –	**name differs** type differs
25	PVDD	PVDD	ELPF	power_in	passive	– / –	**drawn as "ELPF"** type differs
26	AIN1	AIN1	AIN1 to AIN6	input	input	– / –	**drawn as "AIN1 to AIN6"**
27	AIN2	AIN2	AIN1 to AIN6	input	input	– / –	**drawn as "AIN1 to AIN6"**
28	AGND	AGND	AGND	power_out	power_in	– / –	type differs
29	VREFP	VREFP	VREFP	power_in	output	– / –	type differs
30	VREFN	VREFN	VREFN	power_in	output	– / –	type differs
31	AVDD	AVDD	AVDD	power_in	power_in	– / –	match
32	AGND	AGND	AGND	power_out	power_in	– / –	type differs
33	AIN3	AIN3	AIN1 to AIN6	input	input	– / –	**drawn as "AIN1 to AIN6"**
34	AIN4	AIN4	AIN1 to AIN6	input	input	– / –	**drawn as "AIN1 to AIN6"**
35	AIN5	AIN5	AIN1 to AIN6	input	input	– / –	**drawn as "AIN1 to AIN6"**
36	AIN6	AIN6	AIN1 to AIN6	input	input	– / –	**drawn as "AIN1 to AIN6"**
37	RESET	RESET	RESET	unspecified	input	yes / yes (no bar)	**active-low set but no bar drawn** type differs
38	ALSB	ALSB	ALSB	unspecified	input	– / –	type differs
39	SDATA	SDATA	SDATA	unspecified	bidirectional	– / –	type differs
40	SCLK	SCLK	SCLK	unspecified	input	– / –	type differs
41	GPO3	GPO2	GPO0 to GPO3	bidirectional	output	– / –	**name differs; drawn as "GPO0 to GPO3"** type differs
42	GPO2	GPO3	GPO0 to GPO3	bidirectional	output	– / –	**name differs; drawn as "GPO0 to GPO3"** type differs
43	DGND	DGND	DGND	power_out	power_in	– / –	type differs
44	DVDD	DVDD	DVDD	power_in	power_in	– / –	match
45	VS/FIELD	VS/FIELD	VS/FIELD	unspecified	output	– / –	type differs
46	INTRQ	INTRQ	INTRQ	unspecified	output	yes / –	**active-low not set** type differs
47	HS	HS	HS	unspecified	output	– / –	type differs
48	NC	NC	NC	no_connect	no_connect	– / –	match

MATCHES THE REFERENCE

- 43 of 48 pins store the right name. Wrong: 5, 6, 41, 42 (swapped GPOs) and 24 (AGND instead of ELPF). Pin 25 stores PVDD correctly but is drawn as ELPF.
- Both NC pins (15, 48) are flagged no-connect.
- Pin length is 2.54 mm (100 mil), which meets the 100 mil rule. The standard uses 7.62 mm (300 mil).
- Value and MPN carry the full part number ADV7180KST48Z.
- Most electrical types follow the datasheet better than the standard (outputs as output, grounds as power_in).

SCH-01 HIGHEST

Pins 24 and 25 are wrong: ELPF is named AGND and PVDD is drawn as ELPF

Per the datasheet (Figure 11 and Table 12) and the standard symbol, pin 24 is ELPF (the PLL external loop filter) and pin 25 is PVDD (1.8 V PLL supply). In the submitted symbol, pin 24 is named AGND, typed power_in and grouped with the grounds. Pin 25 has the stored name PVDD but is drawn as "ELPF" and typed passive.

PIN	STANDARD / DATASHEET	SUBMITTED	RESULT
24	ELPF (loop filter, input)	AGND, power_in	wrong name and type
25	PVDD (PLL supply, power)	stored PVDD, drawn "ELPF", passive	wrong label and type

STANDARD FILE
ADV7180KST48Z (.kicad_sym)



Standard: Pin 24 is ELPF (PLL loop filter) and pin 25 is PVDD (1.8 V PLL supply), matching the datasheet (Figure 11, Table 12).

SUBMITTED FILE
schematic.glb



Submitted: Pin 25 is drawn as "ELPF" (stored name PVDD, typed passive). Pin 24 is named AGND and placed with the grounds on the right.

> **Issue:** The PLL supply pin is labelled as the loop filter and the loop-filter pin as ground. Wiring the symbol as drawn grounds the loop filter and puts the filter network on the 1.8 V supply pin, so the PLL won't lock.

Bottom of each symbol, each panel scaled to fit (pins 24 and 25 sit on opposite sides).

IMPACT

Wired as drawn, the loop-filter pin goes to ground and the filter network lands on the PLL supply pin. The PLL won't lock and the decoder produces no video.

FIX

- Pin 24: name ELPF, type input. Pin 25: name and label PVDD, type power_in.
- Check every pin against the datasheet's pin-configuration figure, not only the function table, which groups pins.

JIRA

Summary	ADV7180KST48Z symbol: pin 24 named AGND (should be ELPF), pin 25 drawn as ELPF (should be PVDD)
Issue Type	Bug
Priority	Highest
Labels	part-verification, part-library, symbol-error, ADV7180
Components	Part Creation / Symbol Library
Description	Submitted symbol ADV7180KST48Z (library ID 24155, schematic.glb): pin 24 is named AGND (power_in) and pin 25 is drawn as "ELPF" (stored PVDD, passive). Datasheet Figure 11 / Table 12 and the standard Ultra Librarian symbol: pin 24 = ELPF (PLL loop filter), pin 25 = PVDD (1.8 V PLL supply). As drawn, the loop filter would be grounded and the PLL could not lock. Fix names, labels and types.

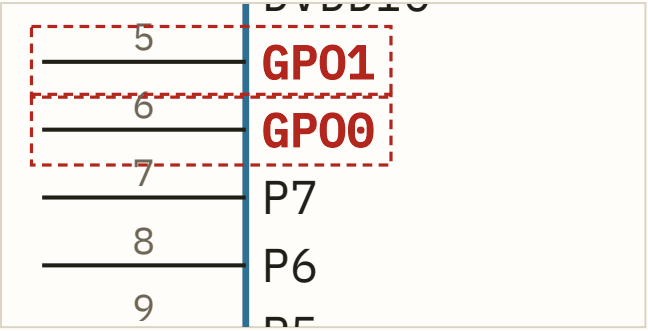
SCH-02 HIGH

GPO0/GPO1 and GPO2/GPO3 are swapped

The datasheet pin configuration (Figure 11) and the standard symbol have pin 5 = GPO1, pin 6 = GPO0, pin 41 = GPO3, pin 42 = GPO2. The submitted symbol stores pin 5 = GPO0, pin 6 = GPO1, pin 41 = GPO2, pin 42 = GPO3: both pairs are swapped. The datasheet's function table lists these as one row, "5, 6, 41, 42: GPO0 to GPO3", and the submitted part assigns the names in that ascending order.

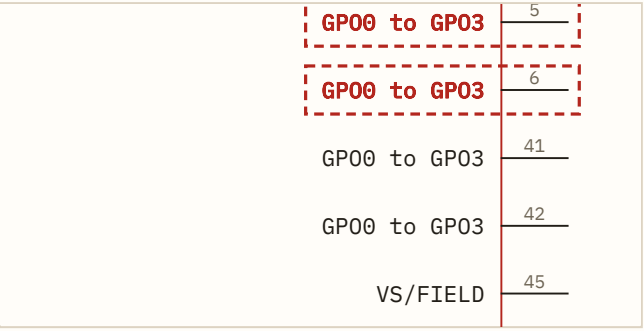
PIN	STANDARD / DATASHEET	SUBMITTED
5	GPO1	GPO0
6	GPO0	GPO1
41	GPO3	GPO2
42	GPO2	GPO3

STANDARD FILE
ADV7180KST48Z (.kicad_sym)



Standard: Pin 5 is GPO1 and pin 6 is GPO0; pin 41 is GPO3 and pin 42 is GPO2. This matches the datasheet pin configuration (Figure 11).

SUBMITTED FILE
schematic.glb



Submitted: Stored as pin 5 GPO0, pin 6 GPO1, pin 41 GPO2, pin 42 GPO3: both pairs swapped. All four are drawn as "GPO0 to GPO3".

Issue: Firmware that drives GPO0 would toggle the pin wired as GPO1, and the same for GPO2/GPO3. The swap follows the datasheet table row "5, 6, 41, 42 → GPO0 to GPO3" read in ascending order.

Both drawn to the same scale, from the files' own pin data. Affected pins.

IMPACT

Firmware that sets GPO0 toggles the pin wired as GPO1, and likewise for GPO2/GPO3. External devices controlled by these outputs get the wrong signals.

FIX

- Set pin 5 GPO1, pin 6 GPO0, pin 41 GPO3, pin 42 GPO2.
- When a datasheet table groups pins ("5, 6, 41, 42 = GPO0 to GPO3"), take the per-pin names from the pin-configuration figure.

JIRA

Summary	ADV7180KST48Z symbol: GPO0/GPO1 (pins 5/6) and GPO2/GPO3 (pins 41/42) swapped
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, symbol-error, ADV7180
Components	Part Creation / Symbol Library
Description	Submitted symbol ADV7180KST48Z (ID 24155): pins 5/6/41/42 stored as GPO0/GPO1/GPO2/GPO3. Datasheet Figure 11 and the standard symbol: 5 = GPO1, 6 = GPO0, 41 = GPO3, 42 = GPO2. The datasheet table groups them as "5, 6, 41, 42 GPO0 to GPO3", which the generator read in ascending order. Take per-pin names from the pin-configuration figure.

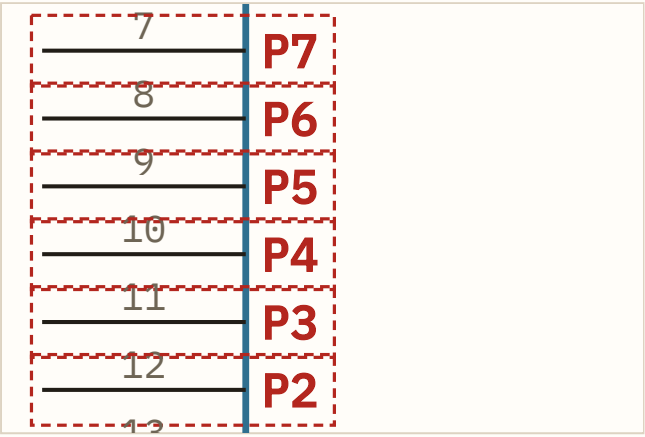
SCH-03 HIGH

16 pins display a grouped label ("P7 to P2", "AIN1 to AIN6", "GPO0 to GPO3")

As on TL16C2550, each pin stores its own name but the drawn label comes from the grouped datasheet row. Pins 7-12 all read "P7 to P2", pins 26, 27 and 33-36 all read "AIN1 to AIN6", and pins 5, 6, 41, 42 all read "GPO0 to GPO3".

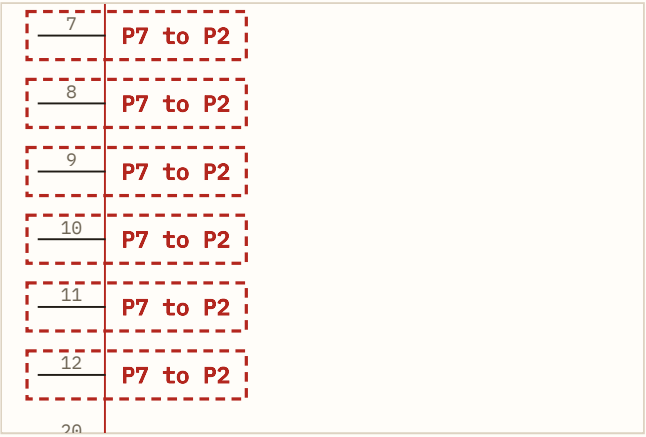
PINS	STANDARD NAMES	LABEL DRAWN
7-12	P7, P6, P5, P4, P3, P2	P7 to P2
26, 27, 33-36	AIN1, AIN2, AIN3-AIN6	AIN1 to AIN6
5, 6, 41, 42	GPO1, GPO0, GPO3, GPO2	GPO0 to GPO3

STANDARD FILE
ADV7180KST48Z (.kicad_sym)



Standard: Each pixel-port pin has its own name: P7, P6, P5, P4, P3, P2.

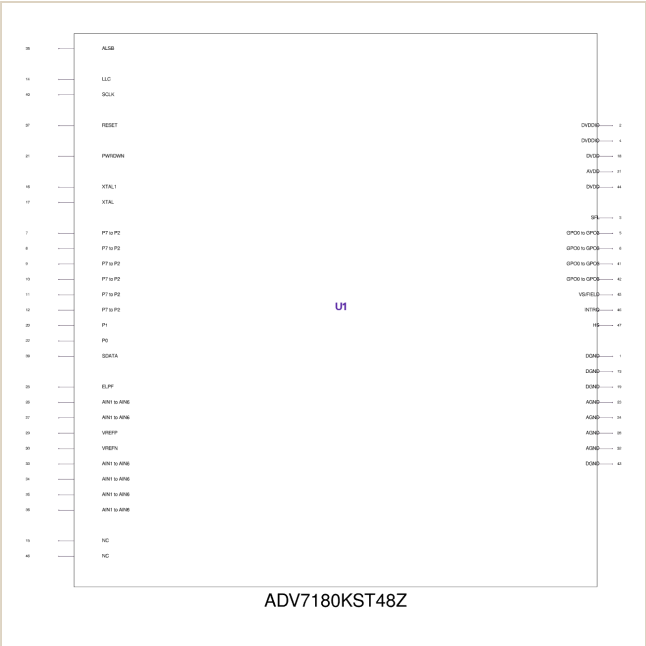
SUBMITTED FILE
schematic.glb



Submitted: All six read "P7 to P2". The same happens for AIN1–AIN6 ("AIN1 to AIN6") and GPO0–GPO3 ("GPO0 to GPO3"): 16 pins in total.

Issue: The data bus bits and video inputs can't be told apart on the symbol.

Both drawn to the same scale, from the files' own pin data. Affected pins.



Submitted symbol as rendered in the editor.

IMPACT

Pixel-bus bits, video inputs and GPOs can't be told apart on the symbol.

FIX

- Render pin labels from the per-pin name (same fix as TL16C2550 SCH-01).

JIRA

Summary	ADV7180KST48Z symbol: 16 pins show grouped labels (P7 to P2, AIN1 to AIN6, GPO0 to GPO3)
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, symbol-error, ADV7180
Components	Part Creation / Symbol Library
Description	Submitted symbol ADV7180KST48Z (ID 24155): drawn labels come from grouped datasheet rows: pins 7-12 "P7 to P2", pins 26/27/33-36 "AIN1 to AIN6", pins 5/6/41/42 "GPO0 to GPO3". Render each pin's own name. Same defect as TL16C2550 (ID 24150).

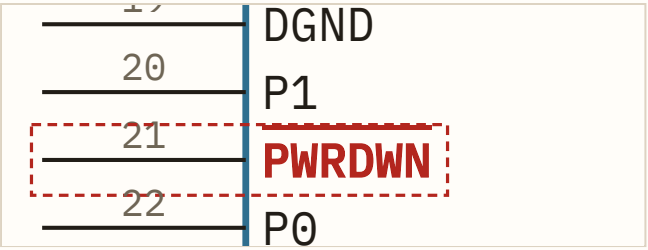
SCH-04 HIGH

Active-low marking missing on PWRDWN and INTRQ, not drawn on RESET

The standard symbol marks PWRDWN (21), RESET (37) and INTRQ (46) as active-low; the datasheet describes PWRDWN and RESET as active-low inputs. The submitted symbol has activeLow = false on PWRDWN and INTRQ. RESET is stored as active-low ("/RESET") but drawn without the bar, as on 74LVC1G99GT.

PIN	STANDARD FILE	SUBMITTED
21 PWRDWN	*PWRDWN	activeLow false
37 RESET	*RESET	activeLow true, no bar drawn
46 INTRQ	*INTRQ	activeLow false

STANDARD FILE
ADV7180KST48Z (.kicad_sym)



Standard: PWRDWN (21), RESET (37) and INTRQ (46) carry an overbar: all three are active-low.

SUBMITTED FILE
schematic.glb



Submitted: PWRDWN and INTRQ are plain labels (activeLow false). RESET is stored as active-low ("/RESET") but drawn without the bar.

Issue: Holding PWRDWN or RESET high to "enable" them keeps the decoder powered down or in reset.

Both drawn to the same scale, from the files' own pin data. Affected pins.

IMPACT

Tying PWRDWN or RESET high to enable the chip keeps it powered down or in reset.

FIX

- Set activeLow on PWRDWN and INTRQ, and render the bar whenever activeLow is true.

JIRA

Summary	ADV7180KST48Z symbol: PWRDWN/INTRQ not active-low; RESET flagged but no bar drawn
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, symbol-error, ADV7180
Components	Part Creation / Symbol Library
Description	Submitted symbol ADV7180KST48Z (ID 24155): PWRDWN (21) and INTRQ (46) have activeLow = false; RESET (37) has activeLow = true but renders without an overbar. Standard symbol marks all three active-low; datasheet confirms PWRDWN and RESET are active-low.

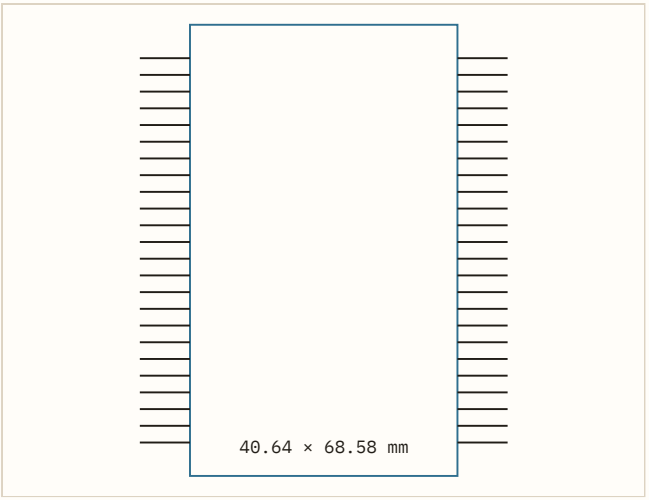
SCH-05 MEDIUM

Symbol body is 86 x 91 mm, 2.8 times the standard, with labels across the outline

The submitted body is 86.41 x 91.49 mm against 40.64 x 68.58 mm for the standard. Pins are split 30 left / 18 right. On the right side the labels are drawn across the body outline (DVDDIO, SFL, INTRQ, HS and the grounds).

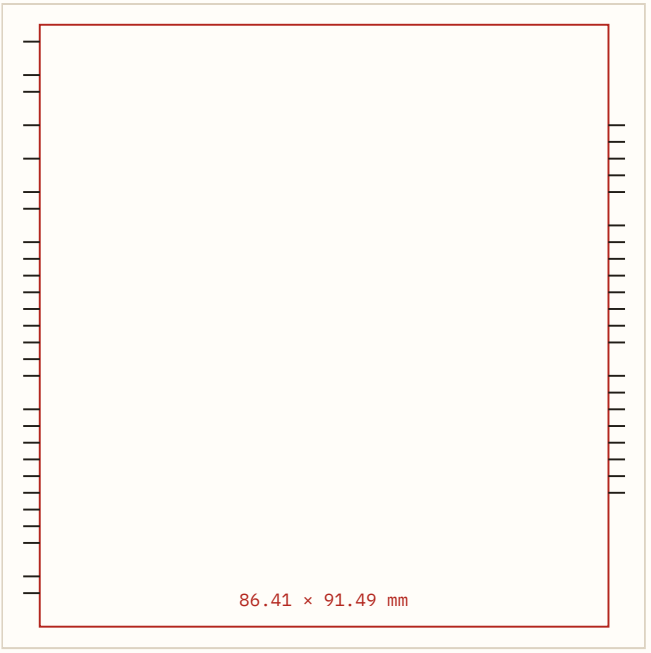
	STANDARD FILE	SUBMITTED
Body	40.64 x 68.58 mm	86.41 x 91.49 mm
Area	2,787 mm²	7,906 mm²
Left / right pins	24 / 24	30 / 18
Pin length	7.62 mm (300 mil)	2.54 mm (100 mil), meets the 100 mil rule

STANDARD FILE
ADV7180KST48Z (.kicad_sym)



Standard: Body 40.64 × 68.58 mm, pins 1–24 down the left and 25–48 up the right.

SUBMITTED FILE
schematic.glb



Submitted: Body 86.41 × 91.49 mm (2.8× the area). 30 pins on the left, 18 on the right, and right-hand labels drawn across the body outline.

Issue: The symbol takes almost half an A4 sheet and its right-side labels collide with the outline.

Both drawn to the same scale, from the files' own pin data.

IMPACT

The symbol fills almost half an A4 sheet, and the overlapping labels are hard to read.

FIX

- Size the body from the pin columns and label lengths; keep labels inside the outline.

JIRA

Summary	ADV7180KST48Z symbol: body 86.41 x 91.49 mm (2.8x standard), right labels overlap outline
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, symbol-layout, ADV7180
Components	Part Creation / Symbol Library
Description	Submitted symbol ADV7180KST48Z (ID 24155): body 86.41 x 91.49 mm vs 40.64 x 68.58 mm standard; 30/18 pin split; right-side labels drawn across the body outline. Size the body to the content and keep labels inside it.

SCH-06 LOW

Category says drivers/receivers/transceivers; footprint name gives no package

The value and MPN are correct (ADV7180KST48Z). The category is "Drivers, Receivers, Transceivers", but the ADV7180 is an SDTV video decoder. The footprint is named after the part number rather than the package (standard: ST_48_ADI).

FIELD	STANDARD FILE	SUBMITTED
Category	-	Drivers, Receivers, Transceivers
Footprint name	ST_48_ADI	ADV7180KST48Z

STANDARD FILE
ADV7180KST48Z (.kicad_sym)

PART PROPERTIES

Value / MPN	ADV7180KST48Z
Category	-
Footprint name	ST_48_ADI

Standard: The footprint is named after the package (ST-48, 7 × 7 mm LQFP).

SUBMITTED FILE
schematic.glb

PART PROPERTIES

Value / MPN	ADV7180KST48Z
Category	**Drivers, Receivers, Transceivers**
Footprint name	**ADV7180KST48Z**

Submitted: The part number is right, but the category says drivers/receivers/transceivers and the footprint name gives no package.

> **Issue:** The ADV7180 is a video decoder; library search by category won't find it.

IMPACT

The part won't be found under video ICs, and the footprint name doesn't reveal the package.

FIX

- Set the category to video decoders; name the footprint after the package (LQFP-48 7 x 7 mm / ST-48).

JIRA

Summary	ADV7180KST48Z part: wrong category (drivers/receivers); footprint name gives no package
Issue Type	Bug
Priority	Low
Labels	part-verification, part-library, metadata, ADV7180
Components	Part Creation / Symbol Library
Description	Submitted part ID 24155: category "Drivers, Receivers, Transceivers"; ADV7180 is a video decoder. Footprint named ADV7180KST48Z (standard ST_48_ADI). Value/MPN correct.

FOOTPRINT (2D)

Reference: ST_48_ADI.kicad_mod (nominal density) · Submitted: twoDthreeD.glb

MATCHES THE REFERENCE

- 48 SMD pads, 0.50 mm pitch on all four sides, square pattern.
- Pin numbering is correct: pin 1 top-left, counter-clockwise.

FP-02 HIGH

Silkscreen lines run over 24 pads

The silkscreen is a full 7.38 x 7.62 mm rectangle. Its top and bottom edges (y = ±3.81) lie on the top and bottom pad rows, whose copper starts at ±3.80. The side edges are 0.11 mm from the left and right pads. The standard uses corner marks only, clear of every pad.

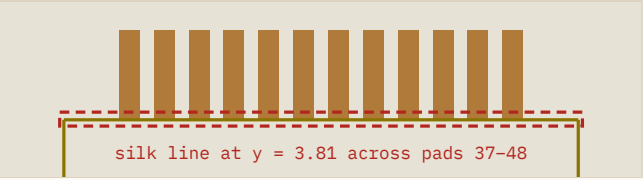
	STANDARD FILE	SUBMITTED
Silk top/bottom edge	corner marks at ±3.72	full line at ±3.81, on pads (copper from ±3.80)
Silk side edge	corner marks	±3.69, 0.11 mm from pads

STANDARD FILE
ST_48_ADI.kicad_mod



Standard: Silkscreen is corner marks at ±3.72 mm, clear of every pad.

SUBMITTED FILE
twoDthreeD.glb



Submitted: The silkscreen is a full rectangle. Its top and bottom edges (y = ±3.81) lie on the top and bottom pad rows, whose copper starts at ±3.80.

Issue: Silk printed on 24 pads (same defect as TL16C2550 FP-03).

Copper pad Silkscreen Problem area · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

Silk printed on 24 pads: either clipped by the fab or ink on the copper.

FIX

- Draw corner marks outside the pad rows, at least 0.2 mm from any pad (same fix as TL16C2550 FP-03).

JIRA

Summary	ADV7180KST48Z footprint: silkscreen crosses 24 pads
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, footprint-error, silkscreen, ADV7180
Components	Part Creation / Footprint Library
Description	Submitted footprint ADV7180KST48Z (ID 24155): silk rectangle edges at y = ±3.81 mm lie on top/bottom pad copper (from ±3.80 mm); sides 0.11 mm from pads. Standard uses corner marks only. Same defect as TL16C2550.

FP-01 MEDIUM

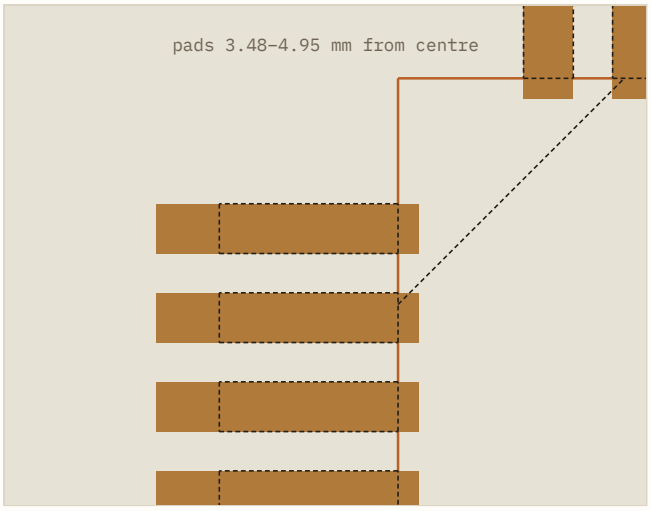
Pads shifted 0.23 mm outward: long toe, short heel

The submitted pad pattern has the right shape: 48 pads, 0.50 mm pitch, square on all four sides, correct numbering. But the pads are 1.30 x 0.29 mm centred 4.45 mm from the centre, against 1.47 x 0.28 mm at 4.22 mm in the standard. The copper starts 0.32 mm further out, so it reaches 0.50 mm past the lead tips but only about 0.2 mm in from the heel.

	STANDARD FILE (MM)	SUBMITTED (MM)
Pad centre	±4.216	±4.45
Pad size	1.473 x 0.279	1.30 x 0.29
Copper span	3.48 to 4.95	3.80 to 5.10

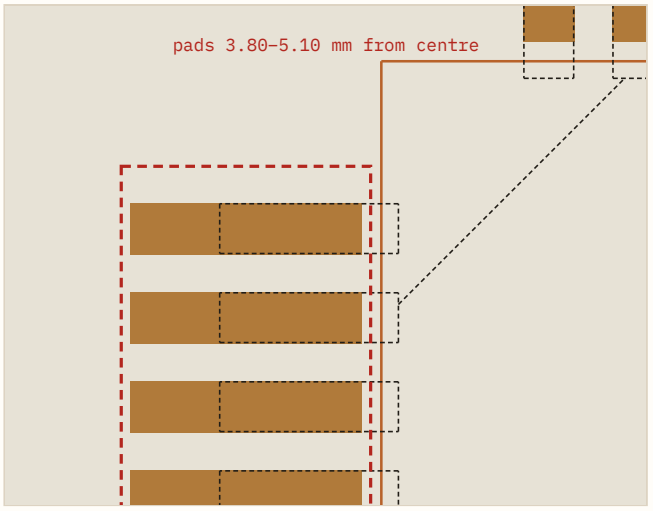
	STANDARD FILE (MM)	SUBMITTED (MM)
Pitch / sides	0.50 / 4	0.50 / 4

STANDARD FILE
ST_48_ADI.kicad_mod



Standard: Pads 1.47 × 0.28 mm centred 4.22 mm from the part centre. Each pad runs from under the lead knee out past the tip (dashed: real leads).

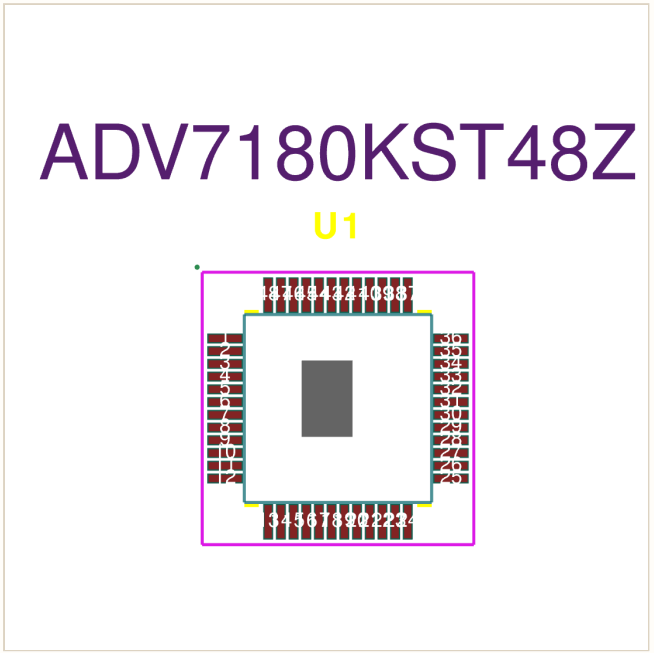
SUBMITTED FILE
twoDthreeD.glb



Submitted: Pads 1.30 × 0.29 mm centred 4.45 mm out. They start 0.32 mm further out than the standard and end 0.15 mm further out, so they reach 0.50 mm past the lead tips (standard 0.36 mm) but only about 0.2 mm in from the heel (standard about 0.5 mm).

Issue: The land is shifted outward: more toe, much less heel. On gull-wing leads the heel fillet carries most of the joint strength, so joints are weaker. All four sides are symmetric, so the pattern itself is the right shape.

Copper pad · Real lead / terminal (standard file) · Body outline (F.Fab) · Problem area · Top view, both drawn to the same scale from the files' own coordinates.



The submitted footprint as it renders in the editor.

IMPACT

FIX

Weaker solder joints: on gull-wing leads the heel fillet carries most of the joint strength.

- Use the standard land: 1.47 x 0.28 mm pads centred at ±4.22 mm (or the ADI recommended pattern for ST-48).

JIRA

Summary	ADV7180KST48Z footprint: pads centred at ±4.45 mm (standard ±4.22), heel extension too short
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, footprint-error, ADV7180
Components	Part Creation / Footprint Library
Description	Submitted footprint ADV7180KST48Z (ID 24155, twoDthreeD.glb): 48 pads 1.30 x 0.29 mm centred at ±4.45 mm; standard ST_48_ADI uses 1.473 x 0.279 mm at ±4.2164 mm. Copper spans 3.80-5.10 mm vs 3.48-4.95 mm, leaving little heel. Shape, pitch and numbering are correct.

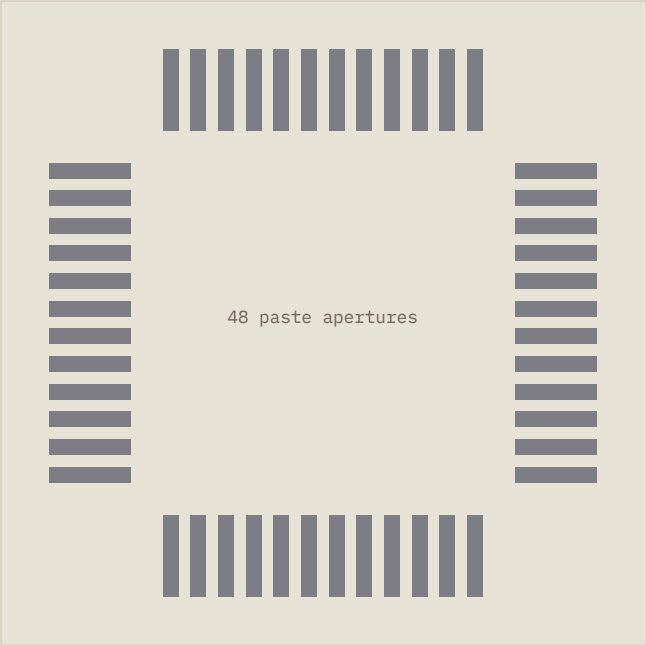
FP-04 MEDIUM

Paste-mask shapes are tagged as copper layer

Same as every other part: each pad's PasteMask node is a copy of the copper record (layerId F.Cu). Nothing is tagged F.Paste.

NODE	EXTRAS.NAME	EXTRAS.LAYERID	RESULT
PasteMask	CopperCirclePad	F.Cu	should be F.Paste

STANDARD FILE
ST_48_ADI.kicad_mod



Standard: Every pad is on F.Paste.

SUBMITTED FILE
twoDthreeD.glb



Submitted: The paste nodes are copies of the copper record (layerId F.Cu).

Issue: Same generator defect as on all other parts.

Paste aperture · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

Fifth part with the same generator defect.

FIX

- Fix once in the generator.

JIRA

Summary	ADV7180KST48Z footprint: PasteMask nodes carry layerId F.Cu (5th part)
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, footprint-error, paste-mask, ADV7180
Components	Part Creation / Footprint Library
Description	Submitted footprint ADV7180KST48Z (ID 24155): PasteMask nodes named CopperCirclePad with layerId F.Cu. Same as IDs 24150, 24152, 24153, 24154.

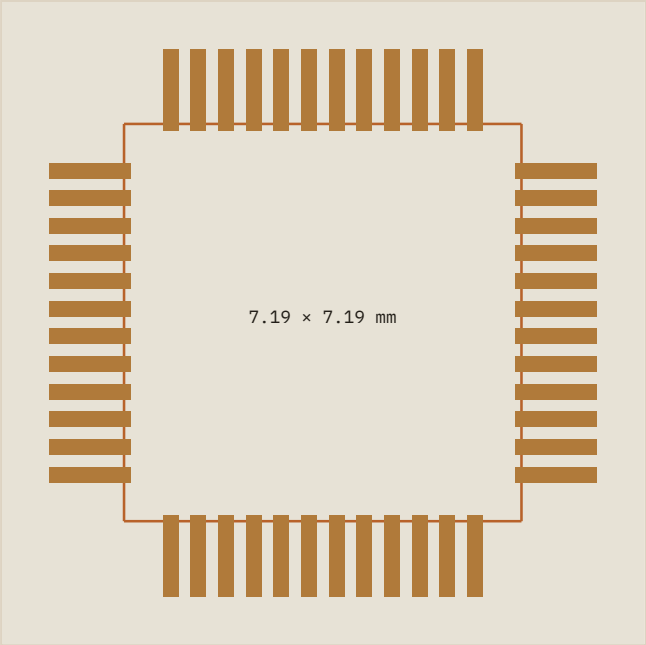
FP-03 LOW

Body outline 7.38 mm, larger than the package maximum

The F.Fab body is 7.38 x 7.38 mm. The datasheet body is 7.00 mm (7.20 max); the standard draws 7.19 mm.

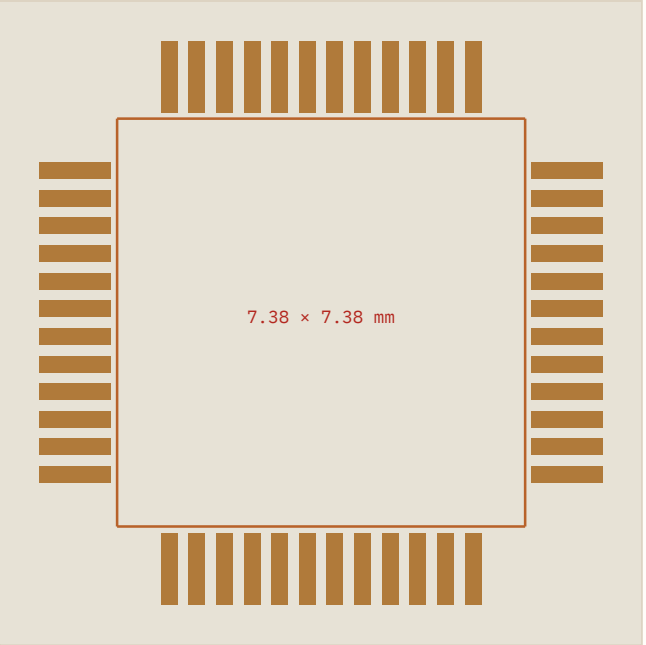
	STANDARD FILE (MM)	SUBMITTED (MM)
F.Fab body	7.19 x 7.19	7.38 x 7.38
Datasheet body	7.00 (max 7.20)	-

STANDARD FILE
ST_48_ADI.kicad_mod



Standard: F.Fab body 7.19 × 7.19 mm (datasheet 7.00 mm, max 7.20).

SUBMITTED FILE
twoDthreeD.glb



Submitted: F.Fab body 7.38 × 7.38 mm.

Issue: The body outline is 0.18 mm larger than the maximum package size. Minor.

Copper pad Body outline (F.Fab) · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

Minor; the assembly drawing shows the part slightly too large.

FIX

- Draw the body at 7.00 mm nominal.

JIRA

Summary	ADV7180KST48Z footprint: fab body 7.38 mm (package 7.00, max 7.20)
Issue Type	Improvement
Priority	Low
Labels	part-verification, part-library, footprint, ADV7180
Components	Part Creation / Footprint Library
Description	Submitted footprint ADV7180KST48Z (ID 24155): F.Fab body 7.38 x 7.38 mm; datasheet 7.00 mm (max 7.20); standard 7.19 mm.

FP-05 LOW

Solder-mask margin declared 0.102 mm, applied 0.051 mm

Pads declare 0.102 mm but the mask shape is the pad plus 0.051 mm per side. The standard sets no override.

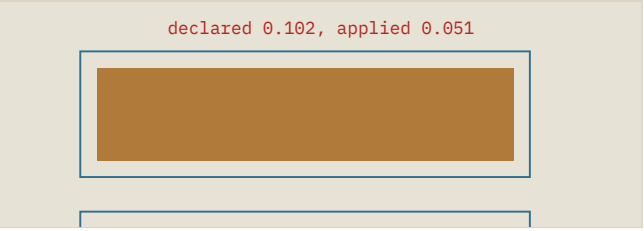
	STANDARD FILE (MM)	SUBMITTED (MM)
Declared margin	none (board default)	0.102
Applied per side	board default	0.051

STANDARD FILE
ST_48_ADI.kicad_mod



Standard: No per-footprint mask margin; the board default applies.

SUBMITTED FILE
twoDthreeD.glb



Submitted: Declares solder_mask_margin 0.102 mm, but the mask shape is the pad plus 0.051 mm per side.

Issue: Same declared-vs-applied mismatch as the other parts.

Solder-mask opening Copper pad · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

Declared and drawn values disagree.

FIX

- Store the margin actually applied, per side.

JIRA

Summary	ADV7180KST48Z footprint: solder mask applied at 0.051 mm/side while 0.102 mm declared
Issue Type	Bug
Priority	Low
Labels	part-verification, part-library, footprint, solder-mask, ADV7180
Components	Part Creation / Footprint Library
Description	Submitted footprint ADV7180KST48Z (ID 24155): solder_mask_margin declared 0.102 mm, mask mesh pad + 0.051 mm per side.

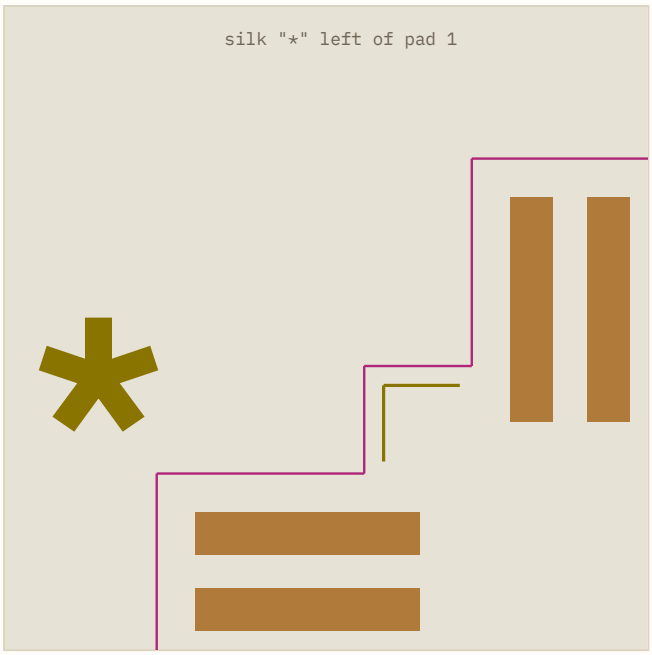
FP-06 LOW

Pin-1 marker sits outside the courtyard

Pin 1 is marked only by a 0.2 mm dot at (-5.55, +5.55) mm, outside the courtyard corner (±5.35). The standard uses a silk "*" just left of pad 1.

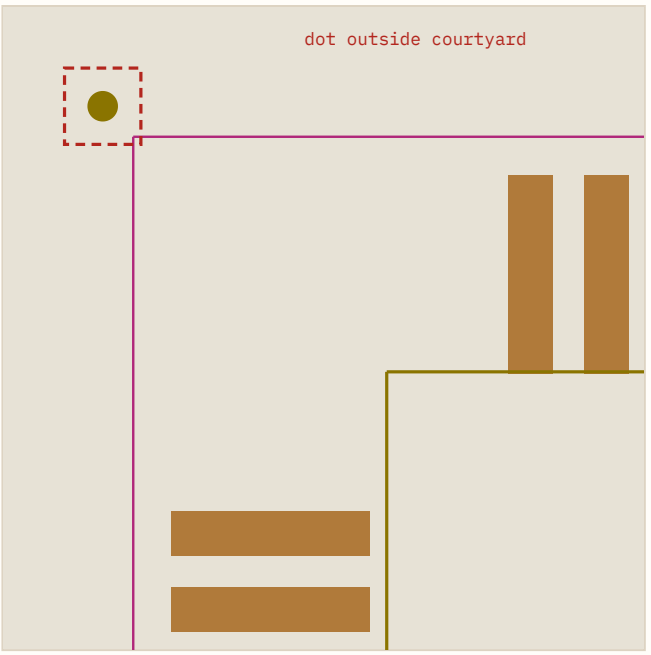
	STANDARD FILE	SUBMITTED
Pin-1 marker	silk "*" beside pad 1	dot outside courtyard

STANDARD FILE
ST_48_ADI.kicad_mod



Standard: Pin 1 is marked with a silk "*" just left of pad 1.

SUBMITTED FILE
twoDthreeD.glb



Submitted: Pin 1 is a 0.2 mm dot at (−5.55, +5.55) mm, outside the courtyard corner (±5.35).

Issue: The marker sits off the part's own area.

Copper pad · Silkscreen · Courtyard · Pin-1 marker · Problem area · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

Easy to lose; orientation errors on a 48-pin QFP are costly.

FIX

- Place the pin-1 mark beside pad 1, inside the courtyard.

FP-07 LOW

Pad numbers are larger than the pads and run into each other

Pad numbers are a fixed 0.58 mm tall; two-digit numbers are 0.77 mm wide, on 0.29 mm pads at 0.50 mm pitch, so neighbouring numbers overlap on all four sides.

	STANDARD	SUBMITTED
Number height	fits the pad	0.58 mm
Two-digit width	fits the pad	0.77 mm
Pad / pitch	0.28 / 0.50 mm	0.29 / 0.50 mm

3D-01 HIGH

No 3D model

twoDthreeD.glb holds only flat footprint layers (tallest mesh 0.8 mm); there is no package body or leads. The Ultra Librarian pack has no STEP model either.

	STANDARD	SUBMITTED
3D model	none in the KiCad pack	none
Real package	ST-48 LQFP, 7 x 7 mm, 1.6 mm max height	-

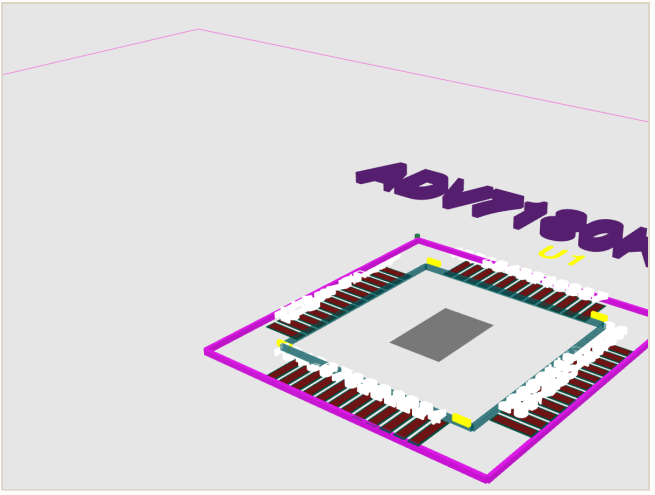
STANDARD FILE
Ultra Librarian KiCad pack

No 3D model in the standard pack.

Datasheet package ST-48: 7 × 7 mm LQFP, 9 × 9 mm over leads, 0.5 mm pitch, 1.6 mm max height.

Standard: The Ultra Librarian pack ships symbol and footprints only; the model has to come from Analog Devices.

SUBMITTED FILE
twoDthreeD.glb rendered in 3D



Submitted: Only flat footprint layers; no package body or leads.

Issue: Neither side supplies a 3D model.

Right: the submitted GLB rendered at an angle.

IMPACT

The 3D view, height and enclosure checks have no real part to use.

FIX

- Attach the ST-48 LQFP model from Analog Devices.

JIRA

Summary	ADV7180KST48Z part: no 3D model
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, 3d-model, ADV7180
Components	Part Creation / 3D Models
Description	Submitted part ADV7180KST48Z (ID 24155): twoDthreeD.glb has 2D layers only. Standard Ultra Librarian pack has no STEP either. Attach the ST-48 (7 x 7 mm LQFP) model.

NOTES ON THE REFERENCE

- The pin names were checked against the ADV7180 datasheet (Rev. I, Figure 11 pin configuration and Table 12), saved as Downloads/files30-7/ADV7180_datasheet_RevI.pdf. The standard symbol matches the datasheet on all 48 pins.
- The standard symbol uses 300 mil (7.62 mm) pins; the IDEEZA rule is 100 mil.
- Many standard pin types are generic: grounds are power_out and most signals unspecified. The submitted types for outputs and grounds are closer to the datasheet; only pins 24 and 25 are wrong (see SCH-01).
- The datasheet's own pin-configuration figure misspells pin 30 as "VFEFN"; the table and both symbols use VREFN.
- The pack contains three density variants (ADI, ADI-M, ADI-L) with the same pad centres; this log compares against the nominal one. It has no 3D model.
- The submitted GLB places the whole part 0.43 mm right and 0.12 mm up on its default board; figures use the part's own coordinates.

PART 6 OF 6

PGA2311P

Part Texas Instruments PGA2311P, stereo audio volume control, PDIP-16

Submitted part library ID 24156

Reference EasyEDA Pro project (LCSC part C1520037: symbol, footprint PDIP-16_L19.3-W6.4-P2.54-LS7.9-BL)

> **Closest to usable of the six parts, but not ready.** The footprint is the right package type (through-hole PDIP-16, correct pitch and numbering). The symbol has the right pins in the right places, but CS and MUTE lost their active-low marking and seven labels are scrambled ("V R IN" for VINR). The footprint's body outline, drill size and pin-1 silkscreen need fixing.

SCHEMATIC SYMBOL

Reference: EasyEDA symbol PGA2311P (in .epro2) · Submitted: schematic.glb

PIN-BY-PIN COMPARISON

16 standard pins: 5 match, 2 differ in electrical type only, 9 with a naming, label or polarity problem

PIN	STANDARD NAME	SUBMITTED NAME	LABEL DRAWN	STANDARD TYPE	SUBMITTED TYPE	ACTIVE-LOW (STD / SUB)	RESULT
1	ZCEN	ZCEN	ZCEN	unspecified	unspecified	– / –	match
2	CS	CS	CS	unspecified	input	yes / –	**active-low not set** type differs
3	SDI	SDI	SDI	unspecified	unspecified	– / –	match
4	VD+	VD+	V + D	unspecified	unspecified	– / –	**drawn as "V + D"**
5	DGND	DGND	DGND	unspecified	power_in	– / –	type differs
6	SCLK	SCLK	SCLK	unspecified	input	– / –	type differs
7	SDO	SDO	SDO	unspecified	unspecified	– / –	match
8	MUTE	MUTE	MUTE	unspecified	unspecified	yes / –	**active-low not set**
9	VINR	VINR	V R IN	unspecified	unspecified	– / –	**drawn as "V R IN"**
10	AGNDR	AGNDR	AGNDR	unspecified	unspecified	– / –	match
11	VOUTR	VOUTR	V R OUT	unspecified	unspecified	– / –	**drawn as "V R OUT"**
12	VA+	VA+	V + A	unspecified	unspecified	– / –	**drawn as "V + A"**
13	VA-	VA-	V - A	unspecified	unspecified	– / –	**drawn as "V - A"**
14	VOUTL	VOUTL	V L OUT	unspecified	unspecified	– / –	**drawn as "V L OUT"**
15	AGNDL	AGNDL	AGNDL	unspecified	unspecified	– / –	match
16	VINL	VINL	V L IN	unspecified	unspecified	– / –	**drawn as "V L IN"**

MATCHES THE REFERENCE

- All 16 pins store the right name and sit in the same place as the standard (1–8 down the left, 16–9 down the right).
- Body 20.5 × 25.6 mm, close to the standard's 20.3 × 22.9 mm.
- Pin length 2.54 mm (100 mil) on both, meeting the 100 mil rule. Pin lines 0.15 mm.
- Value and MPN carry PGA2311P; category Amplifiers matches the standard's tags.

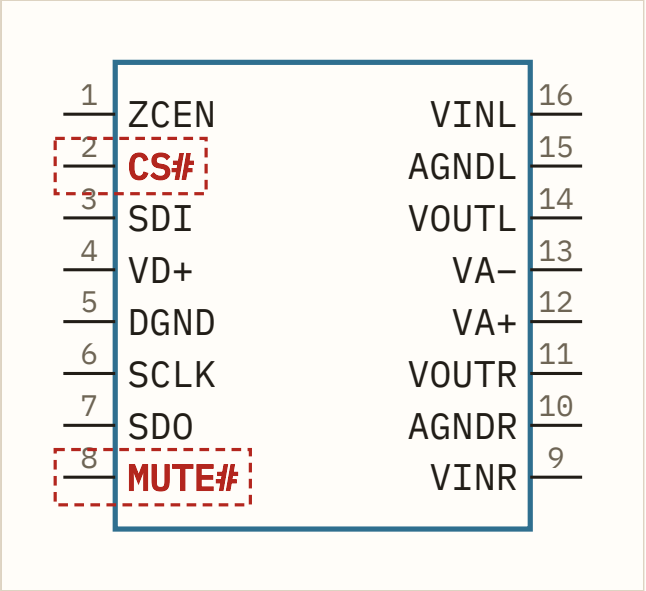
SCH-01 HIGH

CS and MUTE lost their active-low marking

The standard symbol names pins 2 and 8 "CS#" and "MUTE#" (EasyEDA's active-low notation; the TI datasheet shows both with an overbar). The submitted symbol stores and draws them as plain "CS" and "MUTE" with activeLow = false.

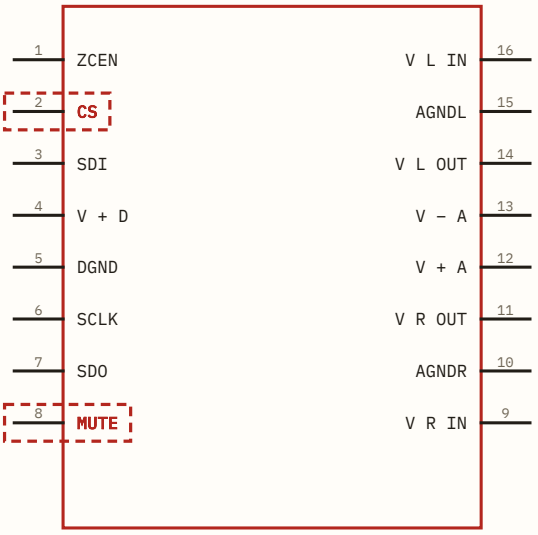
PIN	STANDARD FILE	SUBMITTED
2	CS#	CS, activeLow false
8	MUTE#	MUTE, activeLow false

STANDARD FILE
EasyEDA symbol PGA2311P (LCSC C1520037)



Standard: CS# (2) and MUTE# (8): the "#" suffix is EasyEDA's active-low notation. The TI datasheet shows both with an overbar.

SUBMITTED FILE
schematic.glb



Submitted: Drawn as plain "CS" and "MUTE", with activeLow = false on both.

> **Issue:** CS and MUTE read as active-high. Tying MUTE high to "enable" it would leave the output unmuted, and CS would be driven the wrong way.

Both drawn to the same scale, from the files' own pin data. ⬚ Affected pins.

IMPACT

MUTE reads as active-high: tying it high to "unmute" actually mutes, and the reverse. CS would be driven the wrong way by anyone reading the symbol.

FIX

- Set activeLow on pins 2 and 8 and draw the overbar.

JIRA

Summary	PGA2311P symbol: CS and MUTE not marked active-low
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, symbol-error, PGA2311
Components	Part Creation / Symbol Library
Description	Submitted symbol PGA2311P (library ID 24156, schematic.glb): pins 2 CS and 8 MUTE have activeLow = false and no bar. Standard EasyEDA symbol (LCSC C1520037) names them CS# and MUTE#; datasheet shows both active-low. Set the flag and draw the overbar.

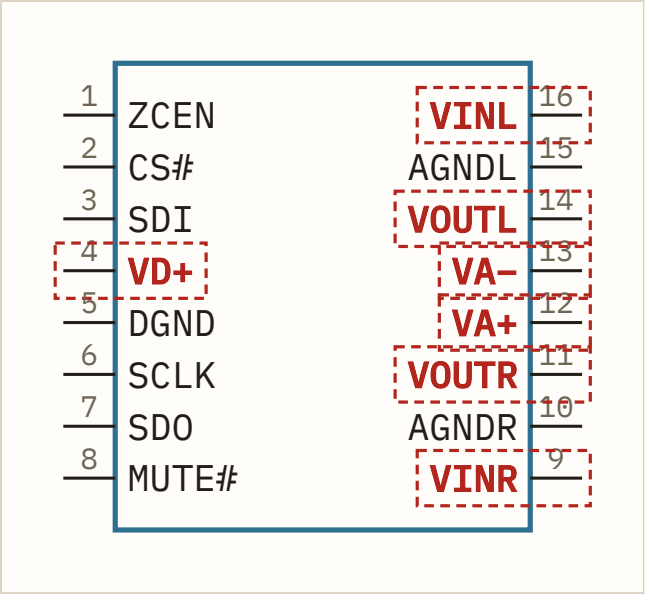
SCH-02 MEDIUM

Seven labels are scrambled ("V R IN" for VINR, "V + D" for VD+)

The stored pin names are correct, but the drawn labels for seven pins come from the datasheet's subscript formatting flattened into spaced letters in the wrong order: VD+ is drawn "V + D", VA+ "V + A", VA– "V – A", VINL "V L IN", VINR "V R IN", VOUTL "V L OUT", VOUTR "V R OUT".

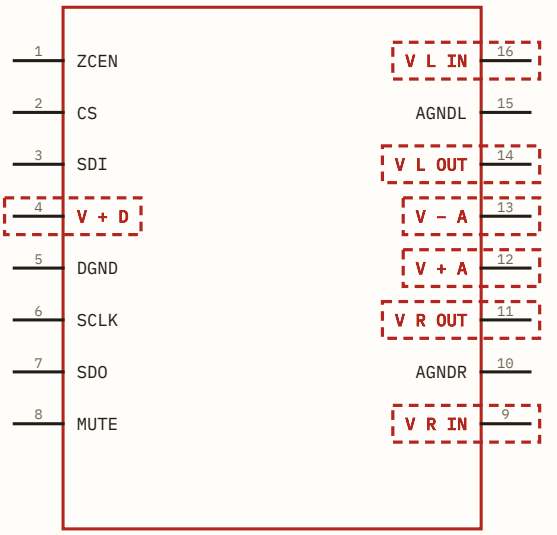
PIN	STANDARD / STORED NAME	LABEL DRAWN
4	VD+	V + D
12	VA+	V + A
13	VA–	V – A
16	VINL	V L IN
9	VINR	V R IN
14	VOUTL	V L OUT
11	VOUTR	V R OUT

STANDARD FILE
EasyEDA symbol PGA2311P (LCSC C1520037)



Standard: Supply and audio pins read VD+, VA+, VA–, VINL, VINR, VOUTL, VOUTR.

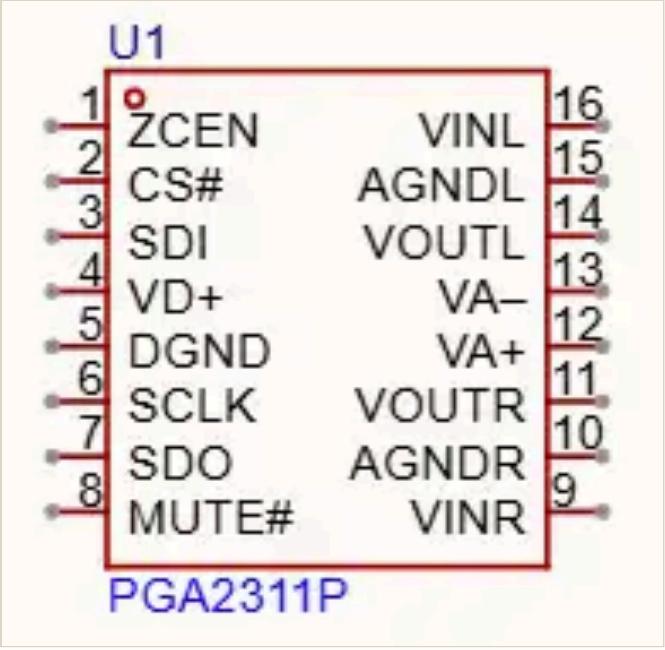
SUBMITTED FILE
schematic.glb



Submitted: The same seven pins are drawn as "V + D", "V + A", "V – A", "V L IN", "V R IN", "V L OUT", "V R OUT": the datasheet's subscripts flattened into spaced letters in the wrong order.

Issue: The labels don't match the datasheet or the stored names, and "V L IN" / "V R IN" are easy to misread when wiring left and right channels.

Both drawn to the same scale, from the files' own pin data. ⬚ Affected pins.



The standard symbol as EasyEDA renders it (from the project preview).

IMPACT

The labels don't match the datasheet; left/right channel pins are easy to misread when wiring.

FIX

- Draw the stored pin name, not the formatted display string (same root cause as the grouped labels on TL16C2550 and ADV7180).

SCH-03 LOW

Five power pins typed unspecified

VD+, VA+, VA– and the analog grounds AGNDR and AGNDL are typed unspecified. DGND is power_in and CS/SCLK are input, which is better than the standard, where every pin is Undefined.

PINS	STANDARD TYPE	SUBMITTED TYPE
4 VD+, 12 VA+, 13 VA–	Undefined	unspecified
10 AGNDR, 15 AGNDL	Undefined	unspecified
5 DGND	Undefined	power_in

STANDARD FILE
EasyEDA symbol PGA2311P (LCSC C1520037)

PART PROPERTIES

4 VD+	Undefined
12 VA+	Undefined
13 VA–	Undefined
10 AGNDR	Undefined
15 AGNDL	Undefined
5 DGND	Undefined
2 CS, 6 SCLK	Undefined

Standard: Every pin type is Undefined in the EasyEDA symbol.

SUBMITTED FILE
schematic.glb

PART PROPERTIES

4 VD+	**unspecified**
12 VA+	**unspecified**
13 VA–	**unspecified**
10 AGNDR	**unspecified**
15 AGNDL	**unspecified**
5 DGND	power_in
2 CS, 6 SCLK	input

Submitted: DGND is power_in and CS/SCLK are input, but the other supply pins (VD+, VA+, VA–) and the analog grounds (AGNDR, AGNDL) are unspecified.

Issue: ERC can't flag an unconnected supply or ground on five power pins.

IMPACT

ERC can't flag a missing supply or ground on these pins.

FIX

- Type VD+, VA+, VA– and the grounds as power_in.

JIRA

Summary	PGA2311P symbol: VD+, VA+, VA-, AGNDR, AGNDL typed unspecified
Issue Type	Improvement
Priority	Low
Labels	part-verification, part-library, symbol-error, PGA2311
Components	Part Creation / Symbol Library
Description	Submitted symbol PGA2311P (ID 24156): supply pins VD+, VA+, VA- and analog grounds AGNDR, AGNDL typed unspecified (DGND is power_in). Type them power_in so ERC can check supplies.

FOOTPRINT (2D)

Reference: EasyEDA footprint PDIP-16_L19.3-W6.4-P2.54-LS7.9-BL (in .epro2) · Submitted: twoDthreeD.glb

MATCHES THE REFERENCE

- Right package type: 16 through-hole pads, 2.54 mm pitch, two rows.
- Pin 1 is a square pad and numbering runs counter-clockwise.
- Row spacing 7.62 mm (300 mil), the PDIP nominal. The standard uses 7.94 mm to allow for lead splay; both work.

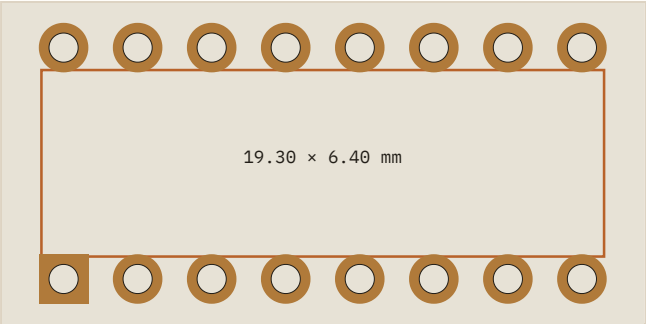
FP-01 MEDIUM

Body outline 5.87 x 20.38 mm instead of 6.4 x 19.3 mm

The submitted F.Fab body is 5.87 x 20.38 mm. The standard's component outline is 19.30 x 6.40 mm, matching the PDIP-16 package. The submitted body is 0.5 mm too narrow and 1.1 mm too long.

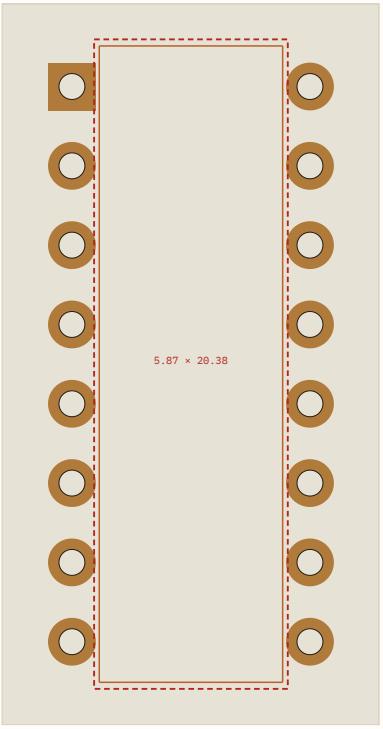
	STANDARD FILE (MM)	SUBMITTED (MM)
Body length	19.30	20.38
Body width	6.40	5.87

STANDARD FILE
EasyEDA footprint PDIP-16_L19.3-W6.4-P2.54-LS7.9-BL



Standard: Body (component shape layer) 19.30 × 6.40 mm, matching the PDIP-16 package (TI N: about 19.3 × 6.35 mm). Drawn with rows horizontal and pin 1 bottom-left.

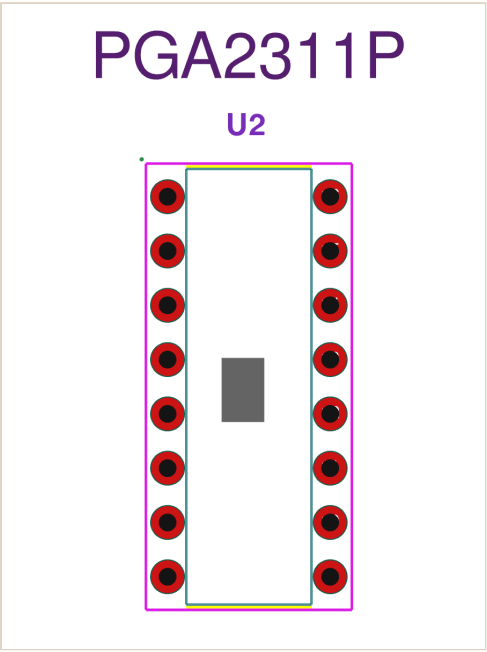
SUBMITTED FILE
twoDthreeD.glb



Submitted: F.Fab body 5.87 × 20.38 mm: 0.5 mm too narrow and 1.1 mm too long. Drawn with rows vertical and pin 1 top-left (a 90° rotation, which is fine).

Issue: The body outline doesn't match the package, so the assembly drawing and the generated 3D body are the wrong size.

Copper pad · Body outline (F.Fab) · Problem area · Top view, both drawn to the same scale from the files' own coordinates.



The submitted footprint as it renders in the editor.

IMPACT

FIX

- Draw the body at 19.3 x 6.35 mm (PDIP-16).

The assembly drawing and any generated 3D body show the wrong size.

JIRA

Summary	PGA2311P footprint: body outline 5.87 x 20.38 mm, package is about 6.4 x 19.3 mm
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, footprint-error, PGA2311
Components	Part Creation / Footprint Library
Description	Submitted footprint PGA2311P (ID 24156, twoDthreeD.glb): F.Fab body 5.87 x 20.38 mm. Standard EasyEDA PDIP-16 outline 19.30 x 6.40 mm. Pads, pitch and numbering are correct.

FP-04 MEDIUM

Paste-mask nodes on through-hole pads, tagged as copper

As on every other part, each pad carries PasteMask nodes copied from the copper record (layerId F.Cu). On a through-hole part there should be no paste at all; the standard has none.

NODE	EXTRAS.NAME	EXTRAS.LAYERID	RESULT
PasteMask	CopperCirclePad	F.Cu	should not exist on TH pads

STANDARD FILE
EasyEDA footprint PDIP-16_L19.3-W6.4-P2.54-LS7.9-BL



Standard: Through-hole pads have no paste, which is correct.

SUBMITTED FILE
twoDthreeD.glb

Submitted: Each pad carries PasteMask nodes copied from the copper record (layerId F.Cu), as on every other part.

Issue: Through-hole pads should have no paste at all; here the paste nodes are also on the wrong layer.

Paste aperture · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

Sixth part with the same generator defect.

FIX

- Fix once in the generator: F.Paste for SMD pads, no paste for through-hole pads.

JIRA

Summary	PGA2311P footprint: PasteMask nodes (layerId F.Cu) on through-hole pads
Issue Type	Bug
Priority	Medium
Labels	part-verification, part-library, footprint-error, paste-mask, PGA2311
Components	Part Creation / Footprint Library
Description	Submitted footprint PGA2311P (ID 24156): through-hole pads carry PasteMask nodes copied from copper (layerId F.Cu). TH pads need no paste. Same generator defect as the other five parts.

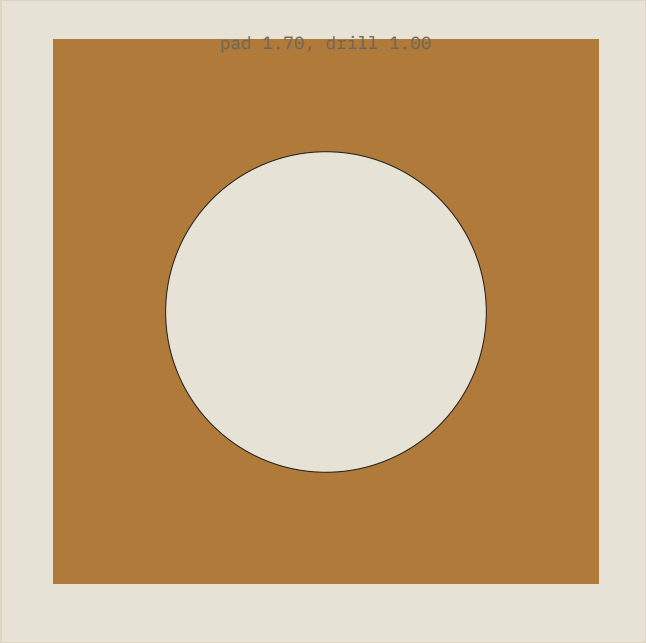
FP-02 LOW

Drill 0.83 mm is tight for PDIP leads

Pads are 1.53 mm with 0.83 mm drills; the standard uses 1.70 mm pads with 1.00 mm drills. PDIP leads are up to 0.56 x 0.36 mm, about 0.67 mm across the diagonal, so the submitted hole leaves only about 0.16 mm of clearance (standard: about 0.33 mm).

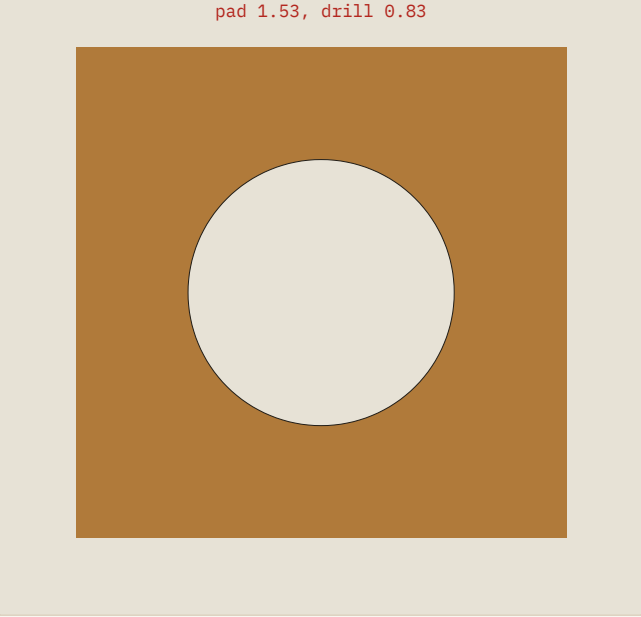
	STANDARD FILE (MM)	SUBMITTED (MM)
Pad	1.70	1.53
Drill	1.00	0.83
Clearance over lead diagonal (0.67)	about 0.33	about 0.16

STANDARD FILE
EasyEDA footprint PDIP-16_L19.3-W6.4-P2.54-LS7.9-BL



Standard: Pads 1.70 mm with 1.00 mm drills.

SUBMITTED FILE
twoDthreeD.glb



Submitted: Pads 1.53 mm with 0.83 mm drills. The PDIP lead is up to 0.56 × 0.36 mm (about 0.67 mm across the diagonal), so the hole leaves only about 0.16 mm of clearance, against 0.33 mm in the standard.

Issue: Tight holes make insertion harder, especially by machine. Workable, but tighter than the standard.

Copper pad · Top view, both drawn to the same scale from the files' own coordinates.

IMPACT

Harder insertion, especially by machine; workable but tight.

FIX

- Use a 0.9-1.0 mm drill with a 1.6-1.7 mm pad, like the standard.

JIRA

Summary	PGA2311P footprint: 0.83 mm drills (standard 1.00 mm) tight for PDIP leads
Issue Type	Improvement
Priority	Low
Labels	part-verification, part-library, footprint, PGA2311
Components	Part Creation / Footprint Library
Description	Submitted footprint PGA2311P (ID 24156): 1.53 mm pads, 0.83 mm drills. Standard 1.70 mm pads, 1.00 mm drills. With PDIP leads up to about 0.67 mm diagonal, clearance is about 0.16 mm vs 0.33 mm.

3D MODEL

Reference: EasyEDA device 3D link (model not in the export) · Submitted: twoDthreeD.glb

3D-01 HIGH

No 3D model

twoDthreeD.glb holds only flat footprint layers and 16 drill cylinders. The standard EasyEDA device links a 3D model (PDIP-16_L19.3-W6.4-P2.54-LS7.9-BL) in the LCSC library; the project export references it but doesn't include the file.

	STANDARD	SUBMITTED
3D model	linked in the EasyEDA library	none

STANDARD FILE

EasyEDA device (3D model referenced, not embedded)

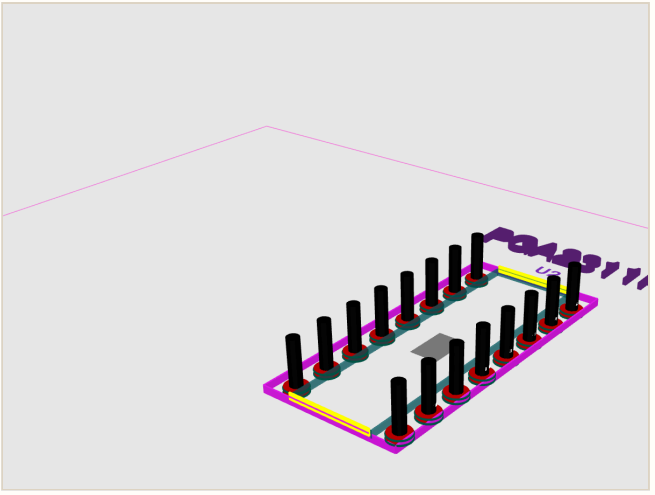
The standard EasyEDA device links a 3D model (PDIP-16_L19.3-W6.4-P2.54-LS7.9-BL), but the project export doesn't include the model file.

Package: PDIP-16, 19.3 × 6.4 mm body.

Standard: The LCSC/EasyEDA library part has a 3D model attached; it only lives in the online library.

SUBMITTED FILE

twoDthreeD.glb rendered in 3D



Submitted: Only flat footprint layers and 16 drill cylinders; no package body.

Issue: No 3D model is stored with the submitted part.

Right: the submitted GLB rendered at an angle.

IMPACT

No 3D view, height or enclosure check for this part.

FIX

- Attach a PDIP-16 3D model (the LCSC model PDIP-16_L19.3-W6.4-P2.54-LS7.9-BL can be exported from EasyEDA).

JIRA

Summary	PGA2311P part: no 3D model
Issue Type	Bug
Priority	High
Labels	part-verification, part-library, 3d-model, PGA2311
Components	Part Creation / 3D Models
Description	Submitted part PGA2311P (ID 24156): twoDthreeD.glb contains 2D layers and drill cylinders only. Standard EasyEDA device links 3D model PDIP-16_L19.3-W6.4-P2.54-LS7.9-BL (not included in the project export). Attach a PDIP-16 model.

Summary	PGA2311P footprint: solder mask applied at 0.051 mm/side while 0.102 mm declared
Issue Type	Bug
Priority	Low
Labels	part-verification, part-library, footprint, solder-mask, PGA2311
Components	Part Creation / Footprint Library
Description	Submitted footprint PGA2311P (ID 24156): solder_mask_margin declared 0.102 mm, mask mesh pad + 0.051 mm per side.

NOTES ON THE REFERENCE

- The standard is an EasyEDA Pro project (.epro2) containing several parts; this comparison uses the PGA2311P device (LCSC C1520037), its symbol and its PDIP-16 footprint. The other parts in the project (AD712KNZ, INA180A2IDBVR, a resistor) were not compared.
- EasyEDA marks active-low pins with a "#" suffix (CS#, MUTE#) instead of an overbar. Pin names otherwise match the TI datasheet pin-out.
- All standard pin types are "Undefined".
- The standard silkscreen outline (±3.20 mm) touches the outer edge of the pad copper (pads reach ±3.12 mm) by about 0.08 mm; EasyEDA normally clips silk over pads at export.
- The standard footprint is drawn with the pin rows horizontal (pin 1 bottom-left); the submitted one with rows vertical (pin 1 top-left). That is a 90° placement rotation, not an error.
- The project's PCB preview image is an empty board outline; the standard images in this section are drawn from the footprint and symbol data, plus EasyEDA's own schematic preview.
- The submitted GLB places the whole part 0.28 mm right and 0.65 mm up on its default board; figures use the part's own coordinates.